1–100 阶有限群分类

1 阶群

只有一个同构类型：1，编号 $[1,1]$。

2 阶群

只有一个同构类型：C_2，编号 $[2,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 2$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 2$。因此 $G = \langle a \rangle \cong C_2$。 □

3 阶群

只有一个同构类型：C_3，编号 $[3,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 3$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 3$。因此 $G = \langle a \rangle \cong C_3$。 □

4 阶群

共有 2 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[4,1]	C_4 $(4,1,4)$ 阿贝尔。
[4,2]	$C_2 \times C_2$ $(4,1,2)$ 阿贝尔。

证明. 这里 $|G| = 4 = 2^2$。非平凡有限 p 群的中心非平凡，故 $|\mathrm{Z}(G)| = p$ 或 p^2。若中心阶为 p，则 $G/\mathrm{Z}(G)$ 循环；将任意两个元素写成 $a^i z, a^j w$ $(z, w \in \mathrm{Z}(G))$，可知它们交换。因此 G 总是阿贝尔群。若有 p^2 阶元素，则为 C_{p^2}；否则取 $a \neq 1$ 及 $b \notin \langle a \rangle$，便有 $G = \langle a \rangle \times \langle b \rangle \cong C_p^2$。两者的最大元素阶不同，故互不同构。 □

5 阶群

只有一个同构类型：C_5，编号 $[5,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 5$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 5$。因此 $G = \langle a \rangle \cong C_5$。 □

6 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[6,1]	S_3 $(1,3,6)$ $R=(2,3)$; $[b,a]=b$
[6,2]	C_6 $(6,1,6)$ 阿贝尔。

证明. 这里 $p=2<q=3$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q=1$。令 $N=C_q \trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q\rtimes C_p$。作用由同态 $C_p\to \mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $2\mid 2$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

7 阶群

只有一个同构类型：C_7，编号 $[7,1]$。

证明. 任取非单位元 $a\in G$。由拉格朗日定理，$\mathrm{ord}(a)\mid 7$，又 $\mathrm{ord}(a)>1$，所以 $\mathrm{ord}(a)=7$。因此 $G=\langle a\rangle\cong C_7$。 □

8 阶群

共有 5 个同构类型，其中 3 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[8,1]	C_8 $(8,1,8)$ 阿贝尔。
[8,2]	$C_2\times C_4$ $(8,1,4)$ 阿贝尔。
[8,3]	D_8 $(2,2,4)$ $R=(2,2,2)$; $[b,a]=c$
[8,4]	Q_8 $(2,2,4)$ $R=(2,2,2)$; $a^2=c$; $b^2=c$; $[b,a]=c$
[8,5]	$C_2\times C_2\times C_2$ $(8,1,2)$ 阿贝尔。

证明. 三个阿贝尔类型为 $C_8, C_4\times C_2, C_2^3$。若 G 非阿贝尔，则不能有 8 阶元素；也不可能所有非单位元都为 2 阶，否则 $(xy)^{-1}=xy=yx$ 会使 G 交换。因此存在 4 阶元素 a，且 $\langle a\rangle$ 指数为 2，从而正规。

取 $b\notin\langle a\rangle$。共轭必须满足 $bab^{-1}=a^{-1}$，否则 a,b 交换。又 $b^2\in\langle a\rangle$，且 b 与 b^2 交换，所以 $b^2=1$ 或 a^2。两种情形分别给出

$$D_8=\langle a,b\mid a^4=b^2=1,\ bab^{-1}=a^{-1}\rangle,$$

$$Q_8=\langle a,b\mid a^4=1,\ b^2=a^2,\ bab^{-1}=a^{-1}\rangle.$$

于是全部五种为 $C_8, C_4\times C_2, C_2^3, D_8, Q_8$。阿贝尔三种由指数区分，非阿贝尔两种由 2 阶元素个数区分。 □

9 阶群

共有 2 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[9,1]	$C_9 \quad (9,1,9) \quad$ 阿贝尔。
[9,2]	$C_3 \times C_3 \quad (9,1,3) \quad$ 阿贝尔。

证明. 这里 $|G| = 9 = 3^2$。非平凡有限 p 群的中心非平凡，故 $|\mathrm{Z}(G)| = p$ 或 p^2。若中心阶为 p，则 $G/\mathrm{Z}(G)$ 循环；将任意两个元素写成 $a^i z, a^j w$ $(z, w \in \mathrm{Z}(G))$，可知它们交换。因此 G 总是阿贝尔群。若有 p^2 阶元素，则为 C_{p^2}；否则取 $a \neq 1$ 及 $b \notin \langle a \rangle$，便有 $G = \langle a \rangle \times \langle b \rangle \cong C_p^2$。两者的最大元素阶不同，故互不同构。 □

10 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[10,1]	$D_{10} \quad (1,5,10)$ $R = (2,5);\ [b,a] = b^3$
[10,2]	$C_{10} \quad (10,1,10) \quad$ 阿贝尔。

证明. 这里 $p = 2 < q = 5$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q = 1$。令 $N = C_q \trianglelefteq G$，取 p 阶子群 $H = C_p$，则 $N \cap H = 1$，$NH = G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $2 \mid 4$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

11 阶群

只有一个同构类型：C_{11}，编号 $[11,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 11$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 11$。因此 $G = \langle a \rangle \cong C_{11}$。 □

12 阶群

共有 5 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[12,1]	$\mathrm{Dic}_3 \quad (2,3,12)$ $R = (2,2,3);\ a^2 = b;\ [c,a] = c$
[12,2]	$C_{12} \quad (12,1,12) \quad$ 阿贝尔。

编号	12 阶群（续）
[12,3]	$A_4 \quad (1,4,6)$ $R=(3,2,2);\ [b,a]=bc;\ [c,a]=b$
[12,4]	$D_{12} \quad (2,3,6)$ $R=(2,2,3);\ [c,a]=c$
[12,5]	$C_2\times C_6 \quad (12,1,6) \quad$ 阿贝尔。

证明. 设 $|G|=12$，则 $n_3=1$ 或 4。

若 $n_3=4$，共轭作用在四个 Sylow 3 子群上给出 $G\to S_4$。其像的阶被 4 整除，核的阶只能为 1 或 3。若核为 3 阶，它就是正规 Sylow 3 子群，矛盾。因此该作用忠实，G 同构于 S_4 的一个 12 阶子群。这样的子群指数为 2；所有换位共轭且生成 S_4，故对应的非平凡同态 $S_4\to C_2$ 必为符号同态。于是 $G\cong A_4$。

若 $n_3=1$，则 $N=C_3\trianglelefteq G$。取 Sylow 2 子群 H，有 $G=N\rtimes H$。H 为 C_4 或 C_2^2，作用取值于 $\mathrm{Aut}(C_3)=C_2$。

- $H=C_4$：平凡作用给出 C_{12}；非平凡作用给出 $C_3\rtimes C_4\cong \mathrm{Dic}_3$。
- $H=C_2^2$：平凡作用给出 $C_6\times C_2$；非平凡作用的核为一个 2 阶子群，所有此类核在 $\mathrm{Aut}(C_2^2)$ 下等价，得到 $S_3\times C_2\cong D_{12}$。

综上，五种为 $C_{12}, C_6\times C_2, D_{12}, A_4, \mathrm{Dic}_3$。其中三个非阿贝尔群的 2 阶元素数依次为 $7,3,1$，因此互不同构。 □

13 阶群

只有一个同构类型：C_{13}，编号 $[13,1]$。

证明. 任取非单位元 $a\in G$。由拉格朗日定理，$\mathrm{ord}(a)\mid 13$，又 $\mathrm{ord}(a)>1$，所以 $\mathrm{ord}(a)=13$。因此 $G=\langle a\rangle\cong C_{13}$。 □

14 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[14,1]	$D_{14} \quad (1,7,14)$ $R=(2,7);\ [b,a]=b^5$
[14,2]	$C_{14} \quad (14,1,14) \quad$ 阿贝尔。

证明. 这里 $p=2<q=7$。Sylow 定理给出 $n_q\mid p$ 且 $n_q\equiv 1 \pmod q$，所以 $n_q=1$。令 $N=C_q\trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q\rtimes C_p$。作用由同态 $C_p\to \mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $2\mid 6$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

15 阶群

只有一个同构类型：C_{15}，编号 $[15,1]$。

证明. 这里 $p=3<q=5$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q=1$。令 $N=C_q \trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q\rtimes C_p$。作用由同态 $C_p\to \mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $3\nmid 4$，作用只能平凡，故只有 $C_q\times C_p\cong C_{15}$。 □

16 阶群

共有 14 个同构类型，其中 5 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[16,1]	C_{16} $(16,1,16)$ 阿贝尔。
[16,2]	$C_4\times C_4$ $(16,1,4)$ 阿贝尔。
[16,3]	$(C_4\times C_2)\rtimes C_2$ $(4,2,4)$ $R=(2,2,2,2);\ a^2=d;\ [b,a]=c$
[16,4]	$C_4\rtimes C_4$ $(4,2,4)$ $R=(2,2,2,2);\ a^2=d;\ b^2=c;\ [b,a]=c$
[16,5]	$C_2\times C_8$ $(16,1,8)$ 阿贝尔。
[16,6]	$C_8\rtimes C_2$ $(4,2,8)$ $R=(2,2,2,2);\ a^2=c;\ c^2=d;\ [b,a]=d$
[16,7]	D_{16} $(2,4,8)$ $R=(2,2,2,2);\ c^2=d;\ [b,a]=c;\ [c,a]=d;\ [c,b]=d$
[16,8]	QD_{16} $(2,4,8)$ $R=(2,2,2,2);\ a^2=d;\ c^2=d;\ [b,a]=c;\ [c,a]=d;\ [c,b]=d$
[16,9]	Q_{16} $(2,4,8)$ $R=(2,2,2,2);\ a^2=d;\ b^2=d;\ c^2=d;\ [b,a]=c;\ [c,a]=d;\ [c,b]=d$
[16,10]	$C_2\times C_2\times C_4$ $(16,1,4)$ 阿贝尔。
[16,11]	$C_2\times D_8$ $(4,2,4)$ $R=(2,2,2,2);\ [b,a]=d$
[16,12]	$C_2\times Q_8$ $(4,2,4)$ $R=(2,2,2,2);\ a^2=d;\ b^2=d;\ [b,a]=d$
[16,13]	$(C_4\times C_2)\rtimes C_2$ $(4,2,4)$ $R=(2,2,2,2);\ c^2=d;\ [b,a]=d$
[16,14]	$C_2\times C_2\times C_2\times C_2$ $(16,1,2)$ 阿贝尔。

17 阶群

只有一个同构类型：C_{17}，编号 $[17,1]$。

证明. 任取非单位元 $a\in G$。由拉格朗日定理，$\mathrm{ord}(a)\mid 17$，又 $\mathrm{ord}(a)>1$，所以 $\mathrm{ord}(a)=17$。因此 $G=\langle a\rangle\cong C_{17}$。 □

18 阶群

共有 5 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[18,1]	D_{18} $(1,9,18)$ $R=(2,3,3)$; $b^3=c$; $[b,a]=bc^2$; $[c,a]=c$
[18,2]	C_{18} $(18,1,18)$ 阿贝尔。
[18,3]	$C_3\times S_3$ $(3,3,6)$ $R=(2,3,3)$; $[c,a]=c$
[18,4]	$(C_3\times C_3)\rtimes C_2$ $(1,9,6)$ $R=(2,3,3)$; $[b,a]=b$; $[c,a]=c$
[18,5]	$C_3\times C_6$ $(18,1,6)$ 阿贝尔。

证明. $n_3\mid 2$ 且 $n_3\equiv 1 \pmod 3$，故 Sylow 3 子群 N 正规，$G=N\rtimes C_2$。

若 $N=C_9$，$\mathrm{Aut}(C_9)=C_6$ 中只有一个非平凡的 2 阶元素，得到 C_{18} 与 D_{18}。

若 $N=C_3^2$，作用由满足 $T^2=I$ 的矩阵给出。在 $\mathbb{F}_3$ 上，多项式 $(X-1)(X+1)$ 无重根，故 T 可对角化。按 -1 特征空间的维数 $0,1,2$，恰得

$$C_6\times C_3,\qquad S_3\times C_3,\qquad C_3^2\rtimes_{-I} C_2.$$

三者分别为阿贝尔群、中心为 C_3 的非阿贝尔群、中心平凡的非阿贝尔群。最后一个群中所有 9 个 N 外元素均为 2 阶；它也不与 D_{18} 同构：后者有 9 阶元素，前者没有。 □

19 阶群

只有一个同构类型：C_{19}，编号 $[19,1]$。

证明. 任取非单位元 $a\in G$。由拉格朗日定理，$\mathrm{ord}(a)\mid 19$，又 $\mathrm{ord}(a)>1$，所以 $\mathrm{ord}(a)=19$。因此 $G=\langle a\rangle\cong C_{19}$。 □

20 阶群

共有 5 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[20,1]	Dic_5 $(2,5,20)$ $R=(2,2,5)$; $a^2=b$; $[c,a]=c^3$
[20,2]	C_{20} $(20,1,20)$ 阿贝尔。
[20,3]	$C_5\rtimes C_4$ $(1,5,20)$ $R=(2,2,5)$; $a^2=b$; $[c,a]=c$; $[c,b]=c^3$
[20,4]	D_{20} $(2,5,10)$ $R=(2,2,5)$; $[c,a]=c^3$
[20,5]	$C_2\times C_{10}$ $(20,1,10)$ 阿贝尔。

证明. 20 阶群的 Sylow 5 子群正规，故可写成 $C_5 \rtimes H$，其中 H 是 4 阶群。取 $H = C_4 = \langle b \rangle$ 时，$\mathrm{Aut}(C_5) = C_4$ 中可取作用阶 $1, 2, 4$，分别产生

$$C_{20}, \quad C_5 \rtimes_{-1} C_4, \quad \langle a, b \mid a^5 = b^4 = 1,\ bab^{-1} = a^2 \rangle.$$

取 $H = C_2^2$ 时，平凡作用给出 $C_{10} \times C_2$，非平凡作用给出 $D_{10} \times C_2 \cong D_{20}$。因此共有五种。$C_4$ 作用中的 $u = 2$ 与 $u = 3$ 通过更换 b 为 b^{-1} 等价；所以不能简单把所有同态逐个计数。非阿贝尔三种中，前两种补子群为 C_4，按作用阶 $2, 4$，其中心阶分别为 $2, 1$；D_{20} 有 11 个 2 阶元素，与前两者区分。 □

21 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z, d, e)；非平凡生成关系
[21,1]	$C_7 \rtimes C_3 \quad (1, 7, 21)$ $R = (3, 7);\ [b, a] = b$
[21,2]	$C_{21} \quad (21, 1, 21) \quad$ 阿贝尔。

证明. 这里 $p = 3 < q = 7$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q = 1$。令 $N = C_q \trianglelefteq G$，取 p 阶子群 $H = C_p$，则 $N \cap H = 1$，$NH = G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $3 \mid 6$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

22 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z, d, e)；非平凡生成关系
[22,1]	$D_{22} \quad (1, 11, 22)$ $R = (2, 11);\ [b, a] = b^9$
[22,2]	$C_{22} \quad (22, 1, 22) \quad$ 阿贝尔。

证明. 这里 $p = 2 < q = 11$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q = 1$。令 $N = C_q \trianglelefteq G$，取 p 阶子群 $H = C_p$，则 $N \cap H = 1$，$NH = G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $2 \mid 10$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

23 阶群

只有一个同构类型：C_{23}，编号 $[23,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 23$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 23$。因此 $G = \langle a \rangle \cong C_{23}$。 □

24 阶群

共有 15 个同构类型，其中 3 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[24,1]	$C_3 \rtimes C_8 \quad (4,3,24)$ $R = (2,2,2,3);\ a^2 = b;\ b^2 = c;\ [d,a] = d$
[24,2]	$C_{24} \quad (24,1,24)$ 阿贝尔。
[24,3]	$\mathrm{SL}(2,3) \quad (2,8,12)$ $R = (3,2,2,2);\ b^2 = d;\ c^2 = d;\ [b,a] = bcd;\ [c,a] = bd;\ [c,b] = d$
[24,4]	$C_3 \rtimes Q_8 \quad (2,6,12)$ $R = (2,2,2,3);\ a^2 = c;\ b^2 = c;\ [b,a] = c;\ [d,a] = d$
[24,5]	$C_4 \times S_3 \quad (4,3,12)$ $R = (2,2,2,3);\ b^2 = c;\ [d,a] = d$
[24,6]	$D_{24} \quad (2,6,12)$ $R = (2,2,2,3);\ b^2 = c;\ [b,a] = c;\ [d,a] = d$
[24,7]	$C_2 \times (C_3 \rtimes C_4) \quad (4,3,12)$ $R = (2,2,2,3);\ a^2 = c;\ [d,a] = d$
[24,8]	$(C_6 \times C_2) \rtimes C_2 \quad (2,6,12)$ $R = (2,2,2,3);\ [b,a] = c;\ [d,a] = d$
[24,9]	$C_2 \times C_{12} \quad (24,1,12)$ 阿贝尔。
[24,10]	$C_3 \times D_8 \quad (6,2,12)$ $R = (2,2,3,2);\ [b,a] = d$
[24,11]	$C_3 \times Q_8 \quad (6,2,12)$ $R = (2,2,3,2);\ a^2 = d;\ b^2 = d;\ [b,a] = d$
[24,12]	$S_4 \quad (1,12,12)$ $R = (2,3,2,2);\ [b,a] = b;\ [c,a] = cd;\ [c,b] = cd;\ [d,a] = cd;\ [d,b] = c$
[24,13]	$C_2 \times A_4 \quad (2,4,6)$ $R = (2,3,2,2);\ [c,b] = cd;\ [d,b] = c$
[24,14]	$C_2 \times C_2 \times S_3 \quad (4,3,6)$ $R = (2,2,2,3);\ [d,a] = d$
[24,15]	$C_2 \times C_2 \times C_6 \quad (24,1,6)$ 阿贝尔。

25 阶群

共有 2 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[25,1]	C_{25} $(25,1,25)$ 阿贝尔。
[25,2]	$C_5\times C_5$ $(25,1,5)$ 阿贝尔。

证明. 这里 $|G|=25=5^2$。非平凡有限 p 群的中心非平凡，故 $|\mathrm{Z}(G)|=p$ 或 p^2。若中心阶为 p，则 $G/\mathrm{Z}(G)$ 循环；将任意两个元素写成 a^iz, a^jw $(z,w\in\mathrm{Z}(G))$，可知它们交换。因此 G 总是阿贝尔群。若有 p^2 阶元素，则为 C_{p^2}；否则取 $a\neq 1$ 及 $b\notin\langle a\rangle$，便有 $G=\langle a\rangle\times\langle b\rangle\cong C_p^2$。两者的最大元素阶不同，故互不同构。 □

26 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[26,1]	D_{26} $(1,13,26)$ $R=(2,13);\ [b,a]=b^{11}$
[26,2]	C_{26} $(26,1,26)$ 阿贝尔。

证明. 这里 $p=2<q=13$。Sylow 定理给出 $n_q\mid p$ 且 $n_q\equiv 1\pmod q$，所以 $n_q=1$。令 $N=C_q\trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q\rtimes C_p$。作用由同态 $C_p\to\mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $2\mid 12$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

27 阶群

共有 5 个同构类型，其中 3 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[27,1]	C_{27} $(27,1,27)$ 阿贝尔。
[27,2]	$C_3\times C_9$ $(27,1,9)$ 阿贝尔。
[27,3]	$(C_3\times C_3)\rtimes C_3$ $(3,3,3)$ $R=(3,3,3);\ [b,a]=c$
[27,4]	$C_9\rtimes C_3$ $(3,3,9)$ $R=(3,3,3);\ a^3=c;\ [b,a]=c$
[27,5]	$C_3\times C_3\times C_3$ $(27,1,3)$ 阿贝尔。

两个非阿贝尔类型也可记为 H_3 与 M_3，其中

$$H_3 = \left\{ \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, z \in \mathbb{F}_3 \right\},$$

$$M_3 = \langle a, b \mid a^9 = b^3 = 1,\ bab^{-1} = a^4 \rangle.$$

下述证明对任意奇素数 p 都成立：将 H_3 中的 $\mathbb{F}_3$ 换成 $\mathbb{F}_p$，并令 $M_p = \langle a, b \mid a^{p^2} = b^p = 1,\ bab^{-1} = a^{1+p} \rangle$。

证明. H_p 的非单位元均为 p 阶，而 M_p 有 p^2 阶元素。穷尽性的关键论证如下。

若 G 非阿贝尔，中心只能为 p 阶；否则 $G/\mathrm{Z}(G)$ 循环。因此 $G/\mathrm{Z}(G) \cong C_p^2$。取提升 x, y，有

$$G = \langle x, y \rangle, \quad G' = \mathrm{Z}(G) \cong C_p, \quad x^p, y^p, [x, y] \in \mathrm{Z}(G).$$

若所有元素的阶都整除 p，适当选择中心生成元即得 H_p。否则取 p^2 阶元素 a，则 $\langle a \rangle$ 指数为 p 且正规；对 $b \notin \langle a \rangle$，非平凡共轭可经更换生成元化为 $a \mapsto a^{1+p}$。又 $b^p \in \langle a^p \rangle$；因 p 为奇数，用 $a^t b$ 替代 b 可使其 p 次幂为 1，即得 M_p。 □

28 阶群

共有 4 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z, d, e)；非平凡生成关系
[28,1]	$C_7 \rtimes C_4 \quad (2, 7, 28)$ $R = (2, 2, 7);\ a^2 = b;\ [c, a] = c^5$
[28,2]	$C_{28} \quad (28, 1, 28) \quad$ 阿贝尔。
[28,3]	$D_{28} \quad (2, 7, 14)$ $R = (2, 2, 7);\ [c, a] = c^5$
[28,4]	$C_2 \times C_{14} \quad (28, 1, 14) \quad$ 阿贝尔。

证明. Sylow 7 子群正规，因为 $n_7 \mid 4$ 且 $n_7 \equiv 1 \pmod 7$。所以 $G = C_7 \rtimes H$，其中 $H = C_4$ 或 C_2^2。作用取值于 $\mathrm{Aut}(C_7) = C_6$，其像只能为 1 或唯一的 C_2。两种 H 各给出平凡与非平凡作用，得到表中的四种群；非平凡作用均为取逆。两个阿贝尔群由指数区分；两个非阿贝尔群中，$\mathrm{Dic}_7 = C_7 \rtimes_{-1} C_4$ 只有一个 2 阶元素，而 D_{28} 有 15 个，因此互不同构。 □

29 阶群

只有一个同构类型：C_{29}，编号 $[29, 1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 29$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 29$。因此 $G = \langle a \rangle \cong C_{29}$。 □

30 阶群

共有 4 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[30,1]	$C_5 \times S_3$　$(5,3,30)$ $R=(2,5,3)$; $[c,a]=c$
[30,2]	$C_3 \times D_{10}$　$(3,5,30)$ $R=(2,3,5)$; $[c,a]=c^3$
[30,3]	D_{30}　$(1,15,30)$ $R=(2,3,5)$; $[b,a]=b$; $[c,a]=c^3$
[30,4]	C_{30}　$(30,1,30)$　阿贝尔。

31 阶群

只有一个同构类型：C_{31}，编号 $[31,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 31$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 31$。因此 $G = \langle a \rangle \cong C_{31}$。 □

32 阶群

共有 51 个同构类型，其中 7 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[32,1]	C_{32}　$(32,1,32)$　阿贝尔。
[32,2]	$(C_4 \times C_2) \rtimes C_4$　$(8,2,4)$ $R=(2,2,2,2,2)$; $a^2=d$; $b^2=e$; $[b,a]=c$
[32,3]	$C_4 \times C_8$　$(32,1,8)$　阿贝尔。
[32,4]	$C_8 \rtimes C_4$　$(8,2,8)$ $R=(2,2,2,2,2)$; $a^2=c$; $b^2=d$; $c^2=e$; $[b,a]=e$
[32,5]	$(C_8 \times C_2) \rtimes C_2$　$(8,2,8)$ $R=(2,2,2,2,2)$; $a^2=d$; $d^2=e$; $[b,a]=c$
[32,6]	$(C_2 \times C_2 \times C_2) \rtimes C_4$　$(2,4,4)$ $R=(2,2,2,2,2)$; $a^2=d$; $[b,a]=c$; $[c,a]=e$; $[d,b]=e$
[32,7]	$(C_8 \rtimes C_2) \rtimes C_2$　$(2,4,8)$ $R=(2,2,2,2,2)$; $a^2=d$; $d^2=e$; $[b,a]=c$; $[c,a]=e$; $[d,b]=e$
[32,8]	$C_2 \,.\, ((C_4 \times C_2) \rtimes C_2) = (C_2 \times C_2) \,.\, (C_4 \times C_2)$　$(2,4,8)$ $R=(2,2,2,2,2)$; $a^2=d$; $b^2=e$; $d^2=e$; $[b,a]=c$; $[c,a]=e$; $[d,b]=e$
[32,9]	$(C_8 \times C_2) \rtimes C_2$　$(4,4,8)$ $R=(2,2,2,2,2)$; $a^2=d$; $c^2=e$; $[b,a]=c$; $[c,a]=e$; $[c,b]=e$
[32,10]	$Q_8 \rtimes C_4$　$(4,4,8)$ $R=(2,2,2,2,2)$; $a^2=d$; $b^2=e$; $c^2=e$; $[b,a]=c$; $[c,a]=e$; $[c,b]=e$

编号	32 阶群（续）
[32,11]	$(C_4 \times C_4) \rtimes C_2 \quad (4,4,8)$ $R = (2,2,2,2,2);\ a^2 = d;\ c^2 = e;\ d^2 = e;\ [b,a] = c;\ [c,a] = e;\ [c,b] = e$
[32,12]	$C_4 \rtimes C_8 \quad (8,2,8)$ $R = (2,2,2,2,2);\ a^2 = d;\ b^2 = c;\ d^2 = e;\ [b,a] = c$
[32,13]	$C_8 \rtimes C_4 \quad (4,4,8)$ $R = (2,2,2,2,2);\ a^2 = d;\ b^2 = c;\ c^2 = e;\ [b,a] = c;\ [c,a] = e$
[32,14]	$C_8 \rtimes C_4 \quad (4,4,8)$ $R = (2,2,2,2,2);\ a^2 = d;\ b^2 = ce;\ c^2 = e;\ [b,a] = c;\ [c,a] = e$
[32,15]	$C_4 \,.\, D_8 = C_4 \,.\, (C_4 \times C_2) \quad (4,4,8)$ $R = (2,2,2,2,2);\ a^2 = d;\ b^2 = c;\ c^2 = e;\ d^2 = e;\ [b,a] = c;\ [c,a] = e$
[32,16]	$C_2 \times C_{16} \quad (32,1,16)$ 阿贝尔。
[32,17]	$C_{16} \rtimes C_2 \quad (8,2,16)$ $R = (2,2,2,2,2);\ a^2 = c;\ c^2 = d;\ d^2 = e;\ [b,a] = e$
[32,18]	$D_{32} \quad (2,8,16)$ $R = (2,2,2,2,2);\ c^2 = de;\ d^2 = e;\ [b,a] = c;\ [c,a] = d;\ [c,b] = d;\ [d,a] = e;\ [d,b] = e$
[32,19]	$\mathrm{QD}_{32} \quad (2,8,16)$ $R = (2,2,2,2,2);\ a^2 = e;\ c^2 = de;\ d^2 = e;\ [b,a] = c;\ [c,a] = d;\ [c,b] = d;\ [d,a] = e;$ $[d,b] = e$
[32,20]	$Q_{32} \quad (2,8,16)$ $R = (2,2,2,2,2);\ a^2 = e;\ b^2 = e;\ c^2 = de;\ d^2 = e;\ [b,a] = c;\ [c,a] = d;\ [c,b] = d;\ [d,a] = e;$ $[d,b] = e$
[32,21]	$C_2 \times C_4 \times C_4 \quad (32,1,4)$ 阿贝尔。
[32,22]	$C_2 \times ((C_4 \times C_2) \rtimes C_2) \quad (8,2,4)$ $R = (2,2,2,2,2);\ a^2 = e;\ [b,a] = d$
[32,23]	$C_2 \times (C_4 \rtimes C_4) \quad (8,2,4)$ $R = (2,2,2,2,2);\ a^2 = e;\ b^2 = d;\ [b,a] = d$
[32,24]	$(C_4 \times C_4) \rtimes C_2 \quad (8,2,4)$ $R = (2,2,2,2,2);\ a^2 = e;\ c^2 = d;\ [b,a] = d$
[32,25]	$C_4 \times D_8 \quad (8,2,4)$ $R = (2,2,2,2,2);\ a^2 = e;\ c^2 = e;\ [b,a] = d$
[32,26]	$C_4 \times Q_8 \quad (8,2,4)$ $R = (2,2,2,2,2);\ a^2 = e;\ b^2 = d;\ c^2 = de;\ [b,a] = d$
[32,27]	$(C_2 \times C_2 \times C_2 \times C_2) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2);\ [b,a] = d;\ [c,a] = e$
[32,28]	$(C_4 \times C_2 \times C_2) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2);\ b^2 = d;\ [b,a] = d;\ [c,a] = e$
[32,29]	$(C_2 \times Q_8) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2);\ a^2 = d;\ b^2 = d;\ [b,a] = d;\ [c,a] = e$
[32,30]	$(C_4 \times C_2 \times C_2) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2);\ c^2 = d;\ [b,a] = d;\ [c,a] = e$
[32,31]	$(C_4 \times C_4) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2);\ b^2 = e;\ c^2 = d;\ [b,a] = d;\ [c,a] = e$
[32,32]	$(C_2 \times C_2) \,.\, (C_2 \times C_2 \times C_2) \quad (4,4,4)$ $R = (2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ c^2 = d;\ [b,a] = d;\ [c,a] = e$
[32,33]	$(C_4 \times C_4) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2);\ b^2 = de;\ c^2 = d;\ [b,a] = d;\ [c,a] = e$

编号	32 阶群（续）
[32,34]	$(C_4 \times C_4) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2);\ b^2 = d;\ c^2 = e;\ [b,a] = d;\ [c,a] = e$
[32,35]	$C_4 \rtimes Q_8 \quad (4,4,4)$ $R = (2,2,2,2,2);\ a^2 = d;\ b^2 = d;\ c^2 = e;\ [b,a] = d;\ [c,a] = e$
[32,36]	$C_2 \times C_2 \times C_8 \quad (32,1,8)$ 阿贝尔。
[32,37]	$C_2 \times (C_8 \rtimes C_2) \quad (8,2,8)$ $R = (2,2,2,2,2);\ a^2 = d;\ d^2 = e;\ [b,a] = e$
[32,38]	$(C_8 \times C_2) \rtimes C_2 \quad (8,2,8)$ $R = (2,2,2,2,2);\ a^2 = d;\ d^2 = e;\ [c,b] = e$
[32,39]	$C_2 \times D_{16} \quad (4,4,8)$ $R = (2,2,2,2,2);\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e$
[32,40]	$C_2 \times \mathrm{QD}_{16} \quad (4,4,8)$ $R = (2,2,2,2,2);\ a^2 = e;\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e$
[32,41]	$C_2 \times Q_{16} \quad (4,4,8)$ $R = (2,2,2,2,2);\ a^2 = e;\ b^2 = e;\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e$
[32,42]	$(C_8 \times C_2) \rtimes C_2 \quad (4,4,8)$ $R = (2,2,2,2,2);\ c^2 = e;\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e$
[32,43]	$C_8 \rtimes (C_2 \times C_2) \quad (2,4,8)$ $R = (2,2,2,2,2);\ d^2 = e;\ [b,a] = d;\ [c,a] = e;\ [d,a] = e;\ [d,b] = e$
[32,44]	$(C_2 \times Q_8) \rtimes C_2 \quad (2,4,8)$ $R = (2,2,2,2,2);\ b^2 = e;\ d^2 = e;\ [b,a] = d;\ [c,a] = e;\ [d,a] = e;\ [d,b] = e$
[32,45]	$C_2 \times C_2 \times C_2 \times C_4 \quad (32,1,4)$ 阿贝尔。
[32,46]	$C_2 \times C_2 \times D_8 \quad (8,2,4)$ $R = (2,2,2,2,2);\ [b,a] = e$
[32,47]	$C_2 \times C_2 \times Q_8 \quad (8,2,4)$ $R = (2,2,2,2,2);\ a^2 = e;\ b^2 = e;\ [b,a] = e$
[32,48]	$C_2 \times ((C_4 \times C_2) \rtimes C_2) \quad (8,2,4)$ $R = (2,2,2,2,2);\ c^2 = e;\ [b,a] = e$
[32,49]	$(C_2 \times C_2 \times C_2) \rtimes (C_2 \times C_2) \quad (2,2,4)$ $R = (2,2,2,2,2);\ [b,a] = e;\ [c,b] = e;\ [d,a] = e$
[32,50]	$(C_2 \times Q_8) \rtimes C_2 \quad (2,2,4)$ $R = (2,2,2,2,2);\ b^2 = e;\ c^2 = e;\ [b,a] = e;\ [c,b] = e;\ [d,a] = e$
[32,51]	$C_2 \times C_2 \times C_2 \times C_2 \times C_2 \quad (32,1,2)$ 阿贝尔。

33 阶群

只有一个同构类型：C_{33}，编号 $[33,1]$。

证明. 这里 $p = 3 < q = 11$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q = 1$。令 $N = C_q \trianglelefteq G$，取 p 阶子群 $H = C_p$，则 $N \cap H = 1$，$NH = G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $3 \nmid 10$，作用只能平凡，故只有 $C_q \times C_p \cong C_{33}$。 □

34 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[34,1]	D_{34}　$(1,17,34)$ $R=(2,17);\ [b,a]=b^{15}$
[34,2]	C_{34}　$(34,1,34)$　阿贝尔。

证明. 这里 $p=2<q=17$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q=1$。令 $N=C_q \trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $2\mid 16$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

35 阶群

只有一个同构类型：C_{35}，编号 $[35,1]$。

证明. 这里 $p=5<q=7$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q=1$。令 $N=C_q \trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $5\nmid 6$，作用只能平凡，故只有 $C_q\times C_p\cong C_{35}$。 □

36 阶群

共有 14 个同构类型，其中 4 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[36,1]	$C_9 \rtimes C_4$　$(2,9,36)$ $R=(2,2,3,3);\ a^2=b;\ c^3=d^2;\ [c,a]=cd;\ [d,a]=d$
[36,2]	C_{36}　$(36,1,36)$　阿贝尔。
[36,3]	$(C_2\times C_2)\rtimes C_9$　$(3,4,18)$ $R=(3,3,2,2);\ a^3=b;\ [c,a]=cd;\ [d,a]=c$
[36,4]	D_{36}　$(2,9,18)$ $R=(2,2,3,3);\ c^3=d^2;\ [c,a]=cd;\ [d,a]=d$
[36,5]	$C_2\times C_{18}$　$(36,1,18)$　阿贝尔。
[36,6]	$C_3\times(C_3\rtimes C_4)$　$(6,3,12)$ $R=(2,3,2,3);\ a^2=c;\ [d,a]=d$
[36,7]	$(C_3\times C_3)\rtimes C_4$　$(2,9,12)$ $R=(2,2,3,3);\ a^2=b;\ [c,a]=c;\ [d,a]=d$
[36,8]	$C_3\times C_{12}$　$(36,1,12)$　阿贝尔。
[36,9]	$(C_3\times C_3)\rtimes C_4$　$(1,9,12)$ $R=(2,2,3,3);\ a^2=b;\ [c,a]=d^2;\ [c,b]=c;\ [d,a]=c^2d;\ [d,b]=d$
[36,10]	$S_3\times S_3$　$(1,9,6)$ $R=(2,2,3,3);\ [c,b]=c;\ [d,a]=d$

编号	36 阶群（续）
[36,11]	$C_3 \times A_4 \quad (3,4,6)$ $R=(3,3,2,2);\ [c,a]=cd;\ [d,a]=c$
[36,12]	$C_6 \times S_3 \quad (6,3,6)$ $R=(2,2,3,3);\ [d,a]=d$
[36,13]	$C_2 \times ((C_3 \times C_3) \rtimes C_2) \quad (2,9,6)$ $R=(2,2,3,3);\ [c,a]=c;\ [d,a]=d$
[36,14]	$C_6 \times C_6 \quad (36,1,6)$ 阿贝尔。

37 阶群

只有一个同构类型：C_{37}，编号 $[37,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 37$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 37$。因此 $G = \langle a \rangle \cong C_{37}$。 □

38 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[38,1]	$D_{38} \quad (1,19,38)$ $R=(2,19);\ [b,a]=b^{17}$
[38,2]	$C_{38} \quad (38,1,38)$ 阿贝尔。

证明. 这里 $p=2<q=19$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q = 1$。令 $N = C_q \trianglelefteq G$，取 p 阶子群 $H = C_p$，则 $N \cap H = 1$，$NH = G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $2 \mid 18$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

39 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[39,1]	$C_{13} \rtimes C_3 \quad (1,13,39)$ $R=(3,13);\ [b,a]=b^2$
[39,2]	$C_{39} \quad (39,1,39)$ 阿贝尔。

证明. 这里 $p=3<q=13$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q = 1$。令 $N = C_q \trianglelefteq G$，取 p 阶子群 $H = C_p$，则 $N \cap H = 1$，$NH = G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $3 \mid 12$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

40 阶群

共有 14 个同构类型，其中 3 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[40,1]	$C_5 \rtimes C_8 \quad (4,5,40)$ $R=(2,2,2,5);\ a^2=b;\ b^2=c;\ [d,a]=d^3$
[40,2]	$C_{40} \quad (40,1,40)$ 阿贝尔。
[40,3]	$C_5 \rtimes C_8 \quad (2,5,40)$ $R=(2,2,2,5);\ a^2=b;\ b^2=c;\ [d,a]=d;\ [d,b]=d^3$
[40,4]	$C_5 \rtimes Q_8 \quad (2,10,20)$ $R=(2,2,2,5);\ a^2=c;\ b^2=c;\ [b,a]=c;\ [d,a]=d^3$
[40,5]	$C_4 \times D_{10} \quad (4,5,20)$ $R=(2,2,2,5);\ b^2=c;\ [d,a]=d^3$
[40,6]	$D_{40} \quad (2,10,20)$ $R=(2,2,2,5);\ b^2=c;\ [b,a]=c;\ [d,a]=d^3$
[40,7]	$C_2 \times (C_5 \rtimes C_4) \quad (4,5,20)$ $R=(2,2,2,5);\ a^2=c;\ [d,a]=d^3$
[40,8]	$(C_{10} \times C_2) \rtimes C_2 \quad (2,10,20)$ $R=(2,2,2,5);\ [b,a]=c;\ [d,a]=d^3$
[40,9]	$C_2 \times C_{20} \quad (40,1,20)$ 阿贝尔。
[40,10]	$C_5 \times D_8 \quad (10,2,20)$ $R=(2,2,5,2);\ [b,a]=d$
[40,11]	$C_5 \times Q_8 \quad (10,2,20)$ $R=(2,2,5,2);\ a^2=d;\ b^2=d;\ [b,a]=d$
[40,12]	$C_2 \times (C_5 \rtimes C_4) \quad (2,5,20)$ $R=(2,2,2,5);\ a^2=c;\ [d,a]=d;\ [d,c]=d^3$
[40,13]	$C_2 \times C_2 \times D_{10} \quad (4,5,10)$ $R=(2,2,2,5);\ [d,a]=d^3$
[40,14]	$C_2 \times C_2 \times C_{10} \quad (40,1,10)$ 阿贝尔。

41 阶群

只有一个同构类型：C_{41}，编号 $[41,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 41$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 41$。因此 $G = \langle a \rangle \cong C_{41}$。 □

42 阶群

共有 6 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[42,1]	$C_7 \rtimes C_6 \quad (1,7,42)$ $R=(2,3,7);\ [c,a]=c^5;\ [c,b]=c$
[42,2]	$C_2 \times (C_7 \rtimes C_3) \quad (2,7,42)$ $R=(2,3,7);\ [c,b]=c$
[42,3]	$C_7 \times S_3 \quad (7,3,42)$ $R=(2,7,3);\ [c,a]=c$
[42,4]	$C_3 \times D_{14} \quad (3,7,42)$ $R=(2,3,7);\ [c,a]=c^5$
[42,5]	$D_{42} \quad (1,21,42)$ $R=(2,3,7);\ [b,a]=b;\ [c,a]=c^5$
[42,6]	$C_{42} \quad (42,1,42)$ 阿贝尔。

43 阶群

只有一个同构类型：C_{43}，编号 $[43,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 43$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 43$。因此 $G = \langle a \rangle \cong C_{43}$。 □

44 阶群

共有 4 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[44,1]	$C_{11} \rtimes C_4 \quad (2,11,44)$ $R=(2,2,11);\ a^2=b;\ [c,a]=c^9$
[44,2]	$C_{44} \quad (44,1,44)$ 阿贝尔。
[44,3]	$D_{44} \quad (2,11,22)$ $R=(2,2,11);\ [c,a]=c^9$
[44,4]	$C_2 \times C_{22} \quad (44,1,22)$ 阿贝尔。

45 阶群

共有 2 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[45,1]	$C_{45} \quad (45,1,45)$ 阿贝尔。
[45,2]	$C_3 \times C_{15} \quad (45,1,15)$ 阿贝尔。

证明. 由 Sylow 定理，$n_5 = 1$ 且 $n_3 = 1$。两个 Sylow 子群都正规，且交为 1，故它们彼此交换，G 是二者的直积。阶为 3^2 的 Sylow 子群只有 C_9 与 C_3^2，阶为 5 的 Sylow 子群为 C_5，因而恰得表中的两种阿贝尔群。 □

46 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[46,1]	D_{46}　$(1,23,46)$ $R=(2,23);\ [b,a]=b^{21}$
[46,2]	C_{46}　$(46,1,46)$　阿贝尔。

证明. 这里 $p=2<q=23$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q=1$。令 $N=C_q \trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $2\mid 22$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

47 阶群

只有一个同构类型：C_{47}，编号 $[47,1]$。

证明. 任取非单位元 $a\in G$。由拉格朗日定理，$\mathrm{ord}(a)\mid 47$，又 $\mathrm{ord}(a)>1$，所以 $\mathrm{ord}(a)=47$。因此 $G=\langle a\rangle\cong C_{47}$。 □

48 阶群

共有 52 个同构类型，其中 5 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[48,1]	$C_3 \rtimes C_{16}$　$(8,3,48)$ $R=(2,2,2,2,3);\ a^2=b;\ b^2=c;\ c^2=d;\ [e,a]=e$
[48,2]	C_{48}　$(48,1,48)$　阿贝尔。
[48,3]	$(C_4\times C_4)\rtimes C_3$　$(1,16,12)$ $R=(3,2,2,2,2);\ b^2=de;\ c^2=d;\ [b,a]=bcde;\ [c,a]=be;\ [d,a]=de;\ [e,a]=d$
[48,4]	$C_8\times S_3$　$(8,3,24)$ $R=(2,2,2,2,3);\ b^2=c;\ c^2=d;\ [e,a]=e$
[48,5]	$C_{24}\rtimes C_2$　$(4,6,24)$ $R=(2,2,2,2,3);\ b^2=c;\ c^2=d;\ [b,a]=d;\ [e,a]=e$
[48,6]	$C_{24}\rtimes C_2$　$(2,12,24)$ $R=(2,2,2,2,3);\ b^2=c;\ c^2=d;\ [b,a]=c;\ [c,a]=d;\ [e,a]=e$
[48,7]	D_{48}　$(2,12,24)$ $R=(2,2,2,2,3);\ b^2=cd;\ c^2=d;\ [b,a]=c;\ [c,a]=d;\ [e,a]=e$
[48,8]	$C_3\rtimes Q_{16}$　$(2,12,24)$ $R=(2,2,2,2,3);\ a^2=d;\ b^2=cd;\ c^2=d;\ [b,a]=c;\ [c,a]=d;\ [e,a]=e$
[48,9]	$C_2\times(C_3\rtimes C_8)$　$(8,3,24)$ $R=(2,2,2,2,3);\ a^2=c;\ c^2=d;\ [e,a]=e$
[48,10]	$(C_3\rtimes C_8)\rtimes C_2$　$(4,6,24)$ $R=(2,2,2,2,3);\ a^2=c;\ c^2=d;\ [b,a]=d;\ [e,a]=e$

编号	48 阶群（续）
[48,11]	$C_4 \times (C_3 \rtimes C_4)$ $(8,3,12)$ $R = (2,2,2,2,3);\ a^2 = c;\ b^2 = d;\ [e,a] = e$
[48,12]	$(C_3 \rtimes C_4) \rtimes C_4$ $(4,6,12)$ $R = (2,2,2,2,3);\ a^2 = d;\ b^2 = c;\ [b,a] = d;\ [e,a] = e$
[48,13]	$C_{12} \rtimes C_4$ $(4,6,12)$ $R = (2,2,2,2,3);\ a^2 = c;\ b^2 = d;\ [b,a] = d;\ [e,a] = e$
[48,14]	$(C_{12} \times C_2) \rtimes C_2$ $(4,6,12)$ $R = (2,2,2,2,3);\ b^2 = c;\ [b,a] = d;\ [e,a] = e$
[48,15]	$(C_3 \times D_8) \rtimes C_2$ $(2,12,24)$ $R = (2,2,2,2,3);\ c^2 = d;\ [b,a] = c;\ [c,a] = d;\ [c,b] = d;\ [e,a] = e$
[48,16]	$(C_3 \rtimes Q_8) \rtimes C_2$ $(2,12,24)$ $R = (2,2,2,2,3);\ a^2 = d;\ c^2 = d;\ [b,a] = c;\ [c,a] = d;\ [c,b] = d;\ [e,a] = e$
[48,17]	$(C_3 \times Q_8) \rtimes C_2$ $(2,12,24)$ $R = (2,2,2,2,3);\ b^2 = d;\ c^2 = d;\ [b,a] = c;\ [c,a] = d;\ [c,b] = d;\ [e,a] = e$
[48,18]	$C_3 \rtimes Q_{16}$ $(2,12,24)$ $R = (2,2,2,2,3);\ a^2 = d;\ b^2 = d;\ c^2 = d;\ [b,a] = c;\ [c,a] = d;\ [c,b] = d;\ [e,a] = e$
[48,19]	$(C_6 \times C_2) \rtimes C_4$ $(4,6,12)$ $R = (2,2,2,2,3);\ a^2 = c;\ [b,a] = d;\ [e,a] = e$
[48,20]	$C_4 \times C_{12}$ $(48,1,12)$ 阿贝尔。
[48,21]	$C_3 \times ((C_4 \times C_2) \rtimes C_2)$ $(12,2,12)$ $R = (2,2,3,2,2);\ a^2 = e;\ [b,a] = d$
[48,22]	$C_3 \times (C_4 \rtimes C_4)$ $(12,2,12)$ $R = (2,2,3,2,2);\ a^2 = e;\ b^2 = d;\ [b,a] = d$
[48,23]	$C_2 \times C_{24}$ $(48,1,24)$ 阿贝尔。
[48,24]	$C_3 \times (C_8 \rtimes C_2)$ $(12,2,24)$ $R = (2,2,3,2,2);\ a^2 = d;\ d^2 = e;\ [b,a] = e$
[48,25]	$C_3 \times D_{16}$ $(6,4,24)$ $R = (2,2,3,2,2);\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e$
[48,26]	$C_3 \times \mathrm{QD}_{16}$ $(6,4,24)$ $R = (2,2,3,2,2);\ a^2 = e;\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e$
[48,27]	$C_3 \times Q_{16}$ $(6,4,24)$ $R = (2,2,3,2,2);\ a^2 = e;\ b^2 = e;\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e$
[48,28]	$C_2 \,.\, S_4 = \mathrm{SL}(2,3) \,.\, C_2$ $(2,24,24)$ $R = (2,3,2,2,2);\ a^2 = e;\ c^2 = e;\ d^2 = e;\ [b,a] = b;\ [c,a] = cde;\ [c,b] = cd;\ [d,a] = cd;$ $[d,b] = ce;\ [d,c] = e$
[48,29]	$\mathrm{GL}(2,3)$ $(2,24,24)$ $R = (2,3,2,2,2);\ c^2 = e;\ d^2 = e;\ [b,a] = b;\ [c,a] = cde;\ [c,b] = cd;\ [d,a] = cd;\ [d,b] = ce;$ $[d,c] = e$
[48,30]	$A_4 \rtimes C_4$ $(2,12,12)$ $R = (2,2,3,2,2);\ a^2 = b;\ [c,a] = c;\ [d,a] = de;\ [d,c] = de;\ [e,a] = de;\ [e,c] = d$
[48,31]	$C_4 \times A_4$ $(4,4,12)$ $R = (2,3,2,2,2);\ a^2 = c;\ [d,b] = de;\ [e,b] = d$
[48,32]	$C_2 \times \mathrm{SL}(2,3)$ $(4,8,12)$ $R = (2,3,2,2,2);\ c^2 = e;\ d^2 = e;\ [c,b] = cde;\ [d,b] = ce;\ [d,c] = e$
[48,33]	$((C_4 \times C_2) \rtimes C_2) \rtimes C_3$ $(4,8,12)$ $R = (2,3,2,2,2);\ a^2 = e;\ c^2 = e;\ d^2 = e;\ [c,b] = cde;\ [d,b] = ce;\ [d,c] = e$

编号	48 阶群（续）
[48,34]	$C_2 \times (C_3 \rtimes Q_8)$ $(4,6,12)$ $R=(2,2,2,2,3);\ a^2=d;\ c^2=d;\ [c,a]=d;\ [e,a]=e$
[48,35]	$C_2 \times C_4 \times S_3$ $(8,3,12)$ $R=(2,2,2,2,3);\ c^2=d;\ [e,a]=e$
[48,36]	$C_2 \times D_{24}$ $(4,6,12)$ $R=(2,2,2,2,3);\ c^2=d;\ [c,a]=d;\ [e,a]=e$
[48,37]	$(C_{12} \times C_2) \rtimes C_2$ $(4,6,12)$ $R=(2,2,2,2,3);\ b^2=d;\ [c,a]=d;\ [e,a]=e$
[48,38]	$D_8 \times S_3$ $(2,6,12)$ $R=(2,2,2,2,3);\ [c,b]=d;\ [e,a]=e$
[48,39]	$(C_4 \times S_3) \rtimes C_2$ $(2,6,12)$ $R=(2,2,2,2,3);\ b^2=d;\ [c,a]=d;\ [c,b]=d;\ [e,a]=e$
[48,40]	$Q_8 \times S_3$ $(2,6,12)$ $R=(2,2,2,2,3);\ b^2=d;\ c^2=d;\ [c,b]=d;\ [e,a]=e$
[48,41]	$(C_4 \times S_3) \rtimes C_2$ $(2,6,12)$ $R=(2,2,2,2,3);\ b^2=d;\ c^2=d;\ [c,a]=d;\ [c,b]=d;\ [e,a]=e$
[48,42]	$C_2 \times C_2 \times (C_3 \rtimes C_4)$ $(8,3,12)$ $R=(2,2,2,2,3);\ a^2=d;\ [e,a]=e$
[48,43]	$C_2 \times ((C_6 \times C_2) \rtimes C_2)$ $(4,6,12)$ $R=(2,2,2,2,3);\ [c,a]=d;\ [e,a]=e$
[48,44]	$C_2 \times C_2 \times C_{12}$ $(48,1,12)$ 阿贝尔。
[48,45]	$C_6 \times D_8$ $(12,2,12)$ $R=(2,2,2,3,2);\ [b,a]=e$
[48,46]	$C_6 \times Q_8$ $(12,2,12)$ $R=(2,2,2,3,2);\ a^2=e;\ b^2=e;\ [b,a]=e$
[48,47]	$C_3 \times ((C_4 \times C_2) \rtimes C_2)$ $(12,2,12)$ $R=(2,2,2,3,2);\ c^2=e;\ [b,a]=e$
[48,48]	$C_2 \times S_4$ $(2,12,12)$ $R=(2,2,3,2,2);\ [c,a]=c;\ [d,a]=de;\ [d,c]=de;\ [e,a]=de;\ [e,c]=d$
[48,49]	$C_2 \times C_2 \times A_4$ $(4,4,6)$ $R=(2,2,3,2,2);\ [d,c]=de;\ [e,c]=d$
[48,50]	$(C_2 \times C_2 \times C_2 \times C_2) \rtimes C_3$ $(1,16,6)$ $R=(3,2,2,2,2);\ [b,a]=bc;\ [c,a]=b;\ [d,a]=de;\ [e,a]=d$
[48,51]	$C_2 \times C_2 \times C_2 \times S_3$ $(8,3,6)$ $R=(2,2,2,2,3);\ [e,a]=e$
[48,52]	$C_2 \times C_2 \times C_2 \times C_6$ $(48,1,6)$ 阿贝尔。

49 阶群

共有 2 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[49,1]	C_{49} $(49,1,49)$ 阿贝尔。
[49,2]	$C_7 \times C_7$ $(49,1,7)$ 阿贝尔。

证明. 这里 $|G| = 49 = 7^2$。非平凡有限 p 群的中心非平凡，故 $|\mathrm{Z}(G)| = p$ 或 p^2。若中心阶为 p，则 $G/\mathrm{Z}(G)$ 循环；将任意两个元素写成 $a^i z, a^j w$ $(z, w \in \mathrm{Z}(G))$，可知它们交换。因此 G 总是阿贝尔群。若有 p^2 阶元素，则为 C_{p^2}；否则取 $a \neq 1$ 及 $b \notin \langle a \rangle$，便有 $G = \langle a \rangle \times \langle b \rangle \cong C_p^2$。两者的最大元素阶不同，故互不同构。 □

50 阶群

共有 5 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z, d, e)；非平凡生成关系
[50,1]	D_{50} $(1, 25, 50)$ $R = (2, 5, 5)$; $b^5 = c$; $[b, a] = b^3c^4$; $[c, a] = c^3$
[50,2]	C_{50} $(50, 1, 50)$ 阿贝尔。
[50,3]	$C_5 \times D_{10}$ $(5, 5, 10)$ $R = (2, 5, 5)$; $[c, a] = c^3$
[50,4]	$(C_5 \times C_5) \rtimes C_2$ $(1, 25, 10)$ $R = (2, 5, 5)$; $[b, a] = b^3$; $[c, a] = c^3$
[50,5]	$C_5 \times C_{10}$ $(50, 1, 10)$ 阿贝尔。

51 阶群

只有一个同构类型：C_{51}，编号 $[51, 1]$。

证明. 这里 $p = 3 < q = 17$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q = 1$。令 $N = C_q \trianglelefteq G$，取 p 阶子群 $H = C_p$，则 $N \cap H = 1$，$NH = G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $3 \nmid 16$，作用只能平凡，故只有 $C_q \times C_p \cong C_{51}$。 □

52 阶群

共有 5 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z, d, e)；非平凡生成关系
[52,1]	$C_{13} \rtimes C_4$ $(2, 13, 52)$ $R = (2, 2, 13)$; $a^2 = b$; $[c, a] = c^{11}$
[52,2]	C_{52} $(52, 1, 52)$ 阿贝尔。
[52,3]	$C_{13} \rtimes C_4$ $(1, 13, 52)$ $R = (2, 2, 13)$; $a^2 = b$; $[c, a] = c^7$; $[c, b] = c^{11}$
[52,4]	D_{52} $(2, 13, 26)$ $R = (2, 2, 13)$; $[c, a] = c^{11}$
[52,5]	$C_2 \times C_{26}$ $(52, 1, 26)$ 阿贝尔。

53 阶群

只有一个同构类型：C_{53}，编号 $[53,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 53$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 53$。因此 $G = \langle a \rangle \cong C_{53}$。 □

54 阶群

共有 15 个同构类型，其中 3 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[54,1]	D_{54} $(1,27,54)$ $R=(2,3,3,3)$; $b^3=c^2d$; $c^3=d^2$; $[b,a]=bc$; $[c,a]=cd$; $[d,a]=d$
[54,2]	C_{54} $(54,1,54)$ 阿贝尔。
[54,3]	$C_3\times D_{18}$ $(3,9,18)$ $R=(2,3,3,3)$; $c^3=d^2$; $[c,a]=cd$; $[d,a]=d$
[54,4]	$C_9\times S_3$ $(9,3,18)$ $R=(2,3,3,3)$; $b^3=c$; $[d,a]=d$
[54,5]	$(C_3\times C_3)\rtimes C_6$ $(1,9,6)$ $R=(2,3,3,3)$; $[c,a]=c$; $[c,b]=d$; $[d,a]=d$
[54,6]	$C_9\rtimes C_6$ $(1,9,18)$ $R=(2,3,3,3)$; $c^3=d^2$; $[c,a]=cd$; $[c,b]=d$; $[d,a]=d$
[54,7]	$(C_9\times C_3)\rtimes C_2$ $(1,27,18)$ $R=(2,3,3,3)$; $b^3=d^2$; $[b,a]=bd$; $[c,a]=c$; $[d,a]=d$
[54,8]	$((C_3\times C_3)\rtimes C_3)\rtimes C_2$ $(3,27,6)$ $R=(2,3,3,3)$; $[b,a]=b$; $[c,a]=c$; $[c,b]=d$
[54,9]	$C_3\times C_{18}$ $(54,1,18)$ 阿贝尔。
[54,10]	$C_2\times((C_3\times C_3)\rtimes C_3)$ $(6,3,6)$ $R=(2,3,3,3)$; $[c,b]=d$
[54,11]	$C_2\times(C_9\rtimes C_3)$ $(6,3,18)$ $R=(2,3,3,3)$; $b^3=d$; $[c,b]=d$
[54,12]	$C_3\times C_3\times S_3$ $(9,3,6)$ $R=(2,3,3,3)$; $[d,a]=d$
[54,13]	$C_3\times((C_3\times C_3)\rtimes C_2)$ $(3,9,6)$ $R=(2,3,3,3)$; $[c,a]=c$; $[d,a]=d$
[54,14]	$(C_3\times C_3\times C_3)\rtimes C_2$ $(1,27,6)$ $R=(2,3,3,3)$; $[b,a]=b$; $[c,a]=c$; $[d,a]=d$
[54,15]	$C_3\times C_3\times C_6$ $(54,1,6)$ 阿贝尔。

55 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[55,1]	$C_{11} \rtimes C_5 \quad (1,11,55)$ $R=(5,11);\ [b,a]=b^3$
[55,2]	$C_{55} \quad (55,1,55) \quad$ 阿贝尔。

证明. 这里 $p=5<q=11$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q=1$。令 $N=C_q \trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $5 \mid 10$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

56 阶群

共有 13 个同构类型，其中 3 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[56,1]	$C_7 \rtimes C_8 \quad (4,7,56)$ $R=(2,2,2,7);\ a^2=b;\ b^2=c;\ [d,a]=d^5$
[56,2]	$C_{56} \quad (56,1,56) \quad$ 阿贝尔。
[56,3]	$C_7 \rtimes Q_8 \quad (2,14,28)$ $R=(2,2,2,7);\ a^2=c;\ b^2=c;\ [b,a]=c;\ [d,a]=d^5$
[56,4]	$C_4 \times D_{14} \quad (4,7,28)$ $R=(2,2,2,7);\ b^2=c;\ [d,a]=d^5$
[56,5]	$D_{56} \quad (2,14,28)$ $R=(2,2,2,7);\ b^2=c;\ [b,a]=c;\ [d,a]=d^5$
[56,6]	$C_2 \times (C_7 \rtimes C_4) \quad (4,7,28)$ $R=(2,2,2,7);\ a^2=c;\ [d,a]=d^5$
[56,7]	$(C_{14} \times C_2) \rtimes C_2 \quad (2,14,28)$ $R=(2,2,2,7);\ [b,a]=c;\ [d,a]=d^5$
[56,8]	$C_2 \times C_{28} \quad (56,1,28) \quad$ 阿贝尔。
[56,9]	$C_7 \times D_8 \quad (14,2,28)$ $R=(2,2,7,2);\ [b,a]=d$
[56,10]	$C_7 \times Q_8 \quad (14,2,28)$ $R=(2,2,7,2);\ a^2=d;\ b^2=d;\ [b,a]=d$
[56,11]	$(C_2 \times C_2 \times C_2) \rtimes C_7 \quad (1,8,14)$ $R=(7,2,2,2);\ [b,a]=bc;\ [c,a]=cd;\ [d,a]=b$
[56,12]	$C_2 \times C_2 \times D_{14} \quad (4,7,14)$ $R=(2,2,2,7);\ [d,a]=d^5$
[56,13]	$C_2 \times C_2 \times C_{14} \quad (56,1,14) \quad$ 阿贝尔。

57 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[57,1]	$C_{19}\rtimes C_3\quad(1,19,57)$ $R=(3,19);\ [b,a]=b^{6}$
[57,2]	$C_{57}\quad(57,1,57)\quad$阿贝尔。

证明. 这里 $p=3<q=19$。Sylow 定理给出 $n_q\mid p$ 且 $n_q\equiv 1\pmod q$，所以 $n_q=1$。令 $N=C_q\trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q\rtimes C_p$。作用由同态 $C_p\to\mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $3\mid 18$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

58 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[58,1]	$D_{58}\quad(1,29,58)$ $R=(2,29);\ [b,a]=b^{27}$
[58,2]	$C_{58}\quad(58,1,58)\quad$阿贝尔。

证明. 这里 $p=2<q=29$。Sylow 定理给出 $n_q\mid p$ 且 $n_q\equiv 1\pmod q$，所以 $n_q=1$。令 $N=C_q\trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q\rtimes C_p$。作用由同态 $C_p\to\mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $2\mid 28$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

59 阶群

只有一个同构类型：C_{59}，编号 $[59,1]$。

证明. 任取非单位元 $a\in G$。由拉格朗日定理，$\mathrm{ord}(a)\mid 59$，又 $\mathrm{ord}(a)>1$，所以 $\mathrm{ord}(a)=59$。因此 $G=\langle a\rangle\cong C_{59}$。 □

60 阶群

共有 13 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[60,1]	$C_5\times(C_3\rtimes C_4)\quad(10,3,60)$ $R=(2,5,2,3);\ a^2=c;\ [d,a]=d$

编号	60 阶群（续）
[60,2]	$C_3 \times (C_5 \rtimes C_4)$ $(6,5,60)$ $R=(2,3,2,5);\ a^2=c;\ [d,a]=d^3$
[60,3]	$C_{15} \rtimes C_4$ $(2,15,60)$ $R=(2,2,3,5);\ a^2=b;\ [c,a]=c;\ [d,a]=d^3$
[60,4]	C_{60} $(60,1,60)$ 阿贝尔。
[60,5]	A_5 $(1,60,30)$ 非可解。
[60,6]	$C_3 \times (C_5 \rtimes C_4)$ $(3,5,60)$ $R=(2,3,2,5);\ a^2=c;\ [d,a]=d;\ [d,c]=d^3$
[60,7]	$C_{15} \rtimes C_4$ $(1,15,60)$ $R=(2,2,3,5);\ a^2=b;\ [c,a]=c;\ [d,a]=d^2;\ [d,b]=d^3$
[60,8]	$S_3 \times D_{10}$ $(1,15,30)$ $R=(2,2,3,5);\ [c,b]=c;\ [d,a]=d^3$
[60,9]	$C_5 \times A_4$ $(5,4,30)$ $R=(3,5,2,2);\ [c,a]=cd;\ [d,a]=c$
[60,10]	$C_6 \times D_{10}$ $(6,5,30)$ $R=(2,2,3,5);\ [d,a]=d^3$
[60,11]	$C_{10} \times S_3$ $(10,3,30)$ $R=(2,2,5,3);\ [d,a]=d$
[60,12]	D_{60} $(2,15,30)$ $R=(2,2,3,5);\ [c,a]=c;\ [d,a]=d^3$
[60,13]	$C_2 \times C_{30}$ $(60,1,30)$ 阿贝尔。

其中 A_5 是偶置换群；它是 1–100 阶范围内唯一的非可解群。其余类型均可解。

证明：A_5 的单性与不可解性. 其共轭类大小为

$$1,\quad 15,\quad 20,\quad 12,\quad 12.$$

正规子群必须是包含单位元的共轭类之并，其阶又必须整除 60。检查上述数的可能和可知，只有 1 与 60 符合。因此 A_5 没有非平凡真正规子群；它又非阿贝尔，所以 $A_5' = A_5$，不可解。 □

61 阶群

只有一个同构类型：C_{61}，编号 $[61,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 61$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 61$。因此 $G = \langle a \rangle \cong C_{61}$。 □

62 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[62,1]	D_{62}　$(1,31,62)$ $R=(2,31)$; $[b,a]=b^{29}$
[62,2]	C_{62}　$(62,1,62)$　阿贝尔。

证明. 这里 $p=2<q=31$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q=1$。令 $N=C_q \trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $2\mid 30$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

63 阶群

共有 4 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[63,1]	$C_7 \rtimes C_9$　$(3,7,63)$ $R=(3,3,7)$; $a^3=b$; $[c,a]=c$
[63,2]	C_{63}　$(63,1,63)$　阿贝尔。
[63,3]	$C_3 \times (C_7 \rtimes C_3)$　$(3,7,21)$ $R=(3,3,7)$; $[c,a]=c$
[63,4]	$C_3 \times C_{21}$　$(63,1,21)$　阿贝尔。

64 阶群

共有 267 个同构类型，其中 11 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[64,1]	C_{64}　$(64,1,64)$　阿贝尔。
[64,2]	$C_8 \times C_8$　$(64,1,8)$　阿贝尔。
[64,3]	$C_8 \rtimes C_8$　$(16,2,8)$ $R=(2,2,2,2,2,2)$; $a^2=c$; $b^2=d$; $c^2=e$; $d^2=f$; $[b,a]=e$
[64,4]	$(C_2\times C_2\times C_2)\rtimes C_8$　$(4,4,8)$ $R=(2,2,2,2,2,2)$; $a^2=d$; $d^2=f$; $[b,a]=c$; $[c,a]=e$; $[d,b]=e$
[64,5]	$(C_4\times C_2)\rtimes C_8$　$(4,4,8)$ $R=(2,2,2,2,2,2)$; $a^2=d$; $b^2=e$; $d^2=f$; $[b,a]=c$; $[c,a]=e$; $[d,b]=e$
[64,6]	$D_8 \rtimes C_8$　$(8,4,8)$ $R=(2,2,2,2,2,2)$; $a^2=d$; $c^2=e$; $d^2=f$; $[b,a]=c$; $[c,a]=e$; $[c,b]=e$
[64,7]	$Q_8 \rtimes C_8$　$(8,4,8)$ $R=(2,2,2,2,2,2)$; $a^2=d$; $b^2=e$; $c^2=e$; $d^2=f$; $[b,a]=c$; $[c,a]=e$; $[c,b]=e$

编号	64 阶群（续）
[64,8]	$(C_4 \rtimes C_4) \rtimes C_4 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ c^2 = f;\ [b,a] = c;\ [c,a] = e;\ [c,b] = f;\ [d,b] = ef$
[64,9]	$(C_2 \times Q_8) \rtimes C_4 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ c^2 = f;\ [b,a] = c;\ [c,a] = e;\ [c,b] = f;\ [d,b] = ef$
[64,10]	$(C_8 \rtimes C_4) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ c^2 = f;\ d^2 = e;\ [b,a] = c;\ [c,a] = e;\ [c,b] = f;\ [d,b] = ef$
[64,11]	$(C_2 \times C_2) \,.\, ((C_4 \times C_2) \rtimes C_2) = (C_4 \times C_2) \,.\, (C_4 \times C_2) \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ c^2 = f;\ d^2 = e;\ [b,a] = c;\ [c,a] = e;\ [c,b] = f;$ $[d,b] = ef$
[64,12]	$(C_4 \rtimes C_8) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ c^2 = f;\ d^2 = ef;\ [b,a] = c;\ [c,a] = e;\ [c,b] = f;\ [d,b] = ef$
[64,13]	$(C_2 \times C_2) \,.\, ((C_4 \times C_2) \rtimes C_2) = (C_4 \times C_2) \,.\, (C_4 \times C_2) \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ c^2 = f;\ d^2 = ef;\ [b,a] = c;\ [c,a] = e;\ [c,b] = f;$ $[d,b] = ef$
[64,14]	$(C_2 \times C_2) \,.\, ((C_4 \times C_2) \rtimes C_2) = (C_4 \times C_2) \,.\, (C_4 \times C_2) \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = f;\ c^2 = f;\ d^2 = ef;\ [b,a] = c;\ [c,a] = e;\ [c,b] = f;$ $[d,b] = ef$
[64,15]	$C_8 \rtimes C_8 \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = c;\ c^2 = e;\ d^2 = f;\ [b,a] = c;\ [c,a] = e$
[64,16]	$C_8 \rtimes C_8 \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = ce;\ c^2 = e;\ d^2 = f;\ [b,a] = c;\ [c,a] = e$
[64,17]	$(C_4 \times C_2) \rtimes C_8 \quad (16,2,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ d^2 = f;\ [b,a] = c$
[64,18]	$(C_8 \times C_2) \rtimes C_4 \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ c^2 = f;\ [b,a] = c;\ [d,b] = f;\ [e,a] = f$
[64,19]	$C_4 \,.\, (C_4 \times C_4) \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ c^2 = f;\ d^2 = f;\ e^2 = f;\ [b,a] = c;\ [d,b] = f;\ [e,a] = f$
[64,20]	$(C_8 \rtimes C_2) \rtimes C_4 \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ c^2 = f;\ [b,a] = c;\ [c,a] = f;\ [e,a] = f$
[64,21]	$(C_4 \rtimes C_4) \rtimes C_4 \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ c^2 = f;\ e^2 = f;\ [b,a] = c;\ [c,a] = f;\ [e,a] = f$
[64,22]	$(C_4 \times C_2) \,.\, D_8 = C_4 \,.\, (C_4 \times C_4) \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ c^2 = f;\ d^2 = f;\ e^2 = f;\ [b,a] = c;\ [c,a] = f;\ [e,a] = f$
[64,23]	$((C_4 \times C_2) \rtimes C_2) \rtimes C_4 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ [b,a] = c;\ [c,a] = f;\ [d,b] = f$
[64,24]	$(C_8 \rtimes C_2) \rtimes C_4 \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ d^2 = f;\ [b,a] = c;\ [c,a] = f;\ [d,b] = f$
[64,25]	$(C_8 \rtimes C_2) \rtimes C_4 \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ e^2 = f;\ [b,a] = c;\ [c,a] = f;\ [d,b] = f$
[64,26]	$C_4 \times C_{16} \quad (64,1,16) \quad$ 阿贝尔。
[64,27]	$C_{16} \rtimes C_4 \quad (16,2,16)$ $R = (2,2,2,2,2,2);\ a^2 = c;\ b^2 = d;\ c^2 = e;\ e^2 = f;\ [b,a] = f$
[64,28]	$C_{16} \rtimes C_4 \quad (4,4,16)$ $R = (2,2,2,2,2,2);\ a^2 = c;\ b^2 = d;\ c^2 = e;\ e^2 = f;\ [b,a] = e;\ [c,b] = f;\ [d,a] = f$
[64,29]	$(C_2 \times C_2) \rtimes C_{16} \quad (16,2,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ d^2 = e;\ e^2 = f;\ [b,a] = c$

编号	64 阶群（续）
[64,30]	$(C_{16} \rtimes C_2) \rtimes C_2 \quad (4,4,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ d^2 = e;\ e^2 = f;\ [b,a] = c;\ [c,a] = f;\ [d,b] = f$
[64,31]	$(C_{16} \rtimes C_2) \rtimes C_2 \quad (8,4,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ c^2 = f;\ d^2 = e;\ e^2 = f;\ [b,a] = c;\ [c,b] = f;\ [d,b] = f$
[64,32]	$((C_4 \times C_2) \rtimes C_2) \rtimes C_4 \quad (2,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ [b,a] = c;\ [c,a] = e;\ [d,b] = e;\ [d,c] = f;\ [e,a] = f$
[64,33]	$((C_4 \times C_2) \rtimes C_2) \rtimes C_4 \quad (2,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = f;\ [b,a] = c;\ [c,a] = e;\ [d,b] = e;\ [d,c] = f;\ [e,a] = f$
[64,34]	$(C_4 \times C_4) \rtimes C_4 \quad (2,8,4)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ c^2 = f;\ [b,a] = c;\ [c,a] = e;\ [c,b] = f;\ [d,b] = ef;\ [d,c] = f;$ $[e,a] = f$
[64,35]	$(C_4 \times C_4) \rtimes C_4 \quad (2,8,4)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = f;\ c^2 = f;\ [b,a] = c;\ [c,a] = e;\ [c,b] = f;\ [d,b] = ef;$ $[d,c] = f;\ [e,a] = f$
[64,36]	$(C_2 \,.\, ((C_4 \times C_2) \rtimes C_2) = (C_2 \times C_2) \,.\, (C_4 \times C_2)) \rtimes C_2 \quad (2,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ c^2 = f;\ d^2 = f;\ [b,a] = c;\ [c,a] = e;\ [c,b] = f;\ [d,b] = ef;$ $[d,c] = f;\ [e,a] = f$
[64,37]	$C_2 \,.\, ((C_2 \times C_2 \times C_2) \rtimes C_4) = (C_4 \times C_2) \,.\, (C_4 \times C_2) \quad (2,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = f;\ c^2 = f;\ d^2 = f;\ [b,a] = c;\ [c,a] = e;\ [c,b] = f;$ $[d,b] = ef;\ [d,c] = f;\ [e,a] = f$
[64,38]	$D_{16} \rtimes C_4 \quad (4,8,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ c^2 = ef;\ e^2 = f;\ [b,a] = c;\ [c,a] = e;\ [c,b] = e;\ [e,a] = f;$ $[e,b] = f$
[64,39]	$Q_{16} \rtimes C_4 \quad (4,8,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = f;\ c^2 = ef;\ e^2 = f;\ [b,a] = c;\ [c,a] = e;\ [c,b] = e;$ $[e,a] = f;\ [e,b] = f$
[64,40]	$(C_{16} \times C_2) \rtimes C_2 \quad (4,8,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ c^2 = ef;\ d^2 = f;\ e^2 = f;\ [b,a] = c;\ [c,a] = e;\ [c,b] = e;$ $[e,a] = f;\ [e,b] = f$
[64,41]	$Q_{16} \rtimes C_4 \quad (2,8,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ c^2 = e;\ e^2 = f;\ [b,a] = c;\ [c,a] = e;\ [c,b] = ef;\ [d,b] = f;$ $[e,a] = f;\ [e,b] = f$
[64,42]	$(C_{16} \rtimes C_2) \rtimes C_2 \quad (2,8,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ c^2 = e;\ d^2 = f;\ e^2 = f;\ [b,a] = c;\ [c,a] = e;\ [c,b] = ef;$ $[d,b] = f;\ [e,a] = f;\ [e,b] = f$
[64,43]	$C_2 \,.\, ((C_8 \times C_2) \rtimes C_2) = C_8 \,.\, (C_4 \times C_2) \quad (2,8,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = f;\ c^2 = e;\ d^2 = f;\ e^2 = f;\ [b,a] = c;\ [c,a] = e;$ $[c,b] = ef;\ [d,b] = f;\ [e,a] = f;\ [e,b] = f$
[64,44]	$C_4 \rtimes C_{16} \quad (16,2,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = c;\ d^2 = e;\ e^2 = f;\ [b,a] = c$
[64,45]	$C_8 \,.\, D_8 = C_4 \,.\, (C_8 \times C_2) \quad (8,4,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = c;\ c^2 = f;\ d^2 = e;\ e^2 = f;\ [b,a] = c;\ [c,a] = f$
[64,46]	$C_{16} \rtimes C_4 \quad (2,8,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = c;\ c^2 = e;\ e^2 = f;\ [b,a] = c;\ [c,a] = e;\ [d,b] = f;$ $[e,a] = f$
[64,47]	$C_{16} \rtimes C_4 \quad (4,8,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = ce;\ c^2 = ef;\ e^2 = f;\ [b,a] = c;\ [c,a] = e;\ [e,a] = f$

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[64,48]	$C_{16} \rtimes C_4 \quad (4,8,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = cef;\ c^2 = ef;\ e^2 = f;\ [b,a] = c;\ [c,a] = e;\ [e,a] = f$
[64,49]	$C_4 \,.\, D_{16} = C_8 \,.\, (C_4 \times C_2) \quad (4,8,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = ce;\ c^2 = ef;\ d^2 = f;\ e^2 = f;\ [b,a] = c;\ [c,a] = e;$ $[e,a] = f$
[64,50]	$C_2 \times C_{32} \quad (64,1,32)$ 阿贝尔。
[64,51]	$C_{32} \rtimes C_2 \quad (16,2,32)$ $R = (2,2,2,2,2,2);\ a^2 = c;\ c^2 = d;\ d^2 = e;\ e^2 = f;\ [b,a] = f$
[64,52]	$D_{64} \quad (2,16,32)$ $R = (2,2,2,2,2,2);\ c^2 = de;\ d^2 = ef;\ e^2 = f;\ [b,a] = c;\ [c,a] = d;\ [c,b] = d;\ [d,a] = e;$ $[d,b] = e;\ [e,a] = f;\ [e,b] = f$
[64,53]	$\mathrm{QD}_{64} \quad (2,16,32)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ c^2 = de;\ d^2 = ef;\ e^2 = f;\ [b,a] = c;\ [c,a] = d;\ [c,b] = d;$ $[d,a] = e;\ [d,b] = e;\ [e,a] = f;\ [e,b] = f$
[64,54]	$Q_{64} \quad (2,16,32)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = f;\ c^2 = de;\ d^2 = ef;\ e^2 = f;\ [b,a] = c;\ [c,a] = d;$ $[c,b] = d;\ [d,a] = e;\ [d,b] = e;\ [e,a] = f;\ [e,b] = f$
[64,55]	$C_4 \times C_4 \times C_4 \quad (64,1,4)$ 阿贝尔。
[64,56]	$C_2 \times ((C_4 \times C_2) \rtimes C_4) \quad (16,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = f;\ [b,a] = d$
[64,57]	$(C_4 \times C_4) \rtimes C_4 \quad (16,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = f;\ c^2 = d;\ [b,a] = d$
[64,58]	$C_4 \times ((C_4 \times C_2) \rtimes C_2) \quad (16,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = f;\ c^2 = e;\ [b,a] = d$
[64,59]	$C_4 \times (C_4 \rtimes C_4) \quad (16,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = f;\ c^2 = de;\ [b,a] = d$
[64,60]	$(C_2 \times C_2) \rtimes ((C_4 \times C_2) \rtimes C_2) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ [b,a] = d;\ [c,a] = e$
[64,61]	$C_4 \rtimes ((C_4 \times C_2) \rtimes C_2) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = d;\ [b,a] = d;\ [c,a] = e$
[64,62]	$(C_4 \times C_2 \times C_2) \rtimes C_4 \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ c^2 = d;\ [b,a] = d;\ [c,a] = e$
[64,63]	$(C_4 \times C_4) \rtimes C_4 \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = e;\ c^2 = d;\ [b,a] = d;\ [c,a] = e$
[64,64]	$(C_4 \times C_4) \rtimes C_4 \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = de;\ c^2 = d;\ [b,a] = d;\ [c,a] = e$
[64,65]	$C_4 \rtimes (C_4 \rtimes C_4) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = d;\ c^2 = e;\ [b,a] = d;\ [c,a] = e$
[64,66]	$(C_2 \times C_2) \rtimes (C_4 \rtimes C_4) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = f;\ [b,a] = d;\ [c,a] = e$
[64,67]	$(C_2 \times C_2) \rtimes ((C_4 \times C_2) \rtimes C_2) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = df;\ [b,a] = d;\ [c,a] = e$
[64,68]	$(C_4 \rtimes C_4) \rtimes C_4 \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = f;\ c^2 = d;\ [b,a] = d;\ [c,a] = e$
[64,69]	$((C_4 \times C_2) \rtimes C_2) \rtimes C_4 \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = df;\ c^2 = d;\ [b,a] = d;\ [c,a] = e$

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[64,70]	$C_4 \rtimes (C_4 \rtimes C_4)$ $(8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = f;\ c^2 = e;\ [b,a] = d;\ [c,a] = e$
[64,71]	$C_4 \rtimes ((C_4 \times C_2) \rtimes C_2)$ $(8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = df;\ c^2 = e;\ [b,a] = d;\ [c,a] = e$
[64,72]	$(C_4 \times C_2) \rtimes Q_8$ $(8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = def;\ c^2 = e;\ [b,a] = d;\ [c,a] = e$
[64,73]	$(C_4 \times C_2) \rtimes D_8$ $(8,8,4)$ $R = (2,2,2,2,2,2);\ [b,a] = d;\ [c,a] = e;\ [c,b] = f$
[64,74]	$(C_2 \times C_2 \times C_2) \rtimes Q_8$ $(8,8,4)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = d;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f$
[64,75]	$(C_4 \times C_2) \rtimes D_8$ $(8,8,4)$ $R = (2,2,2,2,2,2);\ c^2 = d;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f$
[64,76]	$(C_4 \times C_2) \rtimes Q_8$ $(8,8,4)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = d;\ c^2 = d;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f$
[64,77]	$((C_4 \times C_2) \rtimes C_4) \rtimes C_2$ $(8,8,4)$ $R = (2,2,2,2,2,2);\ a^2 = de;\ b^2 = d;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f$
[64,78]	$((C_4 \times C_2) \rtimes C_4) \rtimes C_2$ $(8,8,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ c^2 = d;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f$
[64,79]	$(C_2 \times C_2 \times C_2) \,.\, (C_2 \times C_2 \times C_2)$ $(8,8,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = d;\ c^2 = d;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f$
[64,80]	$(C_2 \times (C_4 \rtimes C_4)) \rtimes C_2$ $(8,8,4)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = de;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f$
[64,81]	$(C_2 \times C_2 \times C_2) \,.\, (C_2 \times C_2 \times C_2)$ $(8,8,4)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = de;\ c^2 = d;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f$
[64,82]	$(C_2 \times C_2 \times C_2) \,.\, (C_2 \times C_2 \times C_2)$ $(8,8,4)$ $R = (2,2,2,2,2,2);\ a^2 = def;\ b^2 = de;\ c^2 = d;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f$
[64,83]	$C_2 \times C_4 \times C_8$ $(64,1,8)$ 阿贝尔。
[64,84]	$C_2 \times (C_8 \rtimes C_4)$ $(16,2,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ d^2 = f;\ [b,a] = f$
[64,85]	$C_4 \times (C_8 \rtimes C_2)$ $(16,2,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ d^2 = f;\ [c,a] = f$
[64,86]	$(C_8 \rtimes C_2) \rtimes C_4$ $(16,2,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = e;\ d^2 = f;\ [c,b] = f$
[64,87]	$C_2 \times ((C_8 \times C_2) \rtimes C_2)$ $(16,2,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ e^2 = f;\ [b,a] = d$
[64,88]	$(C_2 \times C_2) \rtimes (C_8 \rtimes C_2)$ $(8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ e^2 = f;\ [b,a] = d;\ [c,a] = f$
[64,89]	$((C_8 \times C_2) \rtimes C_2) \rtimes C_2$ $(8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ e^2 = f;\ [b,a] = d;\ [c,b] = f$
[64,90]	$C_2 \times ((C_2 \times C_2 \times C_2) \rtimes C_4)$ $(4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ [b,a] = d;\ [d,a] = f;\ [e,b] = f$
[64,91]	$(C_2 \times Q_8) \rtimes C_4$ $(4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ c^2 = f;\ [b,a] = d;\ [d,a] = f;\ [e,b] = f$
[64,92]	$C_2 \times ((C_8 \rtimes C_2) \rtimes C_2)$ $(4,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ e^2 = f;\ [b,a] = d;\ [d,a] = f;\ [e,b] = f$
[64,93]	$C_2 \times (C_2 \,.\, ((C_4 \times C_2) \rtimes C_2) = (C_2 \times C_2) \,.\, (C_4 \times C_2))$ $(4,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = f;\ e^2 = f;\ [b,a] = d;\ [d,a] = f;\ [e,b] = f$

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[64,94]	$(C_2 \,.\, ((C_4 \times C_2) \rtimes C_2) = (C_2 \times C_2) \,.\, (C_4 \times C_2)) \rtimes C_2 \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ c^2 = f;\ e^2 = f;\ [b,a] = d;\ [d,a] = f;\ [e,b] = f$
[64,95]	$C_2 \times ((C_8 \times C_2) \rtimes C_2) \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ d^2 = f;\ [b,a] = d;\ [d,a] = f;\ [d,b] = f$
[64,96]	$C_2 \times (Q_8 \rtimes C_4) \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = f;\ d^2 = f;\ [b,a] = d;\ [d,a] = f;\ [d,b] = f$
[64,97]	$((C_4 \times C_2) \rtimes C_2) \rtimes C_4 \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ c^2 = f;\ d^2 = f;\ [b,a] = d;\ [d,a] = f;\ [d,b] = f$
[64,98]	$Q_8 \rtimes (C_4 \times C_2) \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ d^2 = f;\ [b,a] = d;\ [c,b] = f;\ [d,a] = f;\ [d,b] = f$
[64,99]	$(C_2 \times D_8) \rtimes C_4 \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ c^2 = f;\ d^2 = f;\ [b,a] = d;\ [c,b] = f;\ [d,a] = f;\ [d,b] = f$
[64,100]	$(C_2 \times Q_8) \rtimes C_4 \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = f;\ c^2 = f;\ d^2 = f;\ [b,a] = d;\ [c,b] = f;\ [d,a] = f;$ $[d,b] = f$
[64,101]	$C_2 \times ((C_4 \times C_4) \rtimes C_2) \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [d,a] = f;\ [d,b] = f$
[64,102]	$(C_2 \times Q_8) \rtimes C_4 \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,b] = f;\ [d,a] = f;\ [d,b] = f$
[64,103]	$C_2 \times (C_4 \rtimes C_8) \quad (16,2,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = d;\ e^2 = f;\ [b,a] = d$
[64,104]	$C_4 \rtimes (C_8 \rtimes C_2) \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = d;\ e^2 = f;\ [b,a] = d;\ [c,a] = f$
[64,105]	$(C_4 \rtimes C_8) \rtimes C_2 \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = d;\ e^2 = f;\ [b,a] = d;\ [c,b] = f$
[64,106]	$C_2 \times (C_8 \rtimes C_4) \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = d;\ d^2 = f;\ [b,a] = d;\ [d,a] = f$
[64,107]	$C_2 \times (C_8 \rtimes C_4) \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = df;\ d^2 = f;\ [b,a] = d;\ [d,a] = f$
[64,108]	$(C_8 \times C_2) \rtimes C_4 \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = d;\ c^2 = f;\ d^2 = f;\ [b,a] = d;\ [d,a] = f$
[64,109]	$C_8 \rtimes (C_4 \times C_2) \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = d;\ d^2 = f;\ [b,a] = d;\ [c,b] = f;\ [d,a] = f$
[64,110]	$C_2 \times (C_4 \,.\, D_8 = C_4 \,.\, (C_4 \times C_2)) \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = d;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [d,a] = f$
[64,111]	$(C_4 \,.\, D_8 = C_4 \,.\, (C_4 \times C_2)) \rtimes C_2 \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = d;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,b] = f;\ [d,a] = f$
[64,112]	$(C_4 \times C_2) \rtimes C_8 \quad (16,2,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ c^2 = d;\ e^2 = f;\ [b,a] = d$
[64,113]	$(C_4 \rtimes C_8) \rtimes C_2 \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ c^2 = d;\ e^2 = f;\ [b,a] = d;\ [c,a] = f$
[64,114]	$(C_4 \rtimes C_8) \rtimes C_2 \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ c^2 = d;\ e^2 = f;\ [b,a] = d;\ [c,b] = f$
[64,115]	$C_8 \times D_8 \quad (16,2,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ c^2 = e;\ e^2 = f;\ [b,a] = d$
[64,116]	$(C_2 \times C_2) \rtimes (C_8 \rtimes C_2) \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ c^2 = e;\ e^2 = f;\ [b,a] = d;\ [c,a] = f$

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[64,117]	$C_4 \rtimes (C_8 \rtimes C_2)$ $(8,4,8)$ $R=(2,2,2,2,2,2);\ a^2=e;\ c^2=e;\ e^2=f;\ [b,a]=d;\ [c,a]=f;\ [c,b]=f$
[64,118]	$C_4 \times D_{16}$ $(8,4,8)$ $R=(2,2,2,2,2,2);\ a^2=e;\ c^2=e;\ d^2=f;\ [b,a]=d;\ [d,a]=f;\ [d,b]=f$
[64,119]	$C_4 \times \mathrm{QD}_{16}$ $(8,4,8)$ $R=(2,2,2,2,2,2);\ a^2=e;\ b^2=f;\ c^2=e;\ d^2=f;\ [b,a]=d;\ [d,a]=f;\ [d,b]=f$
[64,120]	$C_4 \times Q_{16}$ $(8,4,8)$ $R=(2,2,2,2,2,2);\ a^2=e;\ b^2=f;\ c^2=ef;\ d^2=f;\ [b,a]=d;\ [d,a]=f;\ [d,b]=f$
[64,121]	$C_8 \rtimes (C_4 \times C_2)$ $(4,4,8)$ $R=(2,2,2,2,2,2);\ a^2=e;\ c^2=e;\ d^2=f;\ [b,a]=d;\ [c,a]=f;\ [d,a]=f;\ [d,b]=f$
[64,122]	$Q_{16} \rtimes C_4$ $(4,4,8)$ $R=(2,2,2,2,2,2);\ a^2=e;\ b^2=f;\ c^2=e;\ d^2=f;\ [b,a]=d;\ [c,a]=f;\ [d,a]=f;$ $[d,b]=f$
[64,123]	$C_8 \rtimes (C_4 \times C_2)$ $(4,4,8)$ $R=(2,2,2,2,2,2);\ a^2=e;\ c^2=ef;\ d^2=f;\ [b,a]=d;\ [c,a]=f;\ [d,a]=f;\ [d,b]=f$
[64,124]	$\mathrm{QD}_{16} \rtimes C_4$ $(8,4,8)$ $R=(2,2,2,2,2,2);\ a^2=e;\ c^2=e;\ d^2=f;\ e^2=f;\ [b,a]=d;\ [d,a]=f;\ [d,b]=f$
[64,125]	$Q_{16} \rtimes C_4$ $(4,4,8)$ $R=(2,2,2,2,2,2);\ a^2=e;\ c^2=e;\ d^2=f;\ e^2=f;\ [b,a]=d;\ [c,a]=f;\ [d,a]=f;$ $[d,b]=f$
[64,126]	$C_8 \times Q_8$ $(16,2,8)$ $R=(2,2,2,2,2,2);\ a^2=e;\ b^2=d;\ c^2=de;\ e^2=f;\ [b,a]=d$
[64,127]	$C_8 \rtimes Q_8$ $(8,4,8)$ $R=(2,2,2,2,2,2);\ a^2=e;\ b^2=d;\ c^2=de;\ e^2=f;\ [b,a]=d;\ [c,a]=f$
[64,128]	$(C_2 \times C_2) \rtimes D_{16}$ $(4,8,8)$ $R=(2,2,2,2,2,2);\ d^2=f;\ [b,a]=d;\ [c,a]=e;\ [d,a]=f;\ [d,b]=f$
[64,129]	$(C_2 \times C_2) \rtimes \mathrm{QD}_{16}$ $(4,8,8)$ $R=(2,2,2,2,2,2);\ b^2=f;\ d^2=f;\ [b,a]=d;\ [c,a]=e;\ [d,a]=f;\ [d,b]=f$
[64,130]	$Q_8 \rtimes D_8$ $(4,8,8)$ $R=(2,2,2,2,2,2);\ c^2=f;\ d^2=f;\ [b,a]=d;\ [c,a]=e;\ [d,a]=f;\ [d,b]=f$
[64,131]	$(C_2 \times C_2) \rtimes \mathrm{QD}_{16}$ $(4,8,8)$ $R=(2,2,2,2,2,2);\ a^2=f;\ d^2=f;\ [b,a]=d;\ [c,a]=e;\ [d,a]=f;\ [d,b]=f$
[64,132]	$(C_2 \times C_2) \rtimes Q_{16}$ $(4,8,8)$ $R=(2,2,2,2,2,2);\ a^2=f;\ b^2=f;\ d^2=f;\ [b,a]=d;\ [c,a]=e;\ [d,a]=f;\ [d,b]=f$
[64,133]	$(Q_8 \rtimes C_4) \rtimes C_2$ $(4,8,8)$ $R=(2,2,2,2,2,2);\ a^2=f;\ c^2=f;\ d^2=f;\ [b,a]=d;\ [c,a]=e;\ [d,a]=f;\ [d,b]=f$
[64,134]	$Q_8 \rtimes D_8$ $(2,8,8)$ $R=(2,2,2,2,2,2);\ d^2=f;\ [b,a]=d;\ [c,a]=e;\ [d,a]=f;\ [d,b]=f;\ [d,c]=f;\ [e,b]=f$
[64,135]	$((C_4 \times C_4) \rtimes C_2) \rtimes C_2$ $(2,8,8)$ $R=(2,2,2,2,2,2);\ c^2=f;\ d^2=f;\ [b,a]=d;\ [c,a]=e;\ [d,a]=f;\ [d,b]=f;\ [d,c]=f;$ $[e,b]=f$
[64,136]	$(C_8 \rtimes C_2) \rtimes (C_2 \times C_2)$ $(2,8,8)$ $R=(2,2,2,2,2,2);\ a^2=f;\ d^2=f;\ [b,a]=d;\ [c,a]=e;\ [d,a]=f;\ [d,b]=f;\ [d,c]=f;$ $[e,b]=f$
[64,137]	$((C_4 \times C_4) \rtimes C_2) \rtimes C_2$ $(2,8,8)$ $R=(2,2,2,2,2,2);\ a^2=f;\ c^2=f;\ d^2=f;\ [b,a]=d;\ [c,a]=e;\ [d,a]=f;\ [d,b]=f;$ $[d,c]=f;\ [e,b]=f$

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[64,138]	$(C_4 \times C_2) \rtimes D_8 \quad (2,8,4)$ $R = (2,2,2,2,2,2);\ [b,a] = d;\ [c,a] = e;\ [d,c] = f;\ [e,b] = f$
[64,139]	$((C_4 \times C_2) \rtimes C_2) \rtimes (C_2 \times C_2) \quad (2,8,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,c] = f;\ [e,b] = f$
[64,140]	$C_4 \rtimes D_{16} \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = d;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;\ [e,c] = f$
[64,141]	$C_4 \rtimes \mathrm{QD}_{16} \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = d;\ c^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;\ [e,c] = f$
[64,142]	$C_4 \rtimes \mathrm{QD}_{16} \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = d;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;\ [e,c] = f$
[64,143]	$C_4 \rtimes Q_{16} \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = d;\ c^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;$ $[e,c] = f$
[64,144]	$(Q_8 \rtimes C_4) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = df;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;\ [e,c] = f$
[64,145]	$(Q_8 \rtimes C_4) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = df;\ c^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;\ [e,c] = f$
[64,146]	$(C_2 \times C_2) \rtimes \mathrm{QD}_{16} \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = d;\ d^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,a] = f$
[64,147]	$(C_2 \times C_2) \rtimes D_{16} \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = df;\ d^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,a] = f$
[64,148]	$(C_2 \times C_2) \rtimes Q_{16} \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = df;\ d^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,a] = f$
[64,149]	$C_8 \rtimes D_8 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = d;\ d^2 = f;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f;\ [d,a] = f$
[64,150]	$C_8 \rtimes D_8 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = df;\ d^2 = f;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f;\ [d,a] = f$
[64,151]	$(Q_8 \rtimes C_4) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = df;\ d^2 = f;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f;\ [d,a] = f$
[64,152]	$(C_2 \,.\, ((C_4 \times C_2) \rtimes C_2) = (C_2 \times C_2) \,.\, (C_4 \times C_2)) \rtimes C_2 \quad (2,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = d;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,a] = f;\ [e,a] = f;$ $[e,c] = f$
[64,153]	$((C_8 \rtimes C_2) \rtimes C_2) \rtimes C_2 \quad (2,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = df;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,a] = f;\ [e,a] = f;$ $[e,c] = f$
[64,154]	$(C_2 \,.\, ((C_4 \times C_2) \rtimes C_2) = (C_2 \times C_2) \,.\, (C_4 \times C_2)) \rtimes C_2 \quad (2,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = df;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,a] = f;$ $[e,a] = f;\ [e,c] = f$
[64,155]	$D_8 \rtimes Q_8 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = d;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;\ [e,c] = f$
[64,156]	$Q_8 \rtimes Q_8 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = d;\ c^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;$ $[e,c] = f$
[64,157]	$D_8 \rtimes Q_8 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = df;\ b^2 = d;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;\ [e,c] = f$

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[64,158]	$Q_8 \rtimes Q_8 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = df;\ b^2 = d;\ c^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;$ $[e,c] = f$
[64,159]	$(C_8 \rtimes C_4) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = df;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;\ [e,c] = f$
[64,160]	$(C_2 \times C_2) \,.\, (C_2 \times D_8) = (C_4 \times C_2) \,.\, (C_2 \times C_2 \times C_2) \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = df;\ c^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;$ $[e,c] = f$
[64,161]	$(C_8 \rtimes C_4) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ c^2 = d;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;\ [e,c] = f$
[64,162]	$(C_8 \rtimes C_4) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ c^2 = df;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;\ [e,c] = f$
[64,163]	$(C_8 \rtimes C_4) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = f;\ c^2 = d;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;\ [e,c] = f$
[64,164]	$(C_8 \rtimes C_4) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ c^2 = d;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;\ [e,c] = f$
[64,165]	$(C_8 \rtimes C_4) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ c^2 = df;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;\ [e,c] = f$
[64,166]	$(C_8 \rtimes C_4) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = f;\ c^2 = d;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f;$ $[e,c] = f$
[64,167]	$(C_4 \rtimes Q_8) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = e;\ c^2 = d;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,a] = f;$ $[e,a] = f$
[64,168]	$(C_2 \times C_2) \,.\, (C_2 \times D_8) = (C_4 \times C_2) \,.\, (C_2 \times C_2 \times C_2) \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = e;\ c^2 = d;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,a] = f;$ $[e,a] = f$
[64,169]	$((C_2 \times C_2) \,.\, (C_2 \times C_2 \times C_2)) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = ef;\ c^2 = d;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,a] = f;$ $[e,a] = f$
[64,170]	$(C_4 \rtimes Q_8) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = e;\ c^2 = d;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f;$ $[d,a] = f;\ [e,a] = f$
[64,171]	$((C_2 \times C_2) \,.\, (C_2 \times C_2 \times C_2)) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = ef;\ c^2 = d;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f;$ $[d,a] = f;\ [e,a] = f$
[64,172]	$(C_2 \times C_2) \,.\, (C_2 \times D_8) = (C_4 \times C_2) \,.\, (C_2 \times C_2 \times C_2) \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = ef;\ c^2 = d;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;$ $[c,b] = f;\ [d,a] = f;\ [e,a] = f$
[64,173]	$C_4 \rtimes \mathrm{QD}_{16} \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = d;\ c^2 = e;\ d^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,a] = f$
[64,174]	$C_4 \rtimes D_{16} \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = df;\ c^2 = e;\ d^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,a] = f$
[64,175]	$C_4 \rtimes Q_{16} \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = df;\ c^2 = e;\ d^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,a] = f$
[64,176]	$(C_2 \times Q_{16}) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = d;\ c^2 = ef;\ d^2 = f;\ [b,a] = d;\ [c,a] = e;\ [d,a] = f$

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[64,177]	$C_8 \rtimes D_8 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ b^2 = d;\ c^2 = e;\ d^2 = f;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f;\ [d,a] = f$
[64,178]	$(C_2 \times Q_{16}) \rtimes C_2 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = d;\ c^2 = e;\ d^2 = f;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f;$ $[d,a] = f$
[64,179]	$C_8 \rtimes Q_8 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = d;\ c^2 = e;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f$
[64,180]	$(C_2 \times C_2) \,.\, (C_2 \times D_8) = (C_4 \times C_2) \,.\, (C_2 \times C_2 \times C_2) \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = df;\ c^2 = e;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f$
[64,181]	$C_8 \rtimes Q_8 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = d;\ c^2 = ef;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [e,a] = f$
[64,182]	$C_8 \rtimes Q_8 \quad (4,8,8)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ b^2 = d;\ c^2 = e;\ e^2 = f;\ [b,a] = d;\ [c,a] = e;\ [c,b] = f;$ $[e,a] = f$
[64,183]	$C_2 \times C_2 \times C_{16} \quad (64,1,16)$ 阿贝尔。
[64,184]	$C_2 \times (C_{16} \rtimes C_2) \quad (16,2,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ d^2 = e;\ e^2 = f;\ [b,a] = f$
[64,185]	$(C_{16} \rtimes C_2) \rtimes C_2 \quad (16,2,16)$ $R = (2,2,2,2,2,2);\ a^2 = d;\ d^2 = e;\ e^2 = f;\ [c,b] = f$
[64,186]	$C_2 \times D_{32} \quad (4,8,16)$ $R = (2,2,2,2,2,2);\ d^2 = ef;\ e^2 = f;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e;\ [e,a] = f;\ [e,b] = f$
[64,187]	$C_2 \times \mathrm{QD}_{32} \quad (4,8,16)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ d^2 = ef;\ e^2 = f;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e;\ [e,a] = f;$ $[e,b] = f$
[64,188]	$C_2 \times Q_{32} \quad (4,8,16)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = f;\ d^2 = ef;\ e^2 = f;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e;$ $[e,a] = f;\ [e,b] = f$
[64,189]	$Q_{32} \rtimes C_2 \quad (4,8,16)$ $R = (2,2,2,2,2,2);\ c^2 = f;\ d^2 = ef;\ e^2 = f;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e;\ [e,a] = f;$ $[e,b] = f$
[64,190]	$C_{16} \rtimes (C_2 \times C_2) \quad (2,8,16)$ $R = (2,2,2,2,2,2);\ d^2 = ef;\ e^2 = f;\ [b,a] = d;\ [c,a] = f;\ [d,a] = e;\ [d,b] = e;\ [e,a] = f;$ $[e,b] = f$
[64,191]	$Q_{32} \rtimes C_2 \quad (2,8,16)$ $R = (2,2,2,2,2,2);\ b^2 = f;\ d^2 = ef;\ e^2 = f;\ [b,a] = d;\ [c,a] = f;\ [d,a] = e;\ [d,b] = e;$ $[e,a] = f;\ [e,b] = f$
[64,192]	$C_2 \times C_2 \times C_4 \times C_4 \quad (64,1,4)$ 阿贝尔。
[64,193]	$C_2 \times C_2 \times ((C_4 \times C_2) \rtimes C_2) \quad (16,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ [b,a] = e$
[64,194]	$C_2 \times C_2 \times (C_4 \rtimes C_4) \quad (16,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = e;\ [b,a] = e$
[64,195]	$C_2 \times ((C_4 \times C_4) \rtimes C_2) \quad (16,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ c^2 = e;\ [b,a] = e$
[64,196]	$C_2 \times C_4 \times D_8 \quad (16,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ c^2 = f;\ [b,a] = e$
[64,197]	$C_2 \times C_4 \times Q_8 \quad (16,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = e;\ c^2 = ef;\ [b,a] = e$

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[64,198]	$C_4 \times ((C_4 \times C_2) \rtimes C_2) \quad (16,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ c^2 = f;\ d^2 = e;\ [b,a] = e$
[64,199]	$D_8 \rtimes (C_4 \times C_2) \quad (4,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ [b,a] = e;\ [c,b] = e;\ [d,a] = e$
[64,200]	$(C_2 \times Q_8) \rtimes C_4 \quad (4,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = e;\ c^2 = e;\ [b,a] = e;\ [c,b] = e;\ [d,a] = e$
[64,201]	$Q_8 \rtimes (C_4 \times C_2) \quad (4,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ d^2 = e;\ [b,a] = e;\ [c,b] = e;\ [d,a] = e$
[64,202]	$C_2 \times ((C_2 \times C_2 \times C_2 \times C_2) \rtimes C_2) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ [b,a] = e;\ [c,a] = f$
[64,203]	$C_2 \times ((C_4 \times C_2 \times C_2) \rtimes C_2) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = e;\ [b,a] = e;\ [c,a] = f$
[64,204]	$C_2 \times ((C_2 \times Q_8) \rtimes C_2) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = e;\ [b,a] = e;\ [c,a] = f$
[64,205]	$C_2 \times ((C_4 \times C_2 \times C_2) \rtimes C_2) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ c^2 = e;\ [b,a] = e;\ [c,a] = f$
[64,206]	$(C_2 \times C_2) \rtimes ((C_4 \times C_2) \rtimes C_2) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ d^2 = e;\ [b,a] = e;\ [c,a] = f$
[64,207]	$C_2 \times ((C_4 \times C_4) \rtimes C_2) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = f;\ c^2 = e;\ [b,a] = e;\ [c,a] = f$
[64,208]	$C_2 \times ((C_2 \times C_2) \,.\, (C_2 \times C_2 \times C_2)) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = f;\ c^2 = e;\ [b,a] = e;\ [c,a] = f$
[64,209]	$C_2 \times ((C_4 \times C_4) \rtimes C_2) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = ef;\ c^2 = e;\ [b,a] = e;\ [c,a] = f$
[64,210]	$((C_4 \times C_4) \rtimes C_2) \rtimes C_2 \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = f;\ d^2 = e;\ [b,a] = e;\ [c,a] = f$
[64,211]	$C_2 \times ((C_4 \times C_4) \rtimes C_2) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = e;\ c^2 = f;\ [b,a] = e;\ [c,a] = f$
[64,212]	$C_2 \times (C_4 \rtimes Q_8) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = e;\ c^2 = f;\ [b,a] = e;\ [c,a] = f$
[64,213]	$C_4 \rtimes ((C_4 \times C_2) \rtimes C_2) \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ c^2 = f;\ d^2 = e;\ [b,a] = e;\ [c,a] = f$
[64,214]	$(C_4 \times C_2) \rtimes Q_8 \quad (8,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ c^2 = f;\ d^2 = e;\ [b,a] = e;\ [c,a] = f$
[64,215]	$(C_2 \times C_2 \times C_2) \rtimes D_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ [b,a] = e;\ [c,a] = f;\ [c,b] = e;\ [d,a] = e$
[64,216]	$(C_4 \times C_2) \rtimes D_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = e;\ [b,a] = e;\ [c,a] = f;\ [c,b] = e;\ [d,a] = e$
[64,217]	$((C_2 \times Q_8) \rtimes C_2) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = e;\ c^2 = e;\ [b,a] = e;\ [c,a] = f;\ [c,b] = e;\ [d,a] = e$
[64,218]	$(C_4 \times C_2) \rtimes D_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [c,b] = e;\ [d,a] = e$
[64,219]	$(C_2 \times Q_8) \rtimes (C_2 \times C_2) \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = f;\ [b,a] = e;\ [c,a] = f;\ [c,b] = e;\ [d,a] = e$
[64,220]	$((C_4 \times C_4) \rtimes C_2) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = f;\ c^2 = e;\ [b,a] = e;\ [c,a] = f;\ [c,b] = e;\ [d,a] = e$

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[64,221]	$((C_4 \times C_2 \times C_2) \rtimes C_2) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = f;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [c,b] = e;\ [d,a] = e$
[64,222]	$((C_2 \times C_2) \,.\, (C_2 \times C_2 \times C_2)) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = f;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [c,b] = e;\ [d,a] = e$
[64,223]	$((C_2 \times Q_8) \rtimes C_2) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = ef;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [c,b] = e;\ [d,a] = e$
[64,224]	$(C_2 \times C_2 \times C_2) \rtimes Q_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ c^2 = f;\ [b,a] = e;\ [c,a] = f;\ [c,b] = e;\ [d,a] = e$
[64,225]	$(C_4 \times C_2) \rtimes Q_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = e;\ c^2 = f;\ [b,a] = e;\ [c,a] = f;\ [c,b] = e;\ [d,a] = e$
[64,226]	$D_8 \times D_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,227]	$D_8 \rtimes D_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,228]	$D_8 \rtimes D_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ c^2 = e;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,229]	$Q_8 \rtimes D_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = e;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,230]	$Q_8 \times D_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = e;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,231]	$Q_8 \rtimes D_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = e;\ c^2 = e;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,232]	$(C_4 \rtimes C_4) \rtimes (C_2 \times C_2) \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = f;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,233]	$((C_4 \times C_4) \rtimes C_2) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = f;\ c^2 = e;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,234]	$((C_4 \times C_4) \rtimes C_2) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = ef;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,235]	$D_8 \rtimes Q_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = ef;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,236]	$((C_4 \times C_4) \rtimes C_2) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = f;\ c^2 = e;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,237]	$((C_4 \times C_4) \rtimes C_2) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ b^2 = ef;\ c^2 = e;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,238]	$Q_8 \rtimes Q_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = e;\ c^2 = f;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,239]	$Q_8 \times Q_8 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = ef;\ b^2 = e;\ c^2 = f;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,240]	$((C_4 \times C_2 \times C_2) \rtimes C_2) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = ef;\ c^2 = f;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [d,b] = e$
[64,241]	$(C_4 \rtimes C_4) \rtimes (C_2 \times C_2) \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ [b,a] = e;\ [c,a] = f;\ [c,b] = ef;\ [d,a] = ef;\ [d,b] = e$
[64,242]	$((C_4 \times C_2) \rtimes C_2) \rtimes (C_2 \times C_2) \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = e;\ [b,a] = e;\ [c,a] = f;\ [c,b] = ef;\ [d,a] = ef;\ [d,b] = e$
[64,243]	$((C_2 \times C_2) \,.\, (C_2 \times C_2 \times C_2)) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ c^2 = e;\ [b,a] = e;\ [c,a] = f;\ [c,b] = ef;\ [d,a] = ef;\ [d,b] = e$

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[64,244]	$((C_4 \times C_4) \rtimes C_2) \rtimes C_2 \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = e;\ c^2 = e;\ [b,a] = e;\ [c,a] = f;\ [c,b] = ef;\ [d,a] = ef;$ $[d,b] = e$
[64,245]	$(C_2 \times C_2) \,.\, (C_2 \times C_2 \times C_2 \times C_2) \quad (4,4,4)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ b^2 = f;\ c^2 = f;\ d^2 = e;\ [b,a] = e;\ [c,a] = f;\ [c,b] = ef;$ $[d,a] = ef;\ [d,b] = e$
[64,246]	$C_2 \times C_2 \times C_2 \times C_8 \quad (64,1,8)$ 阿贝尔。
[64,247]	$C_2 \times C_2 \times (C_8 \rtimes C_2) \quad (16,2,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ e^2 = f;\ [b,a] = f$
[64,248]	$C_2 \times ((C_8 \times C_2) \rtimes C_2) \quad (16,2,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ e^2 = f;\ [c,b] = f$
[64,249]	$(C_8 \times C_2) \rtimes (C_2 \times C_2) \quad (4,2,8)$ $R = (2,2,2,2,2,2);\ a^2 = e;\ e^2 = f;\ [c,b] = f;\ [d,a] = f$
[64,250]	$C_2 \times C_2 \times D_{16} \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ e^2 = f;\ [b,a] = e;\ [e,a] = f;\ [e,b] = f$
[64,251]	$C_2 \times C_2 \times \mathrm{QD}_{16} \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ e^2 = f;\ [b,a] = e;\ [e,a] = f;\ [e,b] = f$
[64,252]	$C_2 \times C_2 \times Q_{16} \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = f;\ e^2 = f;\ [b,a] = e;\ [e,a] = f;\ [e,b] = f$
[64,253]	$C_2 \times ((C_8 \times C_2) \rtimes C_2) \quad (8,4,8)$ $R = (2,2,2,2,2,2);\ c^2 = f;\ e^2 = f;\ [b,a] = e;\ [e,a] = f;\ [e,b] = f$
[64,254]	$C_2 \times (C_8 \rtimes (C_2 \times C_2)) \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ e^2 = f;\ [b,a] = e;\ [c,a] = f;\ [e,a] = f;\ [e,b] = f$
[64,255]	$C_2 \times ((C_2 \times Q_8) \rtimes C_2) \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ b^2 = f;\ e^2 = f;\ [b,a] = e;\ [c,a] = f;\ [e,a] = f;\ [e,b] = f$
[64,256]	$\mathrm{QD}_{16} \rtimes (C_2 \times C_2) \quad (4,4,8)$ $R = (2,2,2,2,2,2);\ d^2 = f;\ e^2 = f;\ [b,a] = e;\ [c,a] = f;\ [e,a] = f;\ [e,b] = f$
[64,257]	$Q_{16} \rtimes (C_2 \times C_2) \quad (2,4,8)$ $R = (2,2,2,2,2,2);\ e^2 = f;\ [b,a] = e;\ [d,c] = f;\ [e,a] = f;\ [e,b] = f$
[64,258]	$\mathrm{QD}_{16} \rtimes (C_2 \times C_2) \quad (2,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ e^2 = f;\ [b,a] = e;\ [d,c] = f;\ [e,a] = f;\ [e,b] = f$
[64,259]	$(C_2 \times Q_{16}) \rtimes C_2 \quad (2,4,8)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = f;\ e^2 = f;\ [b,a] = e;\ [d,c] = f;\ [e,a] = f;\ [e,b] = f$
[64,260]	$C_2 \times C_2 \times C_2 \times C_2 \times C_4 \quad (64,1,4)$ 阿贝尔。
[64,261]	$C_2 \times C_2 \times C_2 \times D_8 \quad (16,2,4)$ $R = (2,2,2,2,2,2);\ [b,a] = f$
[64,262]	$C_2 \times C_2 \times C_2 \times Q_8 \quad (16,2,4)$ $R = (2,2,2,2,2,2);\ a^2 = f;\ b^2 = f;\ [b,a] = f$
[64,263]	$C_2 \times C_2 \times ((C_4 \times C_2) \rtimes C_2) \quad (16,2,4)$ $R = (2,2,2,2,2,2);\ c^2 = f;\ [b,a] = f$
[64,264]	$C_2 \times ((C_2 \times C_2 \times C_2) \rtimes (C_2 \times C_2)) \quad (4,2,4)$ $R = (2,2,2,2,2,2);\ [b,a] = f;\ [c,b] = f;\ [d,a] = f$
[64,265]	$C_2 \times ((C_2 \times Q_8) \rtimes C_2) \quad (4,2,4)$ $R = (2,2,2,2,2,2);\ b^2 = f;\ c^2 = f;\ [b,a] = f;\ [c,b] = f;\ [d,a] = f$
[64,266]	$((C_4 \times C_2) \rtimes C_2) \rtimes (C_2 \times C_2) \quad (4,2,4)$ $R = (2,2,2,2,2,2);\ e^2 = f;\ [b,a] = f;\ [c,b] = f;\ [d,a] = f$

编号	64 阶群（续）
[64,267]	$C_2 \times C_2 \times C_2 \times C_2 \times C_2 \times C_2 \quad (64,1,2)$ 阿贝尔。

65 阶群

只有一个同构类型：C_{65}，编号 $[65,1]$。

证明. 这里 $p = 5 < q = 13$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q = 1$。令 $N = C_q \trianglelefteq G$，取 p 阶子群 $H = C_p$，则 $N \cap H = 1$，$NH = G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $5 \nmid 12$，作用只能平凡，故只有 $C_q \times C_p \cong C_{65}$。 □

66 阶群

共有 4 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[66,1]	$C_{11} \times S_3 \quad (11,3,66)$ $R = (2,11,3);\ [c,a] = c$
[66,2]	$C_3 \times D_{22} \quad (3,11,66)$ $R = (2,3,11);\ [c,a] = c^9$
[66,3]	$D_{66} \quad (1,33,66)$ $R = (2,3,11);\ [b,a] = b;\ [c,a] = c^9$
[66,4]	$C_{66} \quad (66,1,66)$ 阿贝尔。

67 阶群

只有一个同构类型：C_{67}，编号 $[67,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 67$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 67$。因此 $G = \langle a \rangle \cong C_{67}$。 □

68 阶群

共有 5 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[68,1]	$C_{17} \rtimes C_4 \quad (2,17,68)$ $R = (2,2,17);\ a^2 = b;\ [c,a] = c^{15}$
[68,2]	$C_{68} \quad (68,1,68)$ 阿贝尔。
[68,3]	$C_{17} \rtimes C_4 \quad (1,17,68)$ $R = (2,2,17);\ a^2 = b;\ [c,a] = c^{12};\ [c,b] = c^{15}$

编号	68 阶群（续）
[68,4]	$D_{68}\quad (2,17,34)$ $R=(2,2,17);\ [c,a]=c^{15}$
[68,5]	$C_2\times C_{34}\quad (68,1,34)\quad$ 阿贝尔。

69 阶群

只有一个同构类型：C_{69}，编号 $[69,1]$。

证明. 这里 $p=3<q=23$。Sylow 定理给出 $n_q\mid p$ 且 $n_q\equiv 1\pmod q$，所以 $n_q=1$。令 $N=C_q\trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q\rtimes C_p$。作用由同态 $C_p\to \mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $3\nmid 22$，作用只能平凡，故只有 $C_q\times C_p\cong C_{69}$。 □

70 阶群

共有 4 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[70,1]	$C_7\times D_{10}\quad (7,5,70)$ $R=(2,7,5);\ [c,a]=c^{3}$
[70,2]	$C_5\times D_{14}\quad (5,7,70)$ $R=(2,5,7);\ [c,a]=c^{5}$
[70,3]	$D_{70}\quad (1,35,70)$ $R=(2,5,7);\ [b,a]=b^{3};\ [c,a]=c^{5}$
[70,4]	$C_{70}\quad (70,1,70)\quad$ 阿贝尔。

71 阶群

只有一个同构类型：C_{71}，编号 $[71,1]$。

证明. 任取非单位元 $a\in G$。由拉格朗日定理，$\mathrm{ord}(a)\mid 71$，又 $\mathrm{ord}(a)>1$，所以 $\mathrm{ord}(a)=71$。因此 $G=\langle a\rangle\cong C_{71}$。 □

72 阶群

共有 50 个同构类型，其中 6 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[72,1]	$C_9\rtimes C_8\quad (4,9,72)$ $R=(2,2,2,3,3);\ a^2=b;\ b^2=c;\ d^3=e^2;\ [d,a]=de;\ [e,a]=e$
[72,2]	$C_{72}\quad (72,1,72)\quad$ 阿贝尔。

编号	72 阶群（续）
[72,3]	$Q_8 \rtimes C_9 \quad (6,8,36)$ $R = (3,3,2,2,2);\ a^3 = b;\ c^2 = e;\ d^2 = e;\ [c,a] = cde;\ [d,a] = ce;\ [d,c] = e$
[72,4]	$C_9 \rtimes Q_8 \quad (2,18,36)$ $R = (2,2,2,3,3);\ a^2 = c;\ b^2 = c;\ d^3 = e^2;\ [b,a] = c;\ [d,a] = de;\ [e,a] = e$
[72,5]	$C_4 \times D_{18} \quad (4,9,36)$ $R = (2,2,2,3,3);\ b^2 = c;\ d^3 = e^2;\ [d,a] = de;\ [e,a] = e$
[72,6]	$D_{72} \quad (2,18,36)$ $R = (2,2,2,3,3);\ b^2 = c;\ d^3 = e^2;\ [b,a] = c;\ [d,a] = de;\ [e,a] = e$
[72,7]	$C_2 \times (C_9 \rtimes C_4) \quad (4,9,36)$ $R = (2,2,2,3,3);\ a^2 = c;\ d^3 = e^2;\ [d,a] = de;\ [e,a] = e$
[72,8]	$(C_{18} \times C_2) \rtimes C_2 \quad (2,18,36)$ $R = (2,2,2,3,3);\ d^3 = e^2;\ [b,a] = c;\ [d,a] = de;\ [e,a] = e$
[72,9]	$C_2 \times C_{36} \quad (72,1,36)$ 阿贝尔。
[72,10]	$C_9 \times D_8 \quad (18,2,36)$ $R = (2,2,3,2,3);\ c^3 = e;\ [b,a] = d$
[72,11]	$C_9 \times Q_8 \quad (18,2,36)$ $R = (2,2,3,2,3);\ a^2 = d;\ b^2 = d;\ c^3 = e;\ [b,a] = d$
[72,12]	$C_3 \times (C_3 \rtimes C_8) \quad (12,3,24)$ $R = (2,3,2,2,3);\ a^2 = c;\ c^2 = d;\ [e,a] = e$
[72,13]	$(C_3 \times C_3) \rtimes C_8 \quad (4,9,24)$ $R = (2,2,2,3,3);\ a^2 = b;\ b^2 = c;\ [d,a] = d;\ [e,a] = e$
[72,14]	$C_3 \times C_{24} \quad (72,1,24)$ 阿贝尔。
[72,15]	$((C_2 \times C_2) \rtimes C_9) \rtimes C_2 \quad (1,36,36)$ $R = (2,3,3,2,2);\ b^3 = c^2;\ [b,a] = bc;\ [c,a] = c;\ [d,a] = de;\ [d,b] = de;\ [e,a] = de;\ [e,b] = d$
[72,16]	$C_2 \times ((C_2 \times C_2) \rtimes C_9) \quad (6,4,18)$ $R = (2,3,3,2,2);\ b^3 = c;\ [d,b] = de;\ [e,b] = d$
[72,17]	$C_2 \times C_2 \times D_{18} \quad (4,9,18)$ $R = (2,2,2,3,3);\ d^3 = e^2;\ [d,a] = de;\ [e,a] = e$
[72,18]	$C_2 \times C_2 \times C_{18} \quad (72,1,18)$ 阿贝尔。
[72,19]	$(C_3 \times C_3) \rtimes C_8 \quad (2,9,24)$ $R = (2,2,2,3,3);\ a^2 = b;\ b^2 = c;\ [d,a] = e^2;\ [d,b] = d;\ [e,a] = d^2e;\ [e,b] = e$
[72,20]	$(C_3 \rtimes C_4) \times S_3 \quad (2,9,12)$ $R = (2,2,2,3,3);\ b^2 = c;\ [d,b] = d;\ [e,a] = e$
[72,21]	$(C_3 \times C_3) \rtimes (C_4 \times C_2) \quad (2,9,12)$ $R = (2,2,2,3,3);\ a^2 = c;\ b^2 = c;\ [d,b] = d;\ [e,a] = e$
[72,22]	$(C_6 \times S_3) \rtimes C_2 \quad (2,18,12)$ $R = (2,2,2,3,3);\ [b,a] = c;\ [d,b] = d;\ [e,a] = e$
[72,23]	$(C_6 \times S_3) \rtimes C_2 \quad (2,18,12)$ $R = (2,2,2,3,3);\ b^2 = c;\ [b,a] = c;\ [d,b] = d;\ [e,a] = e$
[72,24]	$(C_3 \times C_3) \rtimes Q_8 \quad (2,18,12)$ $R = (2,2,2,3,3);\ a^2 = c;\ b^2 = c;\ [b,a] = c;\ [d,b] = d;\ [e,a] = e$
[72,25]	$C_3 \times \mathrm{SL}(2,3) \quad (6,8,12)$ $R = (3,3,2,2,2);\ c^2 = e;\ d^2 = e;\ [c,a] = cde;\ [d,a] = ce;\ [d,c] = e$
[72,26]	$C_3 \times (C_3 \rtimes Q_8) \quad (6,6,12)$ $R = (2,2,3,2,3);\ a^2 = d;\ b^2 = d;\ [b,a] = d;\ [e,a] = e$

编号	72 阶群（续）
[72,27]	$C_{12} \times S_3 \quad (12,3,12)$ $R = (2,2,3,2,3);\ b^2 = d;\ [e,a] = e$
[72,28]	$C_3 \times D_{24} \quad (6,6,12)$ $R = (2,2,3,2,3);\ b^2 = d;\ [b,a] = d;\ [e,a] = e$
[72,29]	$C_6 \times (C_3 \rtimes C_4) \quad (12,3,12)$ $R = (2,2,3,2,3);\ a^2 = d;\ [e,a] = e$
[72,30]	$C_3 \times ((C_6 \times C_2) \rtimes C_2) \quad (6,6,12)$ $R = (2,2,3,2,3);\ [b,a] = d;\ [e,a] = e$
[72,31]	$(C_3 \times C_3) \rtimes Q_8 \quad (2,18,12)$ $R = (2,2,2,3,3);\ a^2 = c;\ b^2 = c;\ [b,a] = c;\ [d,a] = d;\ [e,a] = e$
[72,32]	$C_4 \times ((C_3 \times C_3) \rtimes C_2) \quad (4,9,12)$ $R = (2,2,2,3,3);\ b^2 = c;\ [d,a] = d;\ [e,a] = e$
[72,33]	$(C_{12} \times C_3) \rtimes C_2 \quad (2,18,12)$ $R = (2,2,2,3,3);\ b^2 = c;\ [b,a] = c;\ [d,a] = d;\ [e,a] = e$
[72,34]	$C_2 \times ((C_3 \times C_3) \rtimes C_4) \quad (4,9,12)$ $R = (2,2,2,3,3);\ a^2 = c;\ [d,a] = d;\ [e,a] = e$
[72,35]	$(C_6 \times C_6) \rtimes C_2 \quad (2,18,12)$ $R = (2,2,2,3,3);\ [b,a] = c;\ [d,a] = d;\ [e,a] = e$
[72,36]	$C_6 \times C_{12} \quad (72,1,12)$ 阿贝尔。
[72,37]	$C_3 \times C_3 \times D_8 \quad (18,2,12)$ $R = (2,2,3,3,2);\ [b,a] = e$
[72,38]	$C_3 \times C_3 \times Q_8 \quad (18,2,12)$ $R = (2,2,3,3,2);\ a^2 = e;\ b^2 = e;\ [b,a] = e$
[72,39]	$(C_3 \times C_3) \rtimes C_8 \quad (1,9,24)$ $R = (2,2,2,3,3);\ a^2 = b;\ b^2 = c;\ [d,a] = d^2e;\ [d,b] = e^2;\ [d,c] = d;\ [e,a] = de;\ [e,b] = d^2e;$ $[e,c] = e$
[72,40]	$(S_3 \times S_3) \rtimes C_2 \quad (1,18,12)$ $R = (2,2,2,3,3);\ [b,a] = c;\ [d,a] = d;\ [d,b] = d^2e;\ [d,c] = d;\ [e,b] = de^2;\ [e,c] = e$
[72,41]	$(C_3 \times C_3) \rtimes Q_8 \quad (1,18,12)$ $R = (2,2,2,3,3);\ a^2 = c;\ b^2 = c;\ [b,a] = c;\ [d,a] = e^2;\ [d,b] = d^2e;\ [d,c] = d;\ [e,a] = d^2e;$ $[e,b] = d^2e^2;\ [e,c] = e$
[72,42]	$C_3 \times S_4 \quad (3,12,12)$ $R = (2,3,3,2,2);\ [c,a] = c;\ [d,a] = de;\ [d,c] = de;\ [e,a] = de;\ [e,c] = d$
[72,43]	$(C_3 \times A_4) \rtimes C_2 \quad (1,36,12)$ $R = (2,3,3,2,2);\ [b,a] = b;\ [c,a] = c;\ [d,a] = de;\ [d,b] = de;\ [e,a] = de;\ [e,b] = d$
[72,44]	$A_4 \times S_3 \quad (1,12,6)$ $R = (2,3,2,2,3);\ [c,b] = d;\ [d,b] = cd;\ [e,a] = e$
[72,45]	$C_2 \times ((C_3 \times C_3) \rtimes C_4) \quad (2,9,12)$ $R = (2,2,2,3,3);\ a^2 = c;\ [d,a] = e^2;\ [d,c] = d;\ [e,a] = d^2e;\ [e,c] = e$
[72,46]	$C_2 \times S_3 \times S_3 \quad (2,9,6)$ $R = (2,2,2,3,3);\ [d,b] = d;\ [e,a] = e$
[72,47]	$C_6 \times A_4 \quad (6,4,6)$ $R = (2,3,3,2,2);\ [d,b] = de;\ [e,b] = d$
[72,48]	$C_2 \times C_6 \times S_3 \quad (12,3,6)$ $R = (2,2,2,3,3);\ [e,a] = e$
[72,49]	$C_2 \times C_2 \times ((C_3 \times C_3) \rtimes C_2) \quad (4,9,6)$ $R = (2,2,2,3,3);\ [d,a] = d;\ [e,a] = e$

编号	72 阶群（续）
[72,50]	$C_2 \times C_6 \times C_6 \quad (72,1,6) \quad$ 阿贝尔。

73 阶群

只有一个同构类型：C_{73}，编号 $[73,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 73$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 73$。因此 $G = \langle a \rangle \cong C_{73}$。 □

74 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[74,1]	$D_{74} \quad (1,37,74)$ $R=(2,37);\ [b,a]=b^{35}$
[74,2]	$C_{74} \quad (74,1,74) \quad$ 阿贝尔。

证明. 这里 $p = 2 < q = 37$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q = 1$。令 $N = C_q \trianglelefteq G$，取 p 阶子群 $H = C_p$，则 $N \cap H = 1$，$NH = G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $2 \mid 36$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

75 阶群

共有 3 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[75,1]	$C_{75} \quad (75,1,75) \quad$ 阿贝尔。
[75,2]	$(C_5 \times C_5) \rtimes C_3 \quad (1,25,15)$ $R=(3,5,5);\ [b,a]=c^2;\ [c,a]=bc^2$
[75,3]	$C_5 \times C_{15} \quad (75,1,15) \quad$ 阿贝尔。

76 阶群

共有 4 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[76,1]	$C_{19}\rtimes C_4 \quad (2,19,76)$ $R=(2,2,19);\ a^2=b;\ [c,a]=c^{17}$
[76,2]	$C_{76} \quad (76,1,76) \quad$ 阿贝尔。
[76,3]	$D_{76} \quad (2,19,38)$ $R=(2,2,19);\ [c,a]=c^{17}$
[76,4]	$C_2\times C_{38} \quad (76,1,38) \quad$ 阿贝尔。

77 阶群

只有一个同构类型：C_{77}，编号 $[77,1]$。

证明. 这里 $p=7<q=11$。Sylow 定理给出 $n_q\mid p$ 且 $n_q\equiv 1\pmod q$，所以 $n_q=1$。令 $N=C_q\trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q\rtimes C_p$。作用由同态 $C_p\to \mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $7\nmid 10$，作用只能平凡，故只有 $C_q\times C_p\cong C_{77}$。 □

78 阶群

共有 6 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[78,1]	$C_{13}\rtimes C_6 \quad (1,13,78)$ $R=(2,3,13);\ [c,a]=c^{11};\ [c,b]=c^2$
[78,2]	$C_2\times (C_{13}\rtimes C_3) \quad (2,13,78)$ $R=(2,3,13);\ [c,b]=c^2$
[78,3]	$C_{13}\times S_3 \quad (13,3,78)$ $R=(2,13,3);\ [c,a]=c$
[78,4]	$C_3\times D_{26} \quad (3,13,78)$ $R=(2,3,13);\ [c,a]=c^{11}$
[78,5]	$D_{78} \quad (1,39,78)$ $R=(2,3,13);\ [b,a]=b;\ [c,a]=c^{11}$
[78,6]	$C_{78} \quad (78,1,78) \quad$ 阿贝尔。

79 阶群

只有一个同构类型：C_{79}，编号 $[79,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 79$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 79$。因此 $G = \langle a \rangle \cong C_{79}$。 □

80 阶群

共有 52 个同构类型，其中 5 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[80,1]	$C_5 \rtimes C_{16}$ $(8,5,80)$ $R = (2,2,2,2,5);\ a^2=b;\ b^2=c;\ c^2=d;\ [e,a]=e^3$
[80,2]	C_{80} $(80,1,80)$ 阿贝尔。
[80,3]	$C_5 \rtimes C_{16}$ $(4,5,80)$ $R = (2,2,2,2,5);\ a^2=b;\ b^2=c;\ c^2=d;\ [e,a]=e;\ [e,b]=e^3$
[80,4]	$C_8 \times D_{10}$ $(8,5,40)$ $R = (2,2,2,2,5);\ b^2=c;\ c^2=d;\ [e,a]=e^3$
[80,5]	$C_{40} \rtimes C_2$ $(4,10,40)$ $R = (2,2,2,2,5);\ b^2=c;\ c^2=d;\ [b,a]=d;\ [e,a]=e^3$
[80,6]	$C_{40} \rtimes C_2$ $(2,20,40)$ $R = (2,2,2,2,5);\ b^2=c;\ c^2=d;\ [b,a]=c;\ [c,a]=d;\ [e,a]=e^3$
[80,7]	D_{80} $(2,20,40)$ $R = (2,2,2,2,5);\ b^2=cd;\ c^2=d;\ [b,a]=c;\ [c,a]=d;\ [e,a]=e^3$
[80,8]	$C_5 \rtimes Q_{16}$ $(2,20,40)$ $R = (2,2,2,2,5);\ a^2=d;\ b^2=cd;\ c^2=d;\ [b,a]=c;\ [c,a]=d;\ [e,a]=e^3$
[80,9]	$C_2 \times (C_5 \rtimes C_8)$ $(8,5,40)$ $R = (2,2,2,2,5);\ a^2=c;\ c^2=d;\ [e,a]=e^3$
[80,10]	$(C_5 \rtimes C_8) \rtimes C_2$ $(4,10,40)$ $R = (2,2,2,2,5);\ a^2=c;\ c^2=d;\ [b,a]=d;\ [e,a]=e^3$
[80,11]	$C_4 \times (C_5 \rtimes C_4)$ $(8,5,20)$ $R = (2,2,2,2,5);\ a^2=c;\ b^2=d;\ [e,a]=e^3$
[80,12]	$(C_5 \rtimes C_4) \rtimes C_4$ $(4,10,20)$ $R = (2,2,2,2,5);\ a^2=d;\ b^2=c;\ [b,a]=d;\ [e,a]=e^3$
[80,13]	$C_{20} \rtimes C_4$ $(4,10,20)$ $R = (2,2,2,2,5);\ a^2=c;\ b^2=d;\ [b,a]=d;\ [e,a]=e^3$
[80,14]	$(C_{20} \times C_2) \rtimes C_2$ $(4,10,20)$ $R = (2,2,2,2,5);\ b^2=c;\ [b,a]=d;\ [e,a]=e^3$
[80,15]	$(C_5 \times D_8) \rtimes C_2$ $(2,20,40)$ $R = (2,2,2,2,5);\ c^2=d;\ [b,a]=c;\ [c,a]=d;\ [c,b]=d;\ [e,a]=e^3$
[80,16]	$(C_5 \rtimes Q_8) \rtimes C_2$ $(2,20,40)$ $R = (2,2,2,2,5);\ a^2=d;\ c^2=d;\ [b,a]=c;\ [c,a]=d;\ [c,b]=d;\ [e,a]=e^3$
[80,17]	$(C_5 \times Q_8) \rtimes C_2$ $(2,20,40)$ $R = (2,2,2,2,5);\ b^2=d;\ c^2=d;\ [b,a]=c;\ [c,a]=d;\ [c,b]=d;\ [e,a]=e^3$
[80,18]	$C_5 \rtimes Q_{16}$ $(2,20,40)$ $R = (2,2,2,2,5);\ a^2=d;\ b^2=d;\ c^2=d;\ [b,a]=c;\ [c,a]=d;\ [c,b]=d;\ [e,a]=e^3$

编号	80 阶群（续）
[80,19]	$(C_{10}\times C_2)\rtimes C_4 \quad (4,10,20)$ $R=(2,2,2,2,5);\ a^2=c;\ [b,a]=d;\ [e,a]=e^3$
[80,20]	$C_4\times C_{20} \quad (80,1,20)$ 阿贝尔。
[80,21]	$C_5\times((C_4\times C_2)\rtimes C_2) \quad (20,2,20)$ $R=(2,2,5,2,2);\ a^2=e;\ [b,a]=d$
[80,22]	$C_5\times(C_4\rtimes C_4) \quad (20,2,20)$ $R=(2,2,5,2,2);\ a^2=e;\ b^2=d;\ [b,a]=d$
[80,23]	$C_2\times C_{40} \quad (80,1,40)$ 阿贝尔。
[80,24]	$C_5\times(C_8\rtimes C_2) \quad (20,2,40)$ $R=(2,2,5,2,2);\ a^2=d;\ d^2=e;\ [b,a]=e$
[80,25]	$C_5\times D_{16} \quad (10,4,40)$ $R=(2,2,5,2,2);\ d^2=e;\ [b,a]=d;\ [d,a]=e;\ [d,b]=e$
[80,26]	$C_5\times \mathrm{QD}_{16} \quad (10,4,40)$ $R=(2,2,5,2,2);\ a^2=e;\ d^2=e;\ [b,a]=d;\ [d,a]=e;\ [d,b]=e$
[80,27]	$C_5\times Q_{16} \quad (10,4,40)$ $R=(2,2,5,2,2);\ a^2=e;\ b^2=e;\ d^2=e;\ [b,a]=d;\ [d,a]=e;\ [d,b]=e$
[80,28]	$C_5\rtimes(C_8\times C_2) \quad (4,5,40)$ $R=(2,2,2,2,5);\ a^2=c;\ b^2=d;\ c^2=d;\ [e,a]=e;\ [e,c]=e^3$
[80,29]	$(C_5\rtimes C_8)\rtimes C_2 \quad (2,10,40)$ $R=(2,2,2,2,5);\ a^2=c;\ b^2=d;\ c^2=d;\ [b,a]=d;\ [e,a]=e;\ [e,c]=e^3$
[80,30]	$C_4\times(C_5\rtimes C_4) \quad (4,5,20)$ $R=(2,2,2,2,5);\ a^2=c;\ b^2=d;\ [e,a]=e;\ [e,c]=e^3$
[80,31]	$C_{20}\rtimes C_4 \quad (2,10,20)$ $R=(2,2,2,2,5);\ a^2=c;\ b^2=d;\ [b,a]=d;\ [e,a]=e;\ [e,c]=e^3$
[80,32]	$C_2\times(C_5\rtimes C_8) \quad (4,5,40)$ $R=(2,2,2,2,5);\ a^2=c;\ c^2=d;\ [e,a]=e;\ [e,c]=e^3$
[80,33]	$(C_5\rtimes C_8)\rtimes C_2 \quad (2,10,40)$ $R=(2,2,2,2,5);\ a^2=c;\ c^2=d;\ [b,a]=d;\ [e,a]=e;\ [e,c]=e^3$
[80,34]	$(C_{10}\times C_2)\rtimes C_4 \quad (2,10,20)$ $R=(2,2,2,2,5);\ a^2=c;\ [b,a]=d;\ [e,a]=e;\ [e,c]=e^3$
[80,35]	$C_2\times(C_5\rtimes Q_8) \quad (4,10,20)$ $R=(2,2,2,2,5);\ a^2=d;\ c^2=d;\ [c,a]=d;\ [e,a]=e^3$
[80,36]	$C_2\times C_4\times D_{10} \quad (8,5,20)$ $R=(2,2,2,2,5);\ c^2=d;\ [e,a]=e^3$
[80,37]	$C_2\times D_{40} \quad (4,10,20)$ $R=(2,2,2,2,5);\ c^2=d;\ [c,a]=d;\ [e,a]=e^3$
[80,38]	$(C_{20}\times C_2)\rtimes C_2 \quad (4,10,20)$ $R=(2,2,2,2,5);\ b^2=d;\ [c,a]=d;\ [e,a]=e^3$
[80,39]	$D_8\times D_{10} \quad (2,10,20)$ $R=(2,2,2,2,5);\ [c,b]=d;\ [e,a]=e^3$
[80,40]	$(C_4\times D_{10})\rtimes C_2 \quad (2,10,20)$ $R=(2,2,2,2,5);\ b^2=d;\ [c,a]=d;\ [c,b]=d;\ [e,a]=e^3$
[80,41]	$Q_8\times D_{10} \quad (2,10,20)$ $R=(2,2,2,2,5);\ b^2=d;\ c^2=d;\ [c,b]=d;\ [e,a]=e^3$
[80,42]	$(C_4\times D_{10})\rtimes C_2 \quad (2,10,20)$ $R=(2,2,2,2,5);\ b^2=d;\ c^2=d;\ [c,a]=d;\ [c,b]=d;\ [e,a]=e^3$

编号	80 阶群（续）
[80,43]	$C_2 \times C_2 \times (C_5 \rtimes C_4)$ $(8,5,20)$ $R=(2,2,2,2,5);\ a^2=d;\ [e,a]=e^3$
[80,44]	$C_2 \times ((C_{10} \times C_2) \rtimes C_2)$ $(4,10,20)$ $R=(2,2,2,2,5);\ [c,a]=d;\ [e,a]=e^3$
[80,45]	$C_2 \times C_2 \times C_{20}$ $(80,1,20)$ 阿贝尔。
[80,46]	$C_{10} \times D_8$ $(20,2,20)$ $R=(2,2,2,5,2);\ [b,a]=e$
[80,47]	$C_{10} \times Q_8$ $(20,2,20)$ $R=(2,2,2,5,2);\ a^2=e;\ b^2=e;\ [b,a]=e$
[80,48]	$C_5 \times ((C_4 \times C_2) \rtimes C_2)$ $(20,2,20)$ $R=(2,2,2,5,2);\ c^2=e;\ [b,a]=e$
[80,49]	$(C_2 \times C_2 \times C_2 \times C_2) \rtimes C_5$ $(1,16,10)$ $R=(5,2,2,2,2);\ [b,a]=be;\ [c,a]=bce;\ [d,a]=bcde;\ [e,a]=bcd$
[80,50]	$C_2 \times C_2 \times (C_5 \rtimes C_4)$ $(4,5,20)$ $R=(2,2,2,2,5);\ a^2=d;\ [e,a]=e;\ [e,d]=e^3$
[80,51]	$C_2 \times C_2 \times C_2 \times D_{10}$ $(8,5,10)$ $R=(2,2,2,2,5);\ [e,a]=e^3$
[80,52]	$C_2 \times C_2 \times C_2 \times C_{10}$ $(80,1,10)$ 阿贝尔。

81 阶群

共有 15 个同构类型，其中 5 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[81,1]	C_{81} $(81,1,81)$ 阿贝尔。
[81,2]	$C_9 \times C_9$ $(81,1,9)$ 阿贝尔。
[81,3]	$(C_9 \times C_3) \rtimes C_3$ $(9,3,9)$ $R=(3,3,3,3);\ a^3=d;\ [b,a]=c$
[81,4]	$C_9 \rtimes C_9$ $(9,3,9)$ $R=(3,3,3,3);\ a^3=d;\ b^3=c;\ [b,a]=c$
[81,5]	$C_3 \times C_{27}$ $(81,1,27)$ 阿贝尔。
[81,6]	$C_{27} \rtimes C_3$ $(9,3,27)$ $R=(3,3,3,3);\ a^3=c;\ c^3=d;\ [b,a]=d$
[81,7]	$(C_3 \times C_3 \times C_3) \rtimes C_3$ $(3,9,9)$ $R=(3,3,3,3);\ [b,a]=c;\ [c,a]=d$
[81,8]	$(C_9 \times C_3) \rtimes C_3$ $(3,9,9)$ $R=(3,3,3,3);\ b^3=d;\ [b,a]=c;\ [c,a]=d$
[81,9]	$(C_9 \times C_3) \rtimes C_3$ $(3,9,9)$ $R=(3,3,3,3);\ b^3=d^2;\ [b,a]=c;\ [c,a]=d$
[81,10]	$C_3 \,.\, ((C_3 \times C_3) \rtimes C_3) = (C_3 \times C_3) \,.\, (C_3 \times C_3)$ $(3,9,9)$ $R=(3,3,3,3);\ a^3=d;\ b^3=d^2;\ [b,a]=c;\ [c,a]=d$
[81,11]	$C_3 \times C_3 \times C_9$ $(81,1,9)$ 阿贝尔。

编号	81 阶群（续）
[81,12]	$C_3 \times ((C_3 \times C_3) \rtimes C_3) \quad (9,3,3)$ $R = (3,3,3,3);\ [b,a] = d$
[81,13]	$C_3 \times (C_9 \rtimes C_3) \quad (9,3,9)$ $R = (3,3,3,3);\ a^3 = d;\ [b,a] = d$
[81,14]	$(C_9 \times C_3) \rtimes C_3 \quad (9,3,9)$ $R = (3,3,3,3);\ c^3 = d;\ [b,a] = d$
[81,15]	$C_3 \times C_3 \times C_3 \times C_3 \quad (81,1,3)$ 阿贝尔。

82 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[82,1]	$D_{82} \quad (1,41,82)$ $R = (2,41);\ [b,a] = b^{39}$
[82,2]	$C_{82} \quad (82,1,82)$ 阿贝尔。

证明. 这里 $p = 2 < q = 41$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q = 1$。令 $N = C_q \trianglelefteq G$，取 p 阶子群 $H = C_p$，则 $N \cap H = 1$，$NH = G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $2 \mid 40$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

83 阶群

只有一个同构类型：C_{83}，编号 $[83,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 83$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 83$。因此 $G = \langle a \rangle \cong C_{83}$。 □

84 阶群

共有 15 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[84,1]	$C_7 \rtimes C_{12} \quad (2,7,84)$ $R = (2,3,2,7);\ a^2 = c;\ [d,a] = d^5;\ [d,b] = d^3$
[84,2]	$C_4 \times (C_7 \rtimes C_3) \quad (4,7,84)$ $R = (2,3,2,7);\ a^2 = c;\ [d,b] = d$
[84,3]	$C_7 \times (C_3 \rtimes C_4) \quad (14,3,84)$ $R = (2,7,2,3);\ a^2 = c;\ [d,a] = d$

编号	84 阶群（续）
[84,4]	$C_3 \times (C_7 \rtimes C_4)$ $(6,7,84)$ $R=(2,3,2,7)$; $a^2=c$; $[d,a]=d^5$
[84,5]	$C_{21} \rtimes C_4$ $(2,21,84)$ $R=(2,2,3,7)$; $a^2=b$; $[c,a]=c$; $[d,a]=d^5$
[84,6]	C_{84} $(84,1,84)$ 阿贝尔。
[84,7]	$C_2 \times (C_7 \rtimes C_6)$ $(2,7,42)$ $R=(2,2,3,7)$; $[d,a]=d^5$; $[d,c]=d^3$
[84,8]	$S_3 \times D_{14}$ $(1,21,42)$ $R=(2,2,3,7)$; $[c,b]=c$; $[d,a]=d^5$
[84,9]	$C_2 \times C_2 \times (C_7 \rtimes C_3)$ $(4,7,42)$ $R=(2,2,3,7)$; $[d,c]=d$
[84,10]	$C_7 \times A_4$ $(7,4,42)$ $R=(3,7,2,2)$; $[c,a]=cd$; $[d,a]=c$
[84,11]	$(C_{14} \times C_2) \rtimes C_3$ $(1,28,42)$ $R=(3,2,2,7)$; $[b,a]=bc$; $[c,a]=b$; $[d,a]=d$
[84,12]	$C_6 \times D_{14}$ $(6,7,42)$ $R=(2,2,3,7)$; $[d,a]=d^5$
[84,13]	$C_{14} \times S_3$ $(14,3,42)$ $R=(2,2,7,3)$; $[d,a]=d$
[84,14]	D_{84} $(2,21,42)$ $R=(2,2,3,7)$; $[c,a]=c$; $[d,a]=d^5$
[84,15]	$C_2 \times C_{42}$ $(84,1,42)$ 阿贝尔。

85 阶群

只有一个同构类型：C_{85}，编号 $[85,1]$。

证明. 这里 $p=5<q=17$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q=1$。令 $N=C_q \trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $5 \nmid 16$，作用只能平凡，故只有 $C_q \times C_p \cong C_{85}$。 □

86 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[86,1]	D_{86} $(1,43,86)$ $R=(2,43)$; $[b,a]=b^{41}$
[86,2]	C_{86} $(86,1,86)$ 阿贝尔。

证明. 这里 $p=2<q=43$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q=1$。令 $N=C_q \trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G \cong C_q \rtimes C_p$。作用由同态

$C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $2 \mid 42$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

87 阶群

只有一个同构类型：C_{87}，编号 $[87,1]$。

证明. 这里 $p = 3 < q = 29$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q = 1$。令 $N = C_q \trianglelefteq G$，取 p 阶子群 $H = C_p$，则 $N \cap H = 1$，$NH = G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $3 \nmid 28$，作用只能平凡，故只有 $C_q \times C_p \cong C_{87}$。 □

88 阶群

共有 12 个同构类型，其中 3 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[88,1]	$C_{11} \rtimes C_8 \quad (4,11,88)$ $R = (2,2,2,11);\ a^2 = b;\ b^2 = c;\ [d,a] = d^9$
[88,2]	$C_{88} \quad (88,1,88) \quad$ 阿贝尔。
[88,3]	$C_{11} \rtimes Q_8 \quad (2,22,44)$ $R = (2,2,2,11);\ a^2 = c;\ b^2 = c;\ [b,a] = c;\ [d,a] = d^9$
[88,4]	$C_4 \times D_{22} \quad (4,11,44)$ $R = (2,2,2,11);\ b^2 = c;\ [d,a] = d^9$
[88,5]	$D_{88} \quad (2,22,44)$ $R = (2,2,2,11);\ b^2 = c;\ [b,a] = c;\ [d,a] = d^9$
[88,6]	$C_2 \times (C_{11} \rtimes C_4) \quad (4,11,44)$ $R = (2,2,2,11);\ a^2 = c;\ [d,a] = d^9$
[88,7]	$(C_{22} \times C_2) \rtimes C_2 \quad (2,22,44)$ $R = (2,2,2,11);\ [b,a] = c;\ [d,a] = d^9$
[88,8]	$C_2 \times C_{44} \quad (88,1,44) \quad$ 阿贝尔。
[88,9]	$C_{11} \times D_8 \quad (22,2,44)$ $R = (2,2,11,2);\ [b,a] = d$
[88,10]	$C_{11} \times Q_8 \quad (22,2,44)$ $R = (2,2,11,2);\ a^2 = d;\ b^2 = d;\ [b,a] = d$
[88,11]	$C_2 \times C_2 \times D_{22} \quad (4,11,22)$ $R = (2,2,2,11);\ [d,a] = d^9$
[88,12]	$C_2 \times C_2 \times C_{22} \quad (88,1,22) \quad$ 阿贝尔。

89 阶群

只有一个同构类型：C_{89}，编号 $[89,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 89$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 89$。因此 $G = \langle a \rangle \cong C_{89}$。 □

90 阶群

共有 10 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[90,1]	$C_5 \times D_{18}$ $(5,9,90)$ $R=(2,5,3,3);\ c^3=d^2;\ [c,a]=cd;\ [d,a]=d$
[90,2]	$C_9 \times D_{10}$ $(9,5,90)$ $R=(2,3,3,5);\ b^3=c;\ [d,a]=d^3$
[90,3]	D_{90} $(1,45,90)$ $R=(2,3,5,3);\ b^3=d^2;\ [b,a]=bd;\ [c,a]=c^3;\ [d,a]=d$
[90,4]	C_{90} $(90,1,90)$ 阿贝尔。
[90,5]	$C_3 \times C_3 \times D_{10}$ $(9,5,30)$ $R=(2,3,3,5);\ [d,a]=d^3$
[90,6]	$C_{15} \times S_3$ $(15,3,30)$ $R=(2,3,5,3);\ [d,a]=d$
[90,7]	$C_3 \times D_{30}$ $(3,15,30)$ $R=(2,3,3,5);\ [c,a]=c;\ [d,a]=d^3$
[90,8]	$C_5 \times ((C_3 \times C_3) \rtimes C_2)$ $(5,9,30)$ $R=(2,5,3,3);\ [c,a]=c;\ [d,a]=d$
[90,9]	$(C_{15} \times C_3) \rtimes C_2$ $(1,45,30)$ $R=(2,3,3,5);\ [b,a]=b;\ [c,a]=c;\ [d,a]=d^3$
[90,10]	$C_3 \times C_{30}$ $(90,1,30)$ 阿贝尔。

91 阶群

只有一个同构类型：C_{91}，编号 $[91,1]$。

证明. 这里 $p = 7 < q = 13$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q = 1$。令 $N = C_q \trianglelefteq G$，取 p 阶子群 $H = C_p$，则 $N \cap H = 1$，$NH = G$，故 $G \cong C_q \rtimes C_p$。作用由同态 $C_p \to \mathrm{Aut}(C_q) \cong C_{q-1}$ 给出。因 $7 \nmid 12$，作用只能平凡，故只有 $C_q \times C_p \cong C_{91}$。 □

92 阶群

共有 4 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[92,1]	$C_{23}\rtimes C_4 \quad (2,23,92)$ $R=(2,2,23);\ a^2=b;\ [c,a]=c^{21}$
[92,2]	$C_{92} \quad (92,1,92) \quad$ 阿贝尔。
[92,3]	$D_{92} \quad (2,23,46)$ $R=(2,2,23);\ [c,a]=c^{21}$
[92,4]	$C_2\times C_{46} \quad (92,1,46) \quad$ 阿贝尔。

93 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[93,1]	$C_{31}\rtimes C_3 \quad (1,31,93)$ $R=(3,31);\ [b,a]=b^{24}$
[93,2]	$C_{93} \quad (93,1,93) \quad$ 阿贝尔。

证明. 这里 $p=3<q=31$。Sylow 定理给出 $n_q\mid p$ 且 $n_q\equiv 1\pmod q$，所以 $n_q=1$。令 $N=C_q\trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q\rtimes C_p$。作用由同态 $C_p\to \mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $3\mid 30$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

94 阶群

共有 2 个同构类型，其中 1 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[94,1]	$D_{94} \quad (1,47,94)$ $R=(2,47);\ [b,a]=b^{45}$
[94,2]	$C_{94} \quad (94,1,94) \quad$ 阿贝尔。

证明. 这里 $p=2<q=47$。Sylow 定理给出 $n_q\mid p$ 且 $n_q\equiv 1\pmod q$，所以 $n_q=1$。令 $N=C_q\trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q\rtimes C_p$。作用由同态 $C_p\to \mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $2\mid 46$，除平凡作用外，非平凡作用的像为 C_{q-1} 中唯一的 p 阶子群。更换 C_p 的生成元可使所有非平凡作用等价，因此恰有循环群与一个非阿贝尔群两种。 □

95 阶群

只有一个同构类型：C_{95}，编号 $[95,1]$。

证明. 这里 $p=5<q=19$。Sylow 定理给出 $n_q \mid p$ 且 $n_q \equiv 1 \pmod q$，所以 $n_q=1$。令 $N=C_q \trianglelefteq G$，取 p 阶子群 $H=C_p$，则 $N\cap H=1$，$NH=G$，故 $G\cong C_q\rtimes C_p$。作用由同态 $C_p\to \mathrm{Aut}(C_q)\cong C_{q-1}$ 给出。因 $5\nmid 18$，作用只能平凡，故只有 $C_q\times C_p\cong C_{95}$。 □

96 阶群

共有 231 个同构类型，其中 7 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[96,1]	$C_3\rtimes C_{32}$ $(16,3,96)$ $R=(2,2,2,2,2,3)$; $a^2=b$; $b^2=c$; $c^2=d$; $d^2=e$; $[f,a]=f$
[96,2]	C_{96} $(96,1,96)$ 阿贝尔。
[96,3]	$((C_4\times C_2)\rtimes C_4)\rtimes C_3$ $(2,32,12)$ $R=(3,2,2,2,2,2)$; $b^2=de$; $c^2=d$; $[b,a]=bcde$; $[c,a]=bef$; $[c,b]=f$; $[d,a]=def$; $[e,a]=df$
[96,4]	$C_{16}\times S_3$ $(16,3,48)$ $R=(2,2,2,2,2,3)$; $b^2=c$; $c^2=d$; $d^2=e$; $[f,a]=f$
[96,5]	$C_3\rtimes(C_{16}\rtimes C_2)$ $(8,6,48)$ $R=(2,2,2,2,2,3)$; $b^2=c$; $c^2=d$; $d^2=e$; $[b,a]=e$; $[f,a]=f$
[96,6]	D_{96} $(2,24,48)$ $R=(2,2,2,2,2,3)$; $b^2=cd$; $c^2=de$; $d^2=e$; $[b,a]=c$; $[c,a]=d$; $[d,a]=e$; $[f,a]=f$
[96,7]	$C_3\rtimes \mathrm{QD}_{32}$ $(2,24,48)$ $R=(2,2,2,2,2,3)$; $b^2=cd$; $c^2=d$; $d^2=e$; $[b,a]=c$; $[c,a]=de$; $[d,a]=e$; $[f,a]=f$
[96,8]	$C_3\rtimes Q_{32}$ $(2,24,48)$ $R=(2,2,2,2,2,3)$; $a^2=e$; $b^2=cd$; $c^2=de$; $d^2=e$; $[b,a]=c$; $[c,a]=d$; $[d,a]=e$; $[f,a]=f$
[96,9]	$C_4\times(C_3\rtimes C_8)$ $(16,3,24)$ $R=(2,2,2,2,2,3)$; $a^2=c$; $b^2=d$; $c^2=e$; $[f,a]=f$
[96,10]	$C_3\rtimes(C_8\rtimes C_4)$ $(8,6,24)$ $R=(2,2,2,2,2,3)$; $a^2=c$; $b^2=d$; $c^2=e$; $[b,a]=e$; $[f,a]=f$
[96,11]	$C_3\rtimes(C_4\rtimes C_8)$ $(8,6,24)$ $R=(2,2,2,2,2,3)$; $a^2=c$; $b^2=d$; $c^2=e$; $[b,a]=d$; $[f,a]=f$
[96,12]	$C_3\rtimes((C_4\times C_4)\rtimes C_2)$ $(4,12,24)$ $R=(2,2,2,2,2,3)$; $b^2=c$; $d^2=e$; $[b,a]=d$; $[c,a]=e$; $[d,a]=e$; $[f,a]=f$
[96,13]	$C_3\rtimes((C_2\times C_2\times C_2)\rtimes C_4)$ $(2,12,12)$ $R=(2,2,2,2,2,3)$; $b^2=c$; $[b,a]=d$; $[c,a]=e$; $[d,b]=e$; $[f,a]=f$
[96,14]	$C_3\rtimes(C_8\rtimes C_4)$ $(4,12,24)$ $R=(2,2,2,2,2,3)$; $a^2=d$; $b^2=c$; $d^2=e$; $[b,a]=d$; $[d,b]=e$; $[f,a]=f$
[96,15]	$C_3\rtimes(C_8\rtimes C_4)$ $(4,12,24)$ $R=(2,2,2,2,2,3)$; $a^2=d$; $b^2=c$; $d^2=e$; $[b,a]=de$; $[d,b]=e$; $[f,a]=f$
[96,16]	$C_3\rtimes((C_8\times C_2)\rtimes C_2)$ $(4,12,24)$ $R=(2,2,2,2,2,3)$; $b^2=c$; $d^2=e$; $[b,a]=d$; $[d,a]=e$; $[d,b]=e$; $[f,a]=f$

编号	96 阶群（续）
[96,17]	$C_3 \rtimes (Q_8 \rtimes C_4)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ b^2 = c;\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e;\ [f,a] = f$
[96,18]	$C_2 \times (C_3 \rtimes C_{16})$ $(16,3,48)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ c^2 = d;\ d^2 = e;\ [f,a] = f$
[96,19]	$C_3 \rtimes (C_{16} \rtimes C_2)$ $(8,6,48)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ c^2 = d;\ d^2 = e;\ [b,a] = e;\ [f,a] = f$
[96,20]	$C_8 \times (C_3 \rtimes C_4)$ $(16,3,24)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ b^2 = d;\ d^2 = e;\ [f,a] = f$
[96,21]	$C_3 \rtimes (C_4 \rtimes C_8)$ $(8,6,24)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ b^2 = c;\ c^2 = e;\ [b,a] = d;\ [f,a] = f$
[96,22]	$C_3 \rtimes (C_8 \rtimes C_4)$ $(8,6,24)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ b^2 = d;\ d^2 = e;\ [b,a] = e;\ [f,a] = f$
[96,23]	$C_3 \rtimes (Q_8 \rtimes C_4)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ b^2 = cd;\ c^2 = e;\ [b,a] = c;\ [c,a] = e;\ [f,a] = f$
[96,24]	$C_3 \rtimes (C_8 \rtimes C_4)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ b^2 = d;\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [f,a] = f$
[96,25]	$C_3 \rtimes (C_8 \rtimes C_4)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ b^2 = de;\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [f,a] = f$
[96,26]	$C_3 \rtimes (C_4 \,.\, D_8 = C_4 \,.\, (C_4 \times C_2))$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ b^2 = d;\ c^2 = e;\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [f,a] = f$
[96,27]	$C_3 \rtimes ((C_8 \times C_2) \rtimes C_2)$ $(8,6,24)$ $R = (2,2,2,2,2,3);\ b^2 = c;\ c^2 = e;\ [b,a] = d;\ [f,a] = f$
[96,28]	$C_3 \rtimes ((C_8 \times C_2) \rtimes C_2)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ b^2 = cd;\ c^2 = e;\ [b,a] = c;\ [c,a] = e;\ [f,a] = f$
[96,29]	$C_3 \rtimes (C_4 \,.\, D_8 = C_4 \,.\, (C_4 \times C_2))$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ b^2 = c;\ c^2 = e;\ d^2 = e;\ [b,a] = d;\ [d,b] = e;\ [f,a] = f$
[96,30]	$C_3 \rtimes ((C_8 \rtimes C_2) \rtimes C_2)$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ b^2 = c;\ c^2 = e;\ [b,a] = d;\ [c,a] = e;\ [d,b] = e;\ [f,a] = f$
[96,31]	$C_3 \rtimes (C_2 \,.\, ((C_4 \times C_2) \rtimes C_2) = (C_2 \times C_2) \,.\, (C_4 \times C_2))$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ b^2 = c;\ c^2 = e;\ [b,a] = d;\ [c,a] = e;\ [d,b] = e;\ [f,a] = f$
[96,32]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ b^2 = c;\ c^2 = e;\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e;\ [f,a] = f$
[96,33]	$C_3 \rtimes D_{32}$ $(2,24,48)$ $R = (2,2,2,2,2,3);\ c^2 = de;\ d^2 = e;\ [b,a] = c;\ [c,a] = d;\ [c,b] = d;\ [d,a] = e;\ [d,b] = e;$ $[f,a] = f$
[96,34]	$C_3 \rtimes \mathrm{QD}_{32}$ $(2,24,48)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ c^2 = de;\ d^2 = e;\ [b,a] = c;\ [c,a] = d;\ [c,b] = d;\ [d,a] = e;$ $[d,b] = e;\ [f,a] = f$
[96,35]	$C_3 \rtimes \mathrm{QD}_{32}$ $(2,24,48)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = de;\ d^2 = e;\ [b,a] = c;\ [c,a] = d;\ [c,b] = d;\ [d,a] = e;$ $[d,b] = e;\ [f,a] = f$
[96,36]	$C_3 \rtimes Q_{32}$ $(2,24,48)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ b^2 = e;\ c^2 = de;\ d^2 = e;\ [b,a] = c;\ [c,a] = d;\ [c,b] = d;$ $[d,a] = e;\ [d,b] = e;\ [f,a] = f$
[96,37]	$C_3 \rtimes ((C_8 \times C_2) \rtimes C_2)$ $(8,6,24)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ c^2 = e;\ [b,a] = d;\ [f,a] = f$

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[96,38]	$C_3 \rtimes ((C_4 \times C_2) \rtimes C_4)$ $(8,6,12)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ b^2 = d;\ [b,a] = e;\ [f,a] = f$
[96,39]	$C_3 \rtimes ((C_8 \times C_2) \rtimes C_2)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e;\ [f,a] = f$
[96,40]	$C_3 \rtimes ((C_8 \rtimes C_2) \rtimes C_2)$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ c^2 = e;\ [b,a] = d;\ [c,b] = e;\ [d,a] = e;\ [f,a] = f$
[96,41]	$C_3 \rtimes ((C_2 \times C_2 \times C_2) \rtimes C_4)$ $(2,12,12)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ [b,a] = d;\ [c,b] = e;\ [d,a] = e;\ [f,a] = f$
[96,42]	$C_3 \rtimes (Q_8 \rtimes C_4)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ b^2 = e;\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e;\ [f,a] = f$
[96,43]	$C_3 \rtimes (C_2 \,.\, ((C_4 \times C_2) \rtimes C_2) = (C_2 \times C_2) \,.\, (C_4 \times C_2))$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ b^2 = e;\ c^2 = e;\ [b,a] = d;\ [c,b] = e;\ [d,a] = e;\ [f,a] = f$
[96,44]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = c;\ c^2 = e;\ d^2 = e;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e;\ [f,a] = f$
[96,45]	$C_3 \times ((C_4 \times C_2) \rtimes C_4)$ $(24,2,12)$ $R = (2,2,3,2,2,2);\ a^2 = e;\ b^2 = f;\ [b,a] = d$
[96,46]	$C_4 \times C_{24}$ $(96,1,24)$ 阿贝尔。
[96,47]	$C_3 \times (C_8 \rtimes C_4)$ $(24,2,24)$ $R = (2,2,3,2,2,2);\ a^2 = d;\ b^2 = e;\ d^2 = f;\ [b,a] = f$
[96,48]	$C_3 \times ((C_8 \times C_2) \rtimes C_2)$ $(24,2,24)$ $R = (2,2,3,2,2,2);\ a^2 = e;\ e^2 = f;\ [b,a] = d$
[96,49]	$C_3 \times ((C_2 \times C_2 \times C_2) \rtimes C_4)$ $(6,4,12)$ $R = (2,2,3,2,2,2);\ a^2 = e;\ [b,a] = d;\ [d,a] = f;\ [e,b] = f$
[96,50]	$C_3 \times ((C_8 \rtimes C_2) \rtimes C_2)$ $(6,4,24)$ $R = (2,2,3,2,2,2);\ a^2 = e;\ e^2 = f;\ [b,a] = d;\ [d,a] = f;\ [e,b] = f$
[96,51]	$C_3 \times (C_2 \,.\, ((C_4 \times C_2) \rtimes C_2) = (C_2 \times C_2) \,.\, (C_4 \times C_2))$ $(6,4,24)$ $R = (2,2,3,2,2,2);\ a^2 = e;\ b^2 = f;\ e^2 = f;\ [b,a] = d;\ [d,a] = f;\ [e,b] = f$
[96,52]	$C_3 \times ((C_8 \times C_2) \rtimes C_2)$ $(12,4,24)$ $R = (2,2,3,2,2,2);\ a^2 = e;\ d^2 = f;\ [b,a] = d;\ [d,a] = f;\ [d,b] = f$
[96,53]	$C_3 \times (Q_8 \rtimes C_4)$ $(12,4,24)$ $R = (2,2,3,2,2,2);\ a^2 = e;\ b^2 = f;\ d^2 = f;\ [b,a] = d;\ [d,a] = f;\ [d,b] = f$
[96,54]	$C_3 \times ((C_4 \times C_4) \rtimes C_2)$ $(12,4,24)$ $R = (2,2,3,2,2,2);\ a^2 = e;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [d,a] = f;\ [d,b] = f$
[96,55]	$C_3 \times (C_4 \rtimes C_8)$ $(24,2,24)$ $R = (2,2,3,2,2,2);\ a^2 = e;\ b^2 = d;\ e^2 = f;\ [b,a] = d$
[96,56]	$C_3 \times (C_8 \rtimes C_4)$ $(12,4,24)$ $R = (2,2,3,2,2,2);\ a^2 = e;\ b^2 = d;\ d^2 = f;\ [b,a] = d;\ [d,a] = f$
[96,57]	$C_3 \times (C_8 \rtimes C_4)$ $(12,4,24)$ $R = (2,2,3,2,2,2);\ a^2 = e;\ b^2 = df;\ d^2 = f;\ [b,a] = d;\ [d,a] = f$
[96,58]	$C_3 \times (C_4 \,.\, D_8 = C_4 \,.\, (C_4 \times C_2))$ $(12,4,24)$ $R = (2,2,3,2,2,2);\ a^2 = e;\ b^2 = d;\ d^2 = f;\ e^2 = f;\ [b,a] = d;\ [d,a] = f$
[96,59]	$C_2 \times C_{48}$ $(96,1,48)$ 阿贝尔。
[96,60]	$C_3 \times (C_{16} \rtimes C_2)$ $(24,2,48)$ $R = (2,2,3,2,2,2);\ a^2 = d;\ d^2 = e;\ e^2 = f;\ [b,a] = f$
[96,61]	$C_3 \times D_{32}$ $(6,8,48)$ $R = (2,2,3,2,2,2);\ d^2 = ef;\ e^2 = f;\ [b,a] = d;\ [d,a] = e;\ [d,b] = e;\ [e,a] = f;\ [e,b] = f$

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[96,62]	$C_3 \times \mathrm{QD}_{32}$ $(6,8,48)$ $R=(2,2,3,2,2,2);\ a^2=f;\ d^2=ef;\ e^2=f;\ [b,a]=d;\ [d,a]=e;\ [d,b]=e;\ [e,a]=f;$ $[e,b]=f$
[96,63]	$C_3 \times Q_{32}$ $(6,8,48)$ $R=(2,2,3,2,2,2);\ a^2=f;\ b^2=f;\ d^2=ef;\ e^2=f;\ [b,a]=d;\ [d,a]=e;\ [d,b]=e;$ $[e,a]=f;\ [e,b]=f$
[96,64]	$(C_4 \times C_4) \rtimes S_3$ $(1,48,24)$ $R=(2,3,2,2,2,2);\ c^2=ef;\ d^2=e;\ [b,a]=b;\ [c,a]=cdef;\ [c,b]=cdef;\ [d,a]=cde;$ $[d,b]=cf;\ [e,a]=f;\ [e,b]=ef;\ [f,b]=e$
[96,65]	$(C_2 \times C_2) \rtimes (C_3 \rtimes C_8)$ $(4,12,24)$ $R=(2,2,2,3,2,2);\ a^2=b;\ b^2=c;\ [d,a]=d;\ [e,a]=ef;\ [e,d]=ef;\ [f,a]=ef;\ [f,d]=e$
[96,66]	$Q_8 \rtimes (C_3 \rtimes C_4)$ $(4,24,24)$ $R=(2,2,3,2,2,2);\ a^2=b;\ d^2=f;\ e^2=f;\ [c,a]=c;\ [d,a]=def;\ [d,c]=de;\ [e,a]=de;$ $[e,c]=df;\ [e,d]=f$
[96,67]	$\mathrm{SL}(2,3) \rtimes C_4$ $(4,24,24)$ $R=(2,2,3,2,2,2);\ a^2=b;\ b^2=f;\ d^2=f;\ e^2=f;\ [c,a]=c;\ [d,a]=def;\ [d,c]=de;$ $[e,a]=de;\ [e,c]=df;\ [e,d]=f$
[96,68]	$C_2 \times ((C_4 \times C_4) \rtimes C_3)$ $(2,16,12)$ $R=(2,3,2,2,2,2);\ c^2=ef;\ d^2=e;\ [c,b]=cdef;\ [d,b]=cf;\ [e,b]=ef;\ [f,b]=e$
[96,69]	$C_4 \times \mathrm{SL}(2,3)$ $(8,8,12)$ $R=(2,3,2,2,2,2);\ a^2=c;\ d^2=f;\ e^2=f;\ [d,b]=def;\ [e,b]=df;\ [e,d]=f$
[96,70]	$((C_2 \times C_2 \times C_2 \times C_2) \rtimes C_2) \rtimes C_3$ $(1,16,12)$ $R=(2,3,2,2,2,2);\ [c,a]=e;\ [c,b]=cd;\ [d,a]=f;\ [d,b]=c;\ [e,b]=ef;\ [f,b]=e$
[96,71]	$((C_4 \times C_4) \rtimes C_2) \rtimes C_3$ $(1,16,12)$ $R=(2,3,2,2,2,2);\ c^2=ef;\ d^2=e;\ [c,a]=e;\ [c,b]=cdef;\ [d,a]=f;\ [d,b]=cf;$ $[e,b]=ef;\ [f,b]=e$
[96,72]	$((C_4 \times C_4) \rtimes C_2) \rtimes C_3$ $(1,16,12)$ $R=(2,3,2,2,2,2);\ c^2=ef;\ d^2=e;\ [c,a]=ef;\ [c,b]=cdef;\ [d,a]=e;\ [d,b]=cf;$ $[e,b]=ef;\ [f,b]=e$
[96,73]	$C_8 \times A_4$ $(8,4,24)$ $R=(2,3,2,2,2,2);\ a^2=c;\ c^2=d;\ [e,b]=ef;\ [f,b]=e$
[96,74]	$((C_8 \times C_2) \rtimes C_2) \rtimes C_3$ $(8,8,24)$ $R=(2,3,2,2,2,2);\ a^2=c;\ c^2=f;\ d^2=f;\ e^2=f;\ [d,b]=def;\ [e,b]=df;\ [e,d]=f$
[96,75]	$C_4 \times (C_3 \rtimes Q_8)$ $(8,6,12)$ $R=(2,2,2,2,2,3);\ a^2=e;\ b^2=e;\ c^2=d;\ [b,a]=e;\ [f,a]=f$
[96,76]	$C_3 \rtimes (C_4 \rtimes Q_8)$ $(4,12,12)$ $R=(2,2,2,2,2,3);\ a^2=e;\ b^2=e;\ c^2=d;\ [b,a]=e;\ [c,a]=d;\ [f,a]=f$
[96,77]	$C_3 \rtimes ((C_2 \times C_2) \,.\, (C_2 \times C_2 \times C_2))$ $(4,12,12)$ $R=(2,2,2,2,2,3);\ a^2=e;\ b^2=d;\ c^2=e;\ [b,a]=e;\ [c,a]=d;\ [f,a]=f$
[96,78]	$C_4 \times C_4 \times S_3$ $(16,3,12)$ $R=(2,2,2,2,2,3);\ b^2=e;\ c^2=d;\ [f,a]=f$
[96,79]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2)$ $(8,6,12)$ $R=(2,2,2,2,2,3);\ b^2=e;\ c^2=d;\ [c,a]=e;\ [f,a]=f$
[96,80]	$C_4 \times D_{24}$ $(8,6,12)$ $R=(2,2,2,2,2,3);\ b^2=e;\ c^2=d;\ [b,a]=e;\ [f,a]=f$
[96,81]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2)$ $(4,12,12)$ $R=(2,2,2,2,2,3);\ b^2=e;\ c^2=d;\ [b,a]=e;\ [c,a]=d;\ [f,a]=f$

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[96,82]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = d;\ c^2 = e;\ [b,a] = e;\ [c,a] = d;\ [f,a] = f$
[96,83]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = d;\ c^2 = e;\ [b,a] = e;\ [c,a] = de;\ [f,a] = f$
[96,84]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2) \quad (4,6,12)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ c^2 = d;\ [c,b] = e;\ [f,a] = f$
[96,85]	$C_3 \rtimes ((C_2 \times Q_8) \rtimes C_2) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ c^2 = d;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,86]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ b^2 = d;\ [c,a] = de;\ [c,b] = e;\ [f,a] = f$
[96,87]	$((C_4 \times C_2) \rtimes C_2) \times S_3 \quad (4,6,12)$ $R = (2,2,2,2,2,3);\ c^2 = d;\ [c,b] = e;\ [f,a] = f$
[96,88]	$C_3 \rtimes (C_4 \times D_8) \quad (4,6,12)$ $R = (2,2,2,2,2,3);\ c^2 = d;\ [b,a] = e;\ [c,b] = e;\ [f,a] = f$
[96,89]	$C_3 \rtimes ((C_2 \times C_2 \times C_2 \times C_2) \rtimes C_2) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ c^2 = d;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,90]	$C_3 \rtimes ((C_4 \times C_2 \times C_2) \rtimes C_2) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = d;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,91]	$C_3 \rtimes ((C_4 \times C_2 \times C_2) \rtimes C_2) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = d;\ [c,a] = de;\ [c,b] = e;\ [f,a] = f$
[96,92]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = d;\ [b,a] = e;\ [c,a] = de;\ [c,b] = e;\ [f,a] = f$
[96,93]	$C_3 \rtimes ((C_4 \times C_2 \times C_2) \rtimes C_2) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ c^2 = de;\ [b,a] = e;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,94]	$C_3 \rtimes (C_4 \times Q_8) \quad (4,6,12)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ b^2 = d;\ c^2 = de;\ [c,b] = d;\ [f,a] = f$
[96,95]	$C_3 \rtimes (C_4 \rtimes Q_8) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ b^2 = e;\ c^2 = d;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,96]	$C_3 \rtimes ((C_2 \times C_2)\,.\,(C_2 \times C_2 \times C_2)) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ b^2 = d;\ c^2 = e;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,97]	$C_3 \rtimes ((C_2 \times C_2)\,.\,(C_2 \times C_2 \times C_2)) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ b^2 = e;\ c^2 = d;\ [c,a] = de;\ [c,b] = e;\ [f,a] = f$
[96,98]	$(C_4 \rtimes C_4) \times S_3 \quad (4,6,12)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = d;\ [c,b] = e;\ [f,a] = f$
[96,99]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2) \quad (4,6,12)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = d;\ [c,a] = e;\ [c,b] = e;\ [f,a] = f$
[96,100]	$C_3 \rtimes (C_4 \times D_8) \quad (4,6,12)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = d;\ [b,a] = e;\ [c,b] = e;\ [f,a] = f$
[96,101]	$C_3 \rtimes ((C_4 \times C_2 \times C_2) \rtimes C_2) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = d;\ c^2 = e;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,102]	$C_3 \rtimes ((C_4 \times C_2 \times C_2) \rtimes C_2) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = d;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,103]	$C_3 \rtimes ((C_2 \times Q_8) \rtimes C_2) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = d;\ c^2 = e;\ [c,a] = de;\ [c,b] = e;\ [f,a] = f$
[96,104]	$C_3 \rtimes ((C_2 \times Q_8) \rtimes C_2) \quad (4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = de;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$

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[96,105]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = d;\ c^2 = e;\ [b,a] = e;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,106]	$C_2 \times C_8 \times S_3$ $(16,3,24)$ $R = (2,2,2,2,2,3);\ c^2 = d;\ d^2 = e;\ [f,a] = f$
[96,107]	$C_2 \times (C_{24} \rtimes C_2)$ $(8,6,24)$ $R = (2,2,2,2,2,3);\ c^2 = d;\ d^2 = e;\ [c,a] = e;\ [f,a] = f$
[96,108]	$C_3 \rtimes ((C_8 \times C_2) \rtimes C_2)$ $(8,6,24)$ $R = (2,2,2,2,2,3);\ c^2 = d;\ d^2 = e;\ [b,a] = e;\ [f,a] = f$
[96,109]	$C_2 \times (C_{24} \rtimes C_2)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ c^2 = d;\ d^2 = e;\ [c,a] = d;\ [d,a] = e;\ [f,a] = f$
[96,110]	$C_2 \times D_{48}$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ c^2 = de;\ d^2 = e;\ [c,a] = d;\ [d,a] = e;\ [f,a] = f$
[96,111]	$C_3 \rtimes ((C_8 \times C_2) \rtimes C_2)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = d;\ d^2 = e;\ [c,a] = d;\ [d,a] = e;\ [f,a] = f$
[96,112]	$C_2 \times (C_3 \rtimes Q_{16})$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ c^2 = de;\ d^2 = e;\ [c,a] = d;\ [d,a] = e;\ [f,a] = f$
[96,113]	$(C_8 \rtimes C_2) \times S_3$ $(4,6,24)$ $R = (2,2,2,2,2,3);\ c^2 = d;\ d^2 = e;\ [c,b] = e;\ [f,a] = f$
[96,114]	$C_3 \rtimes ((C_8 \times C_2) \rtimes C_2)$ $(4,6,24)$ $R = (2,2,2,2,2,3);\ c^2 = d;\ d^2 = e;\ [b,a] = e;\ [c,b] = e;\ [f,a] = f$
[96,115]	$C_3 \rtimes (C_8 \rtimes (C_2 \times C_2))$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ c^2 = d;\ d^2 = e;\ [c,a] = d;\ [c,b] = e;\ [d,a] = e;\ [f,a] = f$
[96,116]	$C_3 \rtimes ((C_2 \times Q_8) \rtimes C_2)$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = d;\ d^2 = e;\ [c,a] = d;\ [c,b] = e;\ [d,a] = e;\ [f,a] = f$
[96,117]	$D_{16} \times S_3$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ d^2 = e;\ [c,b] = d;\ [d,b] = e;\ [d,c] = e;\ [f,a] = f$
[96,118]	$C_3 \rtimes (C_8 \rtimes (C_2 \times C_2))$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ d^2 = e;\ [c,a] = e;\ [c,b] = d;\ [d,b] = e;\ [d,c] = e;\ [f,a] = f$
[96,119]	$C_3 \rtimes ((C_8 \times C_2) \rtimes C_2)$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ d^2 = e;\ [b,a] = e;\ [c,a] = e;\ [c,b] = d;\ [d,b] = e;\ [d,c] = e;\ [f,a] = f$
[96,120]	$\mathrm{QD}_{16} \times S_3$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ c^2 = e;\ d^2 = e;\ [c,b] = d;\ [d,b] = e;\ [d,c] = e;\ [f,a] = f$
[96,121]	$C_3 \rtimes (C_8 \rtimes (C_2 \times C_2))$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ c^2 = e;\ d^2 = e;\ [c,a] = e;\ [c,b] = d;\ [d,b] = e;\ [d,c] = e;\ [f,a] = f$
[96,122]	$C_3 \rtimes ((C_2 \times Q_8) \rtimes C_2)$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ d^2 = e;\ [c,a] = e;\ [c,b] = d;\ [d,b] = e;\ [d,c] = e;\ [f,a] = f$
[96,123]	$C_3 \rtimes ((C_8 \times C_2) \rtimes C_2)$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ c^2 = e;\ d^2 = e;\ [b,a] = e;\ [c,a] = e;\ [c,b] = d;\ [d,b] = e;\ [d,c] = e;$ $[f,a] = f$
[96,124]	$Q_{16} \times S_3$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = e;\ d^2 = e;\ [c,b] = d;\ [d,b] = e;\ [d,c] = e;\ [f,a] = f$
[96,125]	$C_3 \rtimes ((C_2 \times Q_8) \rtimes C_2)$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = e;\ d^2 = e;\ [c,a] = e;\ [c,b] = d;\ [d,b] = e;\ [d,c] = e;$ $[f,a] = f$
[96,126]	$C_3 \rtimes ((C_8 \times C_2) \rtimes C_2)$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = e;\ d^2 = e;\ [b,a] = e;\ [c,a] = e;\ [c,b] = d;\ [d,b] = e;$ $[d,c] = e;\ [f,a] = f$

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[96,127]	$C_2 \times C_2 \times (C_3 \rtimes C_8)$ $(16,3,24)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ d^2 = e;\ [f,a] = f$
[96,128]	$C_2 \times ((C_3 \rtimes C_8) \rtimes C_2)$ $(8,6,24)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ d^2 = e;\ [c,a] = e;\ [f,a] = f$
[96,129]	$C_2 \times C_4 \times (C_3 \rtimes C_4)$ $(16,3,12)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ c^2 = e;\ [f,a] = f$
[96,130]	$C_2 \times ((C_3 \rtimes C_4) \rtimes C_4)$ $(8,6,12)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ c^2 = d;\ [c,a] = e;\ [f,a] = f$
[96,131]	$C_3 \rtimes ((C_2 \times Q_8) \rtimes C_2)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ b^2 = e;\ [b,a] = e;\ [c,a] = d;\ [f,a] = f$
[96,132]	$C_2 \times (C_{12} \rtimes C_4)$ $(8,6,12)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ c^2 = e;\ [c,a] = e;\ [f,a] = f$
[96,133]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2)$ $(8,6,12)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ b^2 = e;\ [c,a] = e;\ [f,a] = f$
[96,134]	$C_2 \times ((C_{12} \times C_2) \rtimes C_2)$ $(8,6,12)$ $R = (2,2,2,2,2,3);\ c^2 = d;\ [c,a] = e;\ [f,a] = f$
[96,135]	$C_4 \times ((C_6 \times C_2) \rtimes C_2)$ $(8,6,12)$ $R = (2,2,2,2,2,3);\ c^2 = d;\ [b,a] = e;\ [f,a] = f$
[96,136]	$C_3 \rtimes ((C_4 \times C_2 \times C_2) \rtimes C_2)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ c^2 = e;\ [b,a] = e;\ [c,a] = d;\ [f,a] = f$
[96,137]	$C_3 \rtimes ((C_4 \times C_2 \times C_2) \rtimes C_2)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ [b,a] = e;\ [c,a] = d;\ [f,a] = f$
[96,138]	$C_2 \times ((C_3 \times D_8) \rtimes C_2)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ d^2 = e;\ [c,a] = d;\ [d,a] = e;\ [d,c] = e;\ [f,a] = f$
[96,139]	$C_3 \rtimes (C_8 \rtimes (C_2 \times C_2))$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ d^2 = e;\ [c,a] = d;\ [c,b] = e;\ [d,a] = e;\ [d,c] = e;\ [f,a] = f$
[96,140]	$C_2 \times ((C_3 \rtimes Q_8) \rtimes C_2)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ d^2 = e;\ [c,a] = d;\ [d,a] = e;\ [d,c] = e;\ [f,a] = f$
[96,141]	$D_8 \times (C_3 \rtimes C_4)$ $(4,6,12)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ [c,b] = e;\ [f,a] = f$
[96,142]	$C_3 \rtimes ((C_4 \times C_2 \times C_2) \rtimes C_2)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,143]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ b^2 = d;\ [c,a] = de;\ [c,b] = d;\ [f,a] = f$
[96,144]	$C_3 \rtimes ((C_2 \times C_2 \times C_2 \times C_2) \rtimes C_2)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,145]	$C_3 \rtimes ((C_4 \times C_2 \times C_2) \rtimes C_2)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,146]	$C_3 \rtimes ((C_4 \times C_2 \times C_2) \rtimes C_2)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ [b,a] = e;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,147]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ [b,a] = e;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,148]	$C_2 \times ((C_3 \times Q_8) \rtimes C_2)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ c^2 = e;\ d^2 = e;\ [c,a] = d;\ [d,a] = e;\ [d,c] = e;\ [f,a] = f$
[96,149]	$C_3 \rtimes ((C_2 \times Q_8) \rtimes C_2)$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = e;\ d^2 = e;\ [c,a] = d;\ [c,b] = e;\ [d,a] = e;\ [d,c] = e;$ $[f,a] = f$

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[96,150]	$C_2 \times (C_3 \rtimes Q_{16})$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ c^2 = e;\ d^2 = e;\ [c,a] = d;\ [d,a] = e;\ [d,c] = e;\ [f,a] = f$
[96,151]	$C_3 \rtimes (C_4 \rtimes Q_8)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ b^2 = d;\ c^2 = d;\ [c,a] = e;\ [c,b] = d;\ [f,a] = f$
[96,152]	$Q_8 \times (C_3 \rtimes C_4)$ $(4,6,12)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ b^2 = e;\ c^2 = e;\ [c,b] = e;\ [f,a] = f$
[96,153]	$C_3 \rtimes ((C_2 \times Q_8) \rtimes C_2)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = e;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,154]	$C_3 \rtimes ((C_4 \times C_4) \rtimes C_2)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = e;\ [b,a] = e;\ [c,a] = d;\ [c,b] = e;\ [f,a] = f$
[96,155]	$C_3 \rtimes ((C_8 \times C_2) \rtimes C_2)$ $(4,6,24)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ d^2 = e;\ [c,b] = e;\ [f,a] = f$
[96,156]	$C_3 \rtimes (C_8 \rtimes (C_2 \times C_2))$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ d^2 = e;\ [c,a] = d;\ [c,b] = e;\ [d,a] = e;\ [d,c] = e;\ [f,a] = f$
[96,157]	$C_3 \rtimes ((C_8 \times C_2) \rtimes C_2)$ $(4,12,24)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ d^2 = e;\ [c,a] = d;\ [d,a] = e;\ [d,c] = e;\ [f,a] = f$
[96,158]	$C_3 \rtimes ((C_2 \times Q_8) \rtimes C_2)$ $(2,12,24)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ d^2 = e;\ [c,a] = d;\ [c,b] = e;\ [d,a] = e;\ [d,c] = e;\ [f,a] = f$
[96,159]	$C_2 \times ((C_6 \times C_2) \rtimes C_4)$ $(8,6,12)$ $R = (2,2,2,2,2,3);\ a^2 = d;\ [c,a] = e;\ [f,a] = f$
[96,160]	$C_3 \rtimes ((C_2 \times C_2 \times C_2 \times C_2) \rtimes C_2)$ $(4,12,12)$ $R = (2,2,2,2,2,3);\ [b,a] = e;\ [c,a] = d;\ [f,a] = f$
[96,161]	$C_2 \times C_4 \times C_{12}$ $(96,1,12)$ 阿贝尔。
[96,162]	$C_6 \times ((C_4 \times C_2) \rtimes C_2)$ $(24,2,12)$ $R = (2,2,2,3,2,2);\ a^2 = f;\ [b,a] = e$
[96,163]	$C_6 \times (C_4 \rtimes C_4)$ $(24,2,12)$ $R = (2,2,2,3,2,2);\ a^2 = f;\ b^2 = e;\ [b,a] = e$
[96,164]	$C_3 \times ((C_4 \times C_4) \rtimes C_2)$ $(24,2,12)$ $R = (2,2,2,3,2,2);\ a^2 = f;\ c^2 = e;\ [b,a] = e$
[96,165]	$C_{12} \times D_8$ $(24,2,12)$ $R = (2,2,2,3,2,2);\ a^2 = f;\ c^2 = f;\ [b,a] = e$
[96,166]	$C_{12} \times Q_8$ $(24,2,12)$ $R = (2,2,2,3,2,2);\ a^2 = f;\ b^2 = e;\ c^2 = ef;\ [b,a] = e$
[96,167]	$C_3 \times ((C_2 \times C_2 \times C_2 \times C_2) \rtimes C_2)$ $(12,4,12)$ $R = (2,2,2,3,2,2);\ [b,a] = e;\ [c,a] = f$
[96,168]	$C_3 \times ((C_4 \times C_2 \times C_2) \rtimes C_2)$ $(12,4,12)$ $R = (2,2,2,3,2,2);\ b^2 = e;\ [b,a] = e;\ [c,a] = f$
[96,169]	$C_3 \times ((C_2 \times Q_8) \rtimes C_2)$ $(12,4,12)$ $R = (2,2,2,3,2,2);\ a^2 = e;\ b^2 = e;\ [b,a] = e;\ [c,a] = f$
[96,170]	$C_3 \times ((C_4 \times C_2 \times C_2) \rtimes C_2)$ $(12,4,12)$ $R = (2,2,2,3,2,2);\ c^2 = e;\ [b,a] = e;\ [c,a] = f$
[96,171]	$C_3 \times ((C_4 \times C_4) \rtimes C_2)$ $(12,4,12)$ $R = (2,2,2,3,2,2);\ b^2 = f;\ c^2 = e;\ [b,a] = e;\ [c,a] = f$
[96,172]	$C_3 \times ((C_2 \times C_2)\,.\,(C_2 \times C_2 \times C_2))$ $(12,4,12)$ $R = (2,2,2,3,2,2);\ a^2 = e;\ b^2 = f;\ c^2 = e;\ [b,a] = e;\ [c,a] = f$
[96,173]	$C_3 \times ((C_4 \times C_4) \rtimes C_2)$ $(12,4,12)$ $R = (2,2,2,3,2,2);\ b^2 = ef;\ c^2 = e;\ [b,a] = e;\ [c,a] = f$

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[96,174]	$C_3 \times ((C_4 \times C_4) \rtimes C_2)$ $(12,4,12)$ $R = (2,2,2,3,2,2);\ b^2 = e;\ c^2 = f;\ [b,a] = e;\ [c,a] = f$
[96,175]	$C_3 \times (C_4 \rtimes Q_8)$ $(12,4,12)$ $R = (2,2,2,3,2,2);\ a^2 = e;\ b^2 = e;\ c^2 = f;\ [b,a] = e;\ [c,a] = f$
[96,176]	$C_2 \times C_2 \times C_{24}$ $(96,1,24)$ 阿贝尔。
[96,177]	$C_6 \times (C_8 \rtimes C_2)$ $(24,2,24)$ $R = (2,2,2,3,2,2);\ a^2 = e;\ e^2 = f;\ [b,a] = f$
[96,178]	$C_3 \times ((C_8 \times C_2) \rtimes C_2)$ $(24,2,24)$ $R = (2,2,2,3,2,2);\ a^2 = e;\ e^2 = f;\ [c,b] = f$
[96,179]	$C_6 \times D_{16}$ $(12,4,24)$ $R = (2,2,2,3,2,2);\ e^2 = f;\ [b,a] = e;\ [e,a] = f;\ [e,b] = f$
[96,180]	$C_6 \times \mathrm{QD}_{16}$ $(12,4,24)$ $R = (2,2,2,3,2,2);\ a^2 = f;\ e^2 = f;\ [b,a] = e;\ [e,a] = f;\ [e,b] = f$
[96,181]	$C_6 \times Q_{16}$ $(12,4,24)$ $R = (2,2,2,3,2,2);\ a^2 = f;\ b^2 = f;\ e^2 = f;\ [b,a] = e;\ [e,a] = f;\ [e,b] = f$
[96,182]	$C_3 \times ((C_8 \times C_2) \rtimes C_2)$ $(12,4,24)$ $R = (2,2,2,3,2,2);\ c^2 = f;\ e^2 = f;\ [b,a] = e;\ [e,a] = f;\ [e,b] = f$
[96,183]	$C_3 \times (C_8 \rtimes (C_2 \times C_2))$ $(6,4,24)$ $R = (2,2,2,3,2,2);\ e^2 = f;\ [b,a] = e;\ [c,a] = f;\ [e,a] = f;\ [e,b] = f$
[96,184]	$C_3 \times ((C_2 \times Q_8) \rtimes C_2)$ $(6,4,24)$ $R = (2,2,2,3,2,2);\ b^2 = f;\ e^2 = f;\ [b,a] = e;\ [c,a] = f;\ [e,a] = f;\ [e,b] = f$
[96,185]	$(C_2 \times C_2) \rtimes (C_3 \rtimes Q_8)$ $(2,24,12)$ $R = (2,2,2,3,2,2);\ a^2 = c;\ b^2 = c;\ [b,a] = c;\ [d,a] = d;\ [e,a] = ef;\ [e,d] = ef;\ [f,a] = ef;$ $[f,d] = e$
[96,186]	$C_4 \times S_4$ $(4,12,12)$ $R = (2,2,2,3,2,2);\ b^2 = c;\ [d,a] = d;\ [e,a] = ef;\ [e,d] = ef;\ [f,a] = ef;\ [f,d] = e$
[96,187]	$C_4 \rtimes S_4$ $(2,24,12)$ $R = (2,2,2,3,2,2);\ b^2 = c;\ [b,a] = c;\ [d,a] = d;\ [e,a] = ef;\ [e,d] = ef;\ [f,a] = ef;$ $[f,d] = e$
[96,188]	$C_2 \times (C_2 \,.\, S_4 = \mathrm{SL}(2,3) \,.\, C_2)$ $(4,24,24)$ $R = (2,2,3,2,2,2);\ a^2 = f;\ d^2 = f;\ e^2 = f;\ [c,a] = c;\ [d,a] = def;\ [d,c] = de;\ [e,a] = de;$ $[e,c] = df;\ [e,d] = f$
[96,189]	$C_2 \times \mathrm{GL}(2,3)$ $(4,24,24)$ $R = (2,2,3,2,2,2);\ d^2 = f;\ e^2 = f;\ [c,a] = c;\ [d,a] = def;\ [d,c] = de;\ [e,a] = de;$ $[e,c] = df;\ [e,d] = f$
[96,190]	$(C_2 \times Q_8) \rtimes S_3$ $(2,24,24)$ $R = (2,2,3,2,2,2);\ d^2 = f;\ e^2 = f;\ [b,a] = f;\ [c,a] = c;\ [d,a] = def;\ [d,c] = de;$ $[e,a] = de;\ [e,c] = df;\ [e,d] = f$
[96,191]	$(C_2 \,.\, S_4 = \mathrm{SL}(2,3) \,.\, C_2) \rtimes C_2$ $(2,24,24)$ $R = (2,2,3,2,2,2);\ a^2 = f;\ b^2 = f;\ d^2 = f;\ e^2 = f;\ [b,a] = f;\ [c,a] = c;\ [d,a] = def;$ $[d,c] = de;\ [e,a] = de;\ [e,c] = df;\ [e,d] = f$
[96,192]	$((C_4 \times C_2) \rtimes C_2) \rtimes S_3$ $(4,24,24)$ $R = (2,2,3,2,2,2);\ b^2 = f;\ d^2 = f;\ e^2 = f;\ [c,a] = c;\ [d,a] = def;\ [d,c] = de;\ [e,a] = de;$ $[e,c] = df;\ [e,d] = f$
[96,193]	$((C_4 \times C_2) \rtimes C_2) \rtimes S_3$ $(2,24,24)$ $R = (2,2,3,2,2,2);\ b^2 = f;\ d^2 = f;\ e^2 = f;\ [b,a] = f;\ [c,a] = c;\ [d,a] = def;\ [d,c] = de;$ $[e,a] = de;\ [e,c] = df;\ [e,d] = f$

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[96,194]	$C_2 \times (A_4 \rtimes C_4)$ $(4,12,12)$ $R = (2,2,2,3,2,2);\ a^2 = c;\ [d,a] = d;\ [e,a] = ef;\ [e,d] = ef;\ [f,a] = ef;\ [f,d] = e$
[96,195]	$(C_2 \times C_2) \rtimes S_4$ $(2,24,12)$ $R = (2,2,2,3,2,2);\ [b,a] = c;\ [d,a] = d;\ [e,a] = ef;\ [e,d] = ef;\ [f,a] = ef;\ [f,d] = e$
[96,196]	$C_2 \times C_4 \times A_4$ $(8,4,12)$ $R = (2,2,3,2,2,2);\ b^2 = d;\ [e,c] = ef;\ [f,c] = e$
[96,197]	$D_8 \times A_4$ $(2,8,12)$ $R = (2,2,3,2,2,2);\ [b,a] = d;\ [e,c] = ef;\ [f,c] = e$
[96,198]	$C_2 \times C_2 \times \mathrm{SL}(2,3)$ $(8,8,12)$ $R = (2,2,3,2,2,2);\ d^2 = f;\ e^2 = f;\ [d,c] = def;\ [e,c] = df;\ [e,d] = f$
[96,199]	$Q_8 \times A_4$ $(2,8,12)$ $R = (2,2,3,2,2,2);\ a^2 = d;\ b^2 = d;\ [b,a] = d;\ [e,c] = ef;\ [f,c] = e$
[96,200]	$C_2 \times (((C_4 \times C_2) \rtimes C_2) \rtimes C_3)$ $(8,8,12)$ $R = (2,2,3,2,2,2);\ b^2 = f;\ d^2 = f;\ e^2 = f;\ [d,c] = def;\ [e,c] = df;\ [e,d] = f$
[96,201]	$((C_2 \times C_2 \times C_2) \rtimes (C_2 \times C_2)) \rtimes C_3$ $(2,8,12)$ $R = (2,2,3,2,2,2);\ a^2 = f;\ b^2 = f;\ d^2 = f;\ e^2 = f;\ [b,a] = f;\ [d,c] = def;\ [e,c] = df;$ $[e,d] = f$
[96,202]	$((C_2 \times Q_8) \rtimes C_2) \rtimes C_3$ $(2,8,12)$ $R = (2,2,3,2,2,2);\ d^2 = f;\ e^2 = f;\ [b,a] = f;\ [d,c] = def;\ [e,c] = df;\ [e,d] = f$
[96,203]	$(C_2 \times C_2 \times Q_8) \rtimes C_3$ $(2,32,12)$ $R = (3,2,2,2,2,2);\ d^2 = f;\ e^2 = f;\ [b,a] = bc;\ [c,a] = b;\ [d,a] = def;\ [e,a] = df;\ [e,d] = f$
[96,204]	$((C_2 \times C_2 \times C_2) \rtimes (C_2 \times C_2)) \rtimes C_3$ $(2,32,12)$ $R = (3,2,2,2,2,2);\ [b,a] = bc;\ [c,a] = b;\ [d,a] = de;\ [d,c] = f;\ [e,a] = d;\ [e,b] = f$
[96,205]	$C_2 \times C_2 \times (C_3 \rtimes Q_8)$ $(8,6,12)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ d^2 = e;\ [d,a] = e;\ [f,a] = f$
[96,206]	$C_2 \times C_2 \times C_4 \times S_3$ $(16,3,12)$ $R = (2,2,2,2,2,3);\ d^2 = e;\ [f,a] = f$
[96,207]	$C_2 \times C_2 \times D_{24}$ $(8,6,12)$ $R = (2,2,2,2,2,3);\ d^2 = e;\ [d,a] = e;\ [f,a] = f$
[96,208]	$C_2 \times ((C_{12} \times C_2) \rtimes C_2)$ $(8,6,12)$ $R = (2,2,2,2,2,3);\ c^2 = e;\ [d,a] = e;\ [f,a] = f$
[96,209]	$C_2 \times D_8 \times S_3$ $(4,6,12)$ $R = (2,2,2,2,2,3);\ [d,c] = e;\ [f,a] = f$
[96,210]	$C_2 \times ((C_4 \times S_3) \rtimes C_2)$ $(4,6,12)$ $R = (2,2,2,2,2,3);\ c^2 = e;\ [d,a] = e;\ [d,c] = e;\ [f,a] = f$
[96,211]	$C_3 \rtimes ((C_2 \times C_2 \times C_2) \rtimes (C_2 \times C_2))$ $(2,6,12)$ $R = (2,2,2,2,2,3);\ [c,b] = e;\ [d,a] = e;\ [f,a] = f$
[96,212]	$C_2 \times Q_8 \times S_3$ $(4,6,12)$ $R = (2,2,2,2,2,3);\ c^2 = e;\ d^2 = e;\ [d,c] = e;\ [f,a] = f$
[96,213]	$C_2 \times ((C_4 \times S_3) \rtimes C_2)$ $(4,6,12)$ $R = (2,2,2,2,2,3);\ c^2 = e;\ d^2 = e;\ [d,a] = e;\ [d,c] = e;\ [f,a] = f$
[96,214]	$C_3 \rtimes ((C_2 \times Q_8) \rtimes C_2)$ $(2,6,12)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = e;\ [c,b] = e;\ [d,a] = e;\ [f,a] = f$
[96,215]	$((C_4 \times C_2) \rtimes C_2) \times S_3$ $(4,6,12)$ $R = (2,2,2,2,2,3);\ c^2 = e;\ [d,b] = e;\ [f,a] = f$
[96,216]	$C_3 \rtimes ((C_2 \times C_2 \times C_2) \rtimes (C_2 \times C_2))$ $(2,6,12)$ $R = (2,2,2,2,2,3);\ d^2 = e;\ [c,b] = e;\ [d,a] = e;\ [f,a] = f$

编号	96 阶群（续）
[96,217]	$C_3 \rtimes ((C_2 \times Q_8) \rtimes C_2)$ $(2,6,12)$ $R = (2,2,2,2,2,3);\ b^2 = e;\ c^2 = e;\ d^2 = e;\ [c,b] = e;\ [d,a] = e;\ [f,a] = f$
[96,218]	$C_2 \times C_2 \times C_2 \times (C_3 \rtimes C_4)$ $(16,3,12)$ $R = (2,2,2,2,2,3);\ a^2 = e;\ [f,a] = f$
[96,219]	$C_2 \times C_2 \times ((C_6 \times C_2) \rtimes C_2)$ $(8,6,12)$ $R = (2,2,2,2,2,3);\ [d,a] = e;\ [f,a] = f$
[96,220]	$C_2 \times C_2 \times C_2 \times C_{12}$ $(96,1,12)$ 阿贝尔。
[96,221]	$C_2 \times C_6 \times D_8$ $(24,2,12)$ $R = (2,2,2,2,3,2);\ [b,a] = f$
[96,222]	$C_2 \times C_6 \times Q_8$ $(24,2,12)$ $R = (2,2,2,2,3,2);\ a^2 = f;\ b^2 = f;\ [b,a] = f$
[96,223]	$C_6 \times ((C_4 \times C_2) \rtimes C_2)$ $(24,2,12)$ $R = (2,2,2,2,3,2);\ c^2 = f;\ [b,a] = f$
[96,224]	$C_3 \times ((C_2 \times C_2 \times C_2) \rtimes (C_2 \times C_2))$ $(6,2,12)$ $R = (2,2,2,2,3,2);\ [b,a] = f;\ [c,b] = f;\ [d,a] = f$
[96,225]	$C_3 \times ((C_2 \times Q_8) \rtimes C_2)$ $(6,2,12)$ $R = (2,2,2,2,3,2);\ b^2 = f;\ c^2 = f;\ [b,a] = f;\ [c,b] = f;\ [d,a] = f$
[96,226]	$C_2 \times C_2 \times S_4$ $(4,12,12)$ $R = (2,2,2,3,2,2);\ [d,a] = d;\ [e,a] = ef;\ [e,d] = ef;\ [f,a] = ef;\ [f,d] = e$
[96,227]	$(C_2 \times C_2) \rtimes S_4$ $(1,48,12)$ $R = (2,3,2,2,2,2);\ [b,a] = b;\ [c,a] = cd;\ [c,b] = cd;\ [d,a] = cd;\ [d,b] = c;\ [e,a] = ef;$ $[e,b] = f;\ [f,a] = ef;\ [f,b] = ef$
[96,228]	$C_2 \times C_2 \times C_2 \times A_4$ $(8,4,6)$ $R = (2,2,2,3,2,2);\ [e,d] = ef;\ [f,d] = e$
[96,229]	$C_2 \times ((C_2 \times C_2 \times C_2 \times C_2) \rtimes C_3)$ $(2,16,6)$ $R = (2,3,2,2,2,2);\ [c,b] = cd;\ [d,b] = c;\ [e,b] = ef;\ [f,b] = e$
[96,230]	$C_2 \times C_2 \times C_2 \times C_2 \times S_3$ $(16,3,6)$ $R = (2,2,2,2,2,3);\ [f,a] = f$
[96,231]	$C_2 \times C_2 \times C_2 \times C_2 \times C_6$ $(96,1,6)$ 阿贝尔。

97 阶群

只有一个同构类型：C_{97}，编号 $[97,1]$。

证明. 任取非单位元 $a \in G$。由拉格朗日定理，$\mathrm{ord}(a) \mid 97$，又 $\mathrm{ord}(a) > 1$，所以 $\mathrm{ord}(a) = 97$。因此 $G = \langle a \rangle \cong C_{97}$。 □

98 阶群

共有 5 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[98,1]	$D_{98}\quad(1,49,98)$ $R=(2,7,7);\ b^7=c;\ [b,a]=b^5c^6;\ [c,a]=c^5$
[98,2]	$C_{98}\quad(98,1,98)\quad$阿贝尔。
[98,3]	$C_7\times D_{14}\quad(7,7,14)$ $R=(2,7,7);\ [c,a]=c^5$
[98,4]	$(C_7\times C_7)\rtimes C_2\quad(1,49,14)$ $R=(2,7,7);\ [b,a]=b^5;\ [c,a]=c^5$
[98,5]	$C_7\times C_{14}\quad(98,1,14)\quad$阿贝尔。

99 阶群

共有 2 个同构类型，其中 2 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[99,1]	$C_{99}\quad(99,1,99)\quad$阿贝尔。
[99,2]	$C_3\times C_{33}\quad(99,1,33)\quad$阿贝尔。

证明. 由 Sylow 定理，$n_{11}=1$ 且 $n_3=1$。两个 Sylow 子群都正规，且交为 1，故它们彼此交换，G 是二者的直积。阶为 3^2 的 Sylow 子群只有 C_9 与 C_3^2，阶为 11 的 Sylow 子群为 C_{11}，因而恰得表中的两种阿贝尔群。 □

100 阶群

共有 16 个同构类型，其中 4 个阿贝尔。

编号	群；不变量 (z,d,e)；非平凡生成关系
[100,1]	$C_{25}\rtimes C_4\quad(2,25,100)$ $R=(2,2,5,5);\ a^2=b;\ c^5=d^4;\ [c,a]=c^3d;\ [d,a]=d^3$
[100,2]	$C_{100}\quad(100,1,100)\quad$阿贝尔。
[100,3]	$C_{25}\rtimes C_4\quad(1,25,100)$ $R=(2,2,5,5);\ a^2=b;\ c^5=d;\ [c,a]=cd;\ [c,b]=c^3d^4;\ [d,a]=d;\ [d,b]=d^3$
[100,4]	$D_{100}\quad(2,25,50)$ $R=(2,2,5,5);\ c^5=d^4;\ [c,a]=c^3d;\ [d,a]=d^3$
[100,5]	$C_2\times C_{50}\quad(100,1,50)\quad$阿贝尔。
[100,6]	$C_5\times(C_5\rtimes C_4)\quad(10,5,20)$ $R=(2,5,2,5);\ a^2=c;\ [d,a]=d^3$
[100,7]	$(C_5\times C_5)\rtimes C_4\quad(2,25,20)$ $R=(2,2,5,5);\ a^2=b;\ [c,a]=c^3;\ [d,a]=d^3$
[100,8]	$C_5\times C_{20}\quad(100,1,20)\quad$阿贝尔。
[100,9]	$C_5\times(C_5\rtimes C_4)\quad(5,5,20)$ $R=(2,5,2,5);\ a^2=c;\ [d,a]=d;\ [d,c]=d^3$

编号	100 阶群（续）
[100,10]	$(C_5 \times C_5) \rtimes C_4 \quad (1, 25, 20)$ $R = (2, 2, 5, 5);\ a^2 = b;\ [c, a] = c^3;\ [d, a] = d^2;\ [d, b] = d^3$
[100,11]	$(C_5 \times C_5) \rtimes C_4 \quad (1, 25, 20)$ $R = (2, 2, 5, 5);\ a^2 = b;\ [c, a] = c;\ [c, b] = c^3;\ [d, a] = d;\ [d, b] = d^3$
[100,12]	$(C_5 \times C_5) \rtimes C_4 \quad (1, 25, 20)$ $R = (2, 2, 5, 5);\ a^2 = b;\ [c, a] = c^2;\ [c, b] = c^3;\ [d, a] = d;\ [d, b] = d^3$
[100,13]	$D_{10} \times D_{10} \quad (1, 25, 10)$ $R = (2, 2, 5, 5);\ [c, b] = c^3;\ [d, a] = d^3$
[100,14]	$C_{10} \times D_{10} \quad (10, 5, 10)$ $R = (2, 2, 5, 5);\ [d, a] = d^3$
[100,15]	$C_2 \times ((C_5 \times C_5) \rtimes C_2) \quad (2, 25, 10)$ $R = (2, 2, 5, 5);\ [c, a] = c^3;\ [d, a] = d^3$
[100,16]	$C_{10} \times C_{10} \quad (100, 1, 10)$ 阿贝尔。

记号与分类依据

群名

C_n 为 n 阶循环群；S_n 为 n 个字母的置换群；A_n 为偶置换群。D_{2m} 的下标是**群的阶**，即

$$D_{2m} = \langle r, s \mid r^m = s^2 = 1,\ srs^{-1} = r^{-1} \rangle.$$

Q_8 为四元数群；一般的双循环群为

$$\mathrm{Dic}_m = \langle a, x \mid a^{2m} = 1,\ x^2 = a^m,\ xax^{-1} = a^{-1} \rangle.$$

当 m 为 2 的幂时，亦记作 Q_{4m}。半二面体群的记号为

$$\mathrm{QD}_{2^k} = \langle r, s \mid r^{2^{k-1}} = s^2 = 1,\ srs^{-1} = r^{2^{k-2}-1} \rangle \quad (k \geq 4).$$

$\mathrm{GL}(2,3)$、$\mathrm{SL}(2,3)$ 分别为 $\mathbb{F}_3$ 上的可逆矩阵群与行列式为 1 的矩阵群。

直积、半直积与扩张

$G \times H$ 是直积。$N \rtimes H$ 是半直积，必须同时指定 $H \to \mathrm{Aut}(N)$ 的作用；表中省略的作用由该项的完整生成关系确定。$N.H$ 表示以 N 为正规子群、商群为 H 的扩张；点号本身不判断是否分裂。**相同的结构简写可能对应不同构的群**，例如 $[16,3]$ 与 $[16,13]$。编号和生成关系区分这些类型。

编号与生成关系

$[n,i]$ 指 SmallGroups 库的 $\mathrm{SmallGroup}(n,i)$。相同阶数下不同编号互不同构。目录共有 1048 个类型，其中 185 个阿贝尔，862 个非阿贝尔可解，以及 A_5。

每个非阿贝尔可解项使用有序生成元 a, b, c, d, e, f 的前若干个。$R = (r_1, \ldots, r_t)$ 是相对阶；任一元素唯一写成

$$a_1^{u_1} \cdots a_t^{u_t}, \qquad 0 \leq u_i < r_i, \qquad |G| = r_1 \cdots r_t.$$

相对阶不是生成元在全群中的实际阶。所有未列出的幂关系为 $a_i^{r_i} = 1$，所有未列出的交换子关系为 $[a_j, a_i] = 1$ $(j > i)$，约定

$$[x, y] = x^{-1}y^{-1}xy.$$

这些默认关系与表中列出的非平凡关系一起构成完整群表示。例如 $R = (2,2,2)$，$a^2 = c$，$[b,a] = c$ 给出 D_8；再加上 $b^2 = c$ 则给出 Q_8。

表中 (z, d, e) 分别为中心阶 $|\mathrm{Z}(G)|$、交换子群阶 $|G'|$ 和指数 $\exp(G)$。这些量用于核对，不能单独区分全部群。

分类依据

较大阶数的完整性引用 SmallGroups 库，而非本文重新证明。数据使用 GAP 4.15.1 与 SmallGrp 1.5.4 生成；全部 862 个非阿贝尔可解类型均由所列关系重新构造并核对编号。阿贝尔类型数另由有限阿贝尔群分类定理独立核对。

https://gap-packages.github.io/smallgrp/doc/chap1.html